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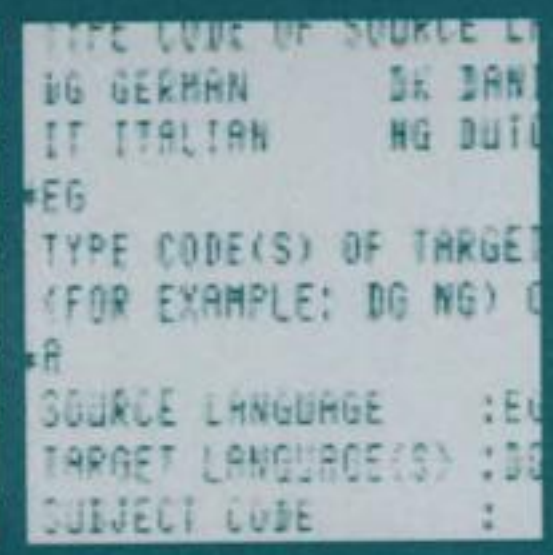

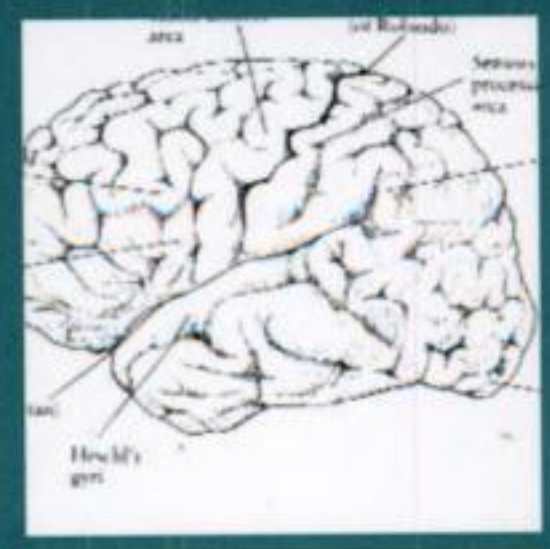

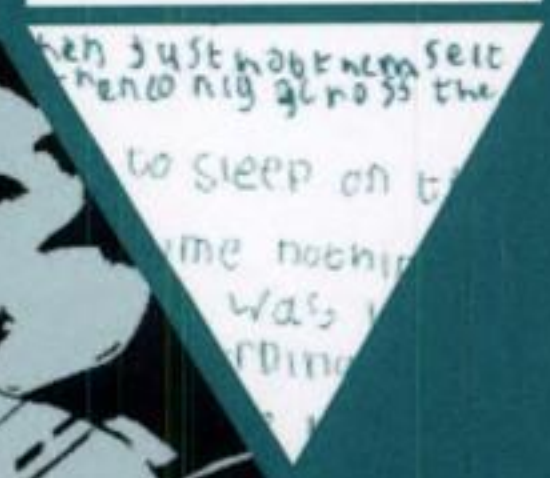

CORPUS LINGUISTICS

Investigating Language Structure and Use

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2 Lexicography

2.1 Investigating lexicographic issues

Lexicography is concerned with the meaning and use of words. Traditionally, lexicographic research investigated the meanings of words and synonyms. In more recent times, such investigations have been extended using corpus-based techniques to study the ways that words are used, considering issues such as:

- How common are different words?
- How common are the different senses for a given word?
- Do words have systematic associations with other words?
- Do words have systematic associations with particular registers or dialects?

This area of study is, of course, central to dictionary making. However, lexicographic research is also central to descriptive and applied linguistics. Many linguists argue that adequate grammatical analyses must incorporate lexicographic information about individual words. Further, as Chapter 4 shows, grammatical and lexicographic patterns interact in systematic ways. For applied linguists, lexicographic studies provide an important source of information for language students and teachers. For instance, where traditional approaches might identify a group of synonymous words, corpus-based lexicographic research attempts to show how related words are used in different ways and are appropriate in different contexts.

Unlike much of linguistics, the field of dictionary making has long been influenced by empirical and corpus-based methods. As early as 1755, for example, Johnson used a corpus of texts to gather

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authentic uses of words, which he then included as examples in his dictionary of English. In the late 1800s, a large group of texts was collected as the basis for the *Oxford English Dictionary*, and as many as 800 volunteers read through the texts and sent in citation slips with quotes using their assigned words.

Such work, however, differed greatly from the use of a corpus-based approach today. First of all, these earlier text collections were not meant to be representative of the language; rather, dictionary makers stressed the normative function of their work, aiming to describe the "proper" use of words. Johnson purposely limited his corpus to works that he regarded "as the wells of English undefiled, as the pure sources of genuine diction" (Johnson 1755, preface, page 7). Even other dictionaries, which were meant to be more complete and objective, were limited by the methods available for collecting and analyzing citations. The entries in the *Oxford English Dictionary*, for example, were dependent upon what the volunteer readers happened to notice, and dictionary makers found that readers paid more attention to unusual uses of words than common ones (Morton 1994, p. 95). Furthermore, spoken language of any sort was rarely included in any dictionary database until the 1970s.

Today, advances in computer technology have given corpus-based lexicographic research several advantages over earlier work. First, computers have made possible the collection and storage of very large corpora from a variety of sources. Complete texts or large chunks of texts can be stored on a computer so that analyses are not limited to sentence-length excerpts. With the ability to store more texts, we can also design corpora that are more representative of the language overall, rather than being limited to a select group of authors. (See Methodology Box 1 for more on representativeness in corpus design.)

In addition, computers facilitate analyses that are more complete and reliable. Unlike human readers, who are likely to miss certain occurrences of a word, computers can find all the instances of a word in a corpus and generate an exhaustive list of them. No occurrences are lost. Furthermore, computers can analyze the patterns of word associations on a far more complex scale than is possible by hand. For example, when a word occurs thousands or

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tens of thousands of times in a corpus, it is unfeasible to ask a person to count and sort all the other words that occur within four words on each side of that word. With a computer, tasks like this can be completed within a couple minutes.

Together, all of these features – the greater size of corpora, their more representative nature, the more thorough and more complex analyses – allow us to investigate a range of lexicographic research questions that were not feasible before. Today, corpus-based lexicographic investigations address six major types of research questions, each of which is illustrated in this chapter. These are:

1. What are the meanings associated with a particular word?

This research question is the traditional focus of lexicography. However, corpus linguistics makes it possible to identify the meanings of words by looking at their occurrences in natural contexts, rather than relying on intuitions about how a word is used or on incomplete citation collections. An investigation of word meanings is introduced in Section 2.2 by looking at the word *deal*.

2. What is the frequency of a word relative to other related words?

Investigations of this sort tell us how often different words are used, allowing us to identify particularly common and uncommon words. This information can be especially useful in designing teaching materials for language students. This research question is introduced in Section 2.3, by considering the frequency of *deal* relative to other words in the LOB Corpus. However, frequency information is more often used to study the distribution of words relative to certain non-linguistic characteristics, as in Question 3.

3. What non-linguistic association patterns does a particular word have (e.g., to registers, historical periods, or dialects)?

Investigating this question can help us understand the association patterns between words and non-linguistic factors. The results can help us to characterize language use patterns in different varieties, for example, describing differences in word use across registers, or following the development of words through time. Section 2.4 focuses on this type of research question by expanding the investigation of *deal* to consider its use across registers.

Sách có bản quyền

4. What words commonly co-occur with a particular word, and what is the distribution of these "collocational" sequences across registers?

This research question focuses on the patterned way that words group together: their "collocations." Such investigations can also be a preliminary step for other research questions. In this chapter, we study collocations as a first step for two other research questions: investigating the distribution of word senses and uses (Question 5), and comparing the use of seemingly synonymous words (Question 6).

5. How are the senses and uses of a word distributed?

This research question focuses on the different senses and uses of words. It considers questions such as: How common are the different senses and grammatical functions of a word? Are those senses and uses distributed in a systematic way across registers? These research questions are exemplified in Section 2.5, which takes another step in the exploration of *deal* by examining its senses, grammatical functions (as a noun versus verb), and their distribution across registers.

6. How are seemingly synonymous words used and distributed in different ways?

Languages have many words that are similar, and dictionary definitions often characterize such words as identical or "synonymous" in meaning. The patterns of use for these words, however, is often very different. Investigating the use and distribution of synonyms in a corpus allows us to determine their contextual preferences, associated with other collocates or associated with register differences. Section 2.6 addresses this research question by comparing the uses of *big*, *large*, and *great* across two registers.

Each of the above research questions can help us understand how people's use of words is systematically patterned. That is, part of the extended meaning of a word is the way in which it is typically used in association with other words or in association with particular non-linguistic contexts.

In addition, for any of the above questions, we can use a corpus-based approach to contrast native-speaker intuitions or

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prescriptions with empirical evidence from authentic language use. In Section 2.5, for example, we will see how the uses of *deal* in a corpus compare with dictionary definitions. Such comparisons are not simply interesting for pointing out where we have misconceptions about language use. Rather, corpus-based evidence allows us to develop dictionaries, handbooks, and teaching materials that are more helpful and accurate for language students of all levels. Such tools describe the ways that language is actually used in natural contexts, and identify the uses that students will most likely encounter, rather than relying on authors' impressions of meaning and use.

As background to the analyses in this chapter, it will be useful if you take a minute to record your intuitions about the research questions listed above. For example, what meanings of the word *deal* can you come up with? Are these meanings equally common across registers? Can you think of any other words that are particularly common co-occurring with *deal*? How about *big*, *large*, and *great*: Can you think of any patterns of use that would systematically distinguish among these nearly synonymous adjectives? By attempting to answer the research questions about use – in this chapter as well as in the following chapters – you will be able to compare the accuracy of your intuitions against the actual patterns of use. In some cases, the observed patterns seem completely obvious once we see them, while in other cases they can be quite unexpected. However, many native speakers are surprised to learn that they often miss out on some of the "obvious" patterns as well as some of the more unexpected ones!

Through illustrations of the six research questions given above, this chapter also addresses important methodological issues in corpus-based research. Information about typical computer software displays and discussions about how to use them are incorporated throughout the examples. Further discussion of methodological issues relating to this chapter is given in Methodology Boxes 1, 6, and 7, which address corpus design, norming frequency counts, and statistical measures of lexical associations.

As discussed further below, lexicographic work requires the use of very large corpora, because word senses and collocational patterns are often much less common than grammatical patterns.

Sách có bản quyền

To facilitate the example analyses in this chapter, however, we have limited the size of the corpora used. We begin with examples from the LOB Corpus, and then we focus on two registers from the Longman–Lancaster Corpus, using approximately 3 million words for each register. Though too limited in scope for compiling a dictionary or making generalizations about English overall, these sample analyses illustrate how investigations of corpora can increase our understanding of word meanings, distributions, and use.

2.2 Investigating the meanings of words

The central question in lexicographic work is the meanings of words. One of the advantages of corpus-based research is that the corpus can be used to show all the contexts in which a word occurs. From these contexts, it is then possible to identify the different meanings associated with a word. Traditionally, citation slips were used for similar purposes. The primary difference is that citation slips represent only those contexts that a human reader happens to notice (in some cases representing only the more unusual uses). In contrast, a corpus-based listing provides an exhaustive listing of all occurrences and contexts.

Many software programs that facilitate the investigation of lexicographic issues are now available. (See the appendix for a description of corpus-analysis software tools.) All of these programs can display the occurrences of a chosen word with its surrounding context. Such displays are called concordance listings. In one common type of display, each occurrence of the chosen word is presented on a single line, with the word in the middle and context on each side. Figure 2.1 shows this kind of display for seven occurrences of *deal* in the LOB Corpus. These displays are referred to as KWIC displays – Key Word in Context. In some cases, it is useful to see more than a single line of context; a KWIC concordance can also display several lines, as illustrated in Figure 2.2.

Even from the short list in Figure 2.1, it is possible to see how KWIC files can reveal the different meanings of words. In that list, the surrounding contexts show three different meanings of *deal*. The first and fifth entries convey a sense of handling some problem, e.g., dealing with a "mass sit-down." The second and seventh entries concern business transactions – a property deal and a deal

Sách có bản quyền

and secret plans prepared to	deal	with the mass sit-down	1
of companies and put one property	deal	through each. Mr.	2
. In particular, a good	deal	of concern has been	3
hangs a tale – and a great	deal	of money. Neville	4
where his new measures to	deal	with Britain's	5
just a matter of working a good	deal	harder before we really	6
. "I'm mixed up in a	deal	involving millions	7

Figure 2.1 Sample "Key Word in Context" (KWIC) concordance listings for *deal*

those "stagging" on a new issue and
property dealers who form a chain of companies
and put one property
– – –> deal
through each.
Mr Harold Wilson, shadow Chancellor,

in the event, these rising costs have already
overtaken the airlines. In particular
a good
– – –> deal
of concern has been caused in European
air transport by the decision of

Figure 2.2 KWIC entries with extended context

involving millions. Finally, the remaining entries use *deal* as an amount, with *good* or *great* preceding it – a good deal of concern, a great deal of money, and working a good deal harder.

Using a computer, it is relatively easy to get a list of all the occurrences of a particular word in context, and, through the context, see all the meanings associated with the word. In fact, the complete concordance listing for *deal* shows that it has many other meanings in addition to the three illustrated here, including:

– "to have as a topic"
  (*Two articles in astronomy deal in turn with stellar evolution and the determination of stellar distances*)
– "a kind of treatment"
  (*So let us see them get a fair and square deal*)
– "a type of wood"
  (*He went to the far end of the deal table and sat against it . . .*)

Sách có bản quyền

Although the use of concordances generated from text corpora provides a more solid empirical basis than traditional lexicographic approaches, concordances can actually provide too much data. For example, a KWIC concordance for *deal* in an 8-million-word sub-sample from the Longman–Lancaster Corpus generated over 1,500 entries. Simply determining the number of different senses in a database of this size is a daunting task; to accurately group different uses or rank them in order of importance is not really feasible without the use of additional tools. In the following sections we describe some of these additional tools, and we then return to the different senses of the word *deal* in Section 2.5.

2.3 Investigating the frequency of words

Which are the most common words in English? Which are uncommon? Where does a particular word fit on the continuum of very common to very uncommon? The answers to such questions provide the first step in understanding the patterns of use associated with a word.

Most concordancing programs can generate a frequency list of all the words in a corpus. Such a list simply shows the number of times that each word occurs in the corpus. For example, Figure 2.3 shows a section of the frequency list for ten selected texts from the LOB Corpus, generated by a program called TACT (Text-Analysis

a	478
abandoned	2
abide	1
ability	2
able	3
abnormalities	1
aboriginal	1
abortive	1
about	37
above	15
abroad	1

Figure 2.3 Frequency list of words generated by TACT for ten selected texts from the LOB Corpus

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deal	182
dealing	52
deals	25
dealt	31

Figure 2.4 Frequency list of forms of *deal* generated by TACT (from the LOB Corpus: 1 million words)

Computing Tools). This list is in alphabetical order, although frequency lists can also be generated in the order of occurrence of words or in order of frequency.

The partial frequency list in Figure 2.3 shows that in this sample of texts from the LOB Corpus, the word *a* occurs 478 times, while the word *abandoned* occurs only twice. Figure 2.4 provides a similar listing for forms of the word *deal* – the focus of our sample investigation – in the complete LOB Corpus; this list shows that *deal* occurs 182 times, *dealing* 52 times, *deals* 25 times, and *dealt* 31 times.

When studying a word, it is often useful to consider the different forms of the word collectively. That is, though the computer identifies *deal*, *deals*, *dealing*, and *dealt* as different words, we may want to discuss them all together or investigate their frequency as a group. The term "lemma" is used to mean the base form of a word, disregarding grammatical changes such as tense and plurality. Thus, the frequency of the lemma DEAL – including *deal*, *deals*, *dealing*, and *dealt* – in the LOB Corpus is 290. (To distinguish between word forms and the lemma, we will use small capital letters to refer to the lemma DEAL.)

Without any comparison, it is difficult to know what it means that DEAL occurs 290 times in the LOB. Is this particularly common? rare? or just average frequency? One use of frequency lists is to compare the frequency of a target word to other words. When we compare DEAL to function words – for example, articles, prepositions, and pronouns – we see that DEAL is relatively rare. The definite article *the* occurs over 50,000 times in the LOB Corpus, the preposition *of* occurs over 35,000 times, while the pronouns *he* and *I* occur about 9,000 times and 7,800 times, respectively.

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When the frequency of DEAL is compared to other content words, we find that it is neither particularly common nor particularly rare. The very common verb MAKE occurs 2,417 times in the LOB Corpus, far more frequently than DEAL. On the other hand, SIGH is very rare, occurring only 16 times. Compared to the lemma APPROACH, which occurs a total of 185 times, DEAL is only somewhat more common.

The frequency list from the LOB Corpus also highlights one of the challenges for corpus-based lexicographic research: to study the meaning and use of words, we need a very large corpus. That is, a 1-million-word corpus will not provide sufficient data for many words to allow for meaningful generalizations. The frequencies are relatively reliable for the most common words in a corpus, but to analyze the senses and association patterns of words we need a very large number of occurrences. For example, it would be impossible to say very much about a word such as SIGH based on its use in the LOB Corpus, other than to note that it is rare.

Furthermore, in a small corpus, the inclusion of only moderately common or rare words can depend a great deal on the different topics represented in the texts of the corpus. The frequency list for the LOB shows, for example, that there is just one occurrence of *clowns* and none of *clown*. There are 10 occurrences of *elephant*, but none of *giraffe*, *zebra*, or *hippopotamus*. If the LOB had happened to include a story about a clown or an article about giraffes, these figures would be very different. With a very large corpus containing many different texts, however, a wider range of topics should be represented, so that the frequencies of words are less influenced by individual texts. (See Methodology Box 1 for more about the issues of size and representativeness in corpus design.)

The analysis of word meaning is further complicated by the fact that many word forms in English have multiple grammatical functions. For example, the word *deals* can be used as either a third person singular verb or a plural noun. Similarly, both *deal* and *dealing* can be used as a verb or a noun. When the words in a corpus have not been annotated for their grammatical category, the corpus is referred to as "untagged." Frequency lists based on an untagged corpus are limited in usefulness, because they do not tell us which grammatical uses are common or rare. In fact, we

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deal_nn	115
Deal_np	1
deals_nns	5
deal_vb	66
dealing_vbg	51
deals_vbz	20
dealt_vbd	14
dealt_vbn	17

Figure 2.5 Frequency list of tagged forms of *deal* generated by TACT (from the complete LOB Corpus)

might consider *deal* as a noun to be a different "word" from *deal* as a verb; from this perspective, a list from an untagged corpus cannot even tell us the frequencies of "words"!

To determine which occurrences of *deal* are nouns versus verbs, we must look at the forms in context, determine their grammatical category, and keep count. While such a solution is not overwhelming for the 182 occurrences of *deal* in the LOB Corpus, it would be very time consuming. This approach would be even more tedious for extremely common words, such as *look* (which occurs around 500 times per million words); and it would be unworkable as the standard procedure for compiling a dictionary, where the use of all words must be analyzed in a very large corpus.

A much more efficient solution is to use a "tagged" corpus, that is, a corpus in which each word has been coded, or "tagged," for its grammatical category. (See Methodology Boxes 4 and 5 for more on tagging procedures.) Using a tagged corpus, we could do automatic frequency counts for each grammatical word separately.

Figure 2.5 displays a frequency list generated by TACT showing the distribution of grammatical forms of the lemma DEAL in the tagged LOB Corpus. The grammatical category of each word follows the word itself. Thus, there are 115 occurrences of *deal* as a singular noun ("nn"), one occurrence as a proper name ("np"), and 66 occurrences as a verb ("vb"). The word *deals* occurs 5 times as a plural noun ("nns") and 20 times as a third person verb ("vbz"). *Dealing* occurs once as a singular noun ("nn") and 51 times as a present participle verb ("vbg"). Finally, *dealt* occurs 14 times

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Table 2.1 *Frequency of* DEAL *as a noun in selected registers from the LOB Corpus (excluding proper nouns), per 100,000 words*

Register	Approx. number of words in sample	Raw count for DEAL	Normed count for DEAL (per 100,000 words)
press reportage	88,000	14	15.9
press reviews	34,000	4	11.8
press editorials	54,000	4	7.4
religion	34,000	5	14.7
learned/scientific	160,000	16	10.0
popular lore	88,000	11	12.5
belles lettres	154,000	24	15.6
general fiction	58,000	5	8.6

as a simple past tense verb ("vbd") and 17 times as a past participle ("vbn"). With this information from the tagged corpus, we can go on to explore the occurrences of DEAL in more detail, looking at the distribution of its verb and noun forms and comparing them across registers.

2.4 Distribution of DEAL across registers

In this section, we explore the frequency of DEAL as a noun and verb across registers. In particular, we show that overall characterizations of a word can be misleading, because words are often used in very different ways in different registers.

We begin with an analysis of the lemma DEAL as a noun in the tagged LOB Corpus, including only those forms we want to focus on: singular and plural nouns (i.e., *deal* and *deals*). (We have not included the one occurrence of *deal* as a proper noun, when it is used as a place name.)

Since the LOB Corpus is composed of different registers, such as "learned and scientific writing," "belles lettres," and "adventure and western fiction," we can compare the frequency of DEAL across registers. Table 2.1 presents the frequency distribution by register for DEAL as a singular or plural noun.

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Table 2.1 includes both raw counts and normed counts. The raw counts show the actual number of occurrences of the word in each register; for example, press reportage has 14 occurrences, press reviews have 4 tokens, and the learned/scientific category has 16 occurrences. However, these text categories have different numbers of words: press reportage has around 88,000 words, press reviews has only 34,000, and the learned/scientific category has 160,000 words. For this reason, a comparison of raw counts cannot be used to conclude that a word is more common in one register than another. That is, analyzing a larger sample of text from a register is likely to uncover more occurrences of the target word. In the present case, we would expect raw counts for DEAL in press reviews to be much lower than the counts from learned/scientific, simply because we analyzed a much smaller sample of text for press reviews. But this does not necessarily mean that DEAL is less common in press reviews. Rather, we need a measurement of how often a reader will come across occurrences of DEAL as a noun, that is, a rate of how often DEAL occurs in a fixed amount of text.

Such measurements are referred to as "normed" frequencies. Normed counts convert the number of occurrences of a word to a standard scale, in this case per 100,000 words. When the counts are normed, press reportage has 15.9 occurrences per 100,000 words, press reviews have 11.8 occurrences per 100,000 words, and the learned/scientific category has 10 occurrences per 100,000 words. This example shows how raw counts can give a misleading impression of frequency relationships. For example, the raw counts show many more occurrences of DEAL as a noun in learned/scientific than in press reviews, but the normed counts show that DEAL is actually slightly more common in the press reviews. Thus, only normed counts provide a fair basis for comparison across registers. (See Methodology Box 6 for more on norming.)

When the occurrences of DEAL as a noun are broken down by register, the problem of corpus size for lexicographic work is even more apparent. Table 2.1 shows that four of the eight categories have five or fewer occurrences. None of the registers have many occurrences; belles lettres is the highest with 24. Clearly, this corpus is too small to allow detailed analysis of the use of DEAL as a noun.

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Table 2.2 *Frequency of* DEAL *as a noun and verb in sub-samples from two registers in the Longman–Lancaster Corpus*

	Approx. number of words in sample	Raw counts		Normed per million words	
		Noun	verb	Noun	Verb
Total sample	4,000,000	366	482	90	119
Fiction	2,000,000	217	127	107	63
Academic prose	2,000,000	149	355	74	176

The normed counts do suggest, however, that DEAL may be distributed differentially across registers. As Table 2.1 shows, there is a great deal of variation in the relative frequency of DEAL. We therefore turn to a larger corpus to investigate these patterns in more detail.

Table 2.2 shows the occurrences of DEAL as a noun and verb in two registers (fiction and academic prose) from a considerably larger corpus: a 4-million-word sample from the Longman–Lancaster Corpus. Here there are well over 800 occurrences of *deal/deals*, providing a much more solid base for analyzing the uses of this word. Because these register samples are much larger than those from the LOB Corpus, frequency counts are normed to counts per million words (rather than per 100,000 words).

Several interesting patterns are evident in this frequency table. First, the normed counts for the total sample show that *deal/deals* as a verb is only slightly more common than *deal/deals* as a noun (119 per million words compared to 90 per million words). However, when we consider the frequencies for each register, a different picture emerges. In academic prose, *deal/deals* functioning as a verb is more than twice as common as the noun use (176 versus 74 occurrences per million words). Fiction shows the opposite pattern, with the noun use being considerably more common than the verb use (107 versus 63 occurrences per million words).

These patterns of use for DEAL highlight another important point in corpus design: a corpus restricted to any one register will not represent language use in other registers. That is, it is impossible to generalize from one register to the patterns that will be found in

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other registers. With respect to the present example, the relative frequency of DEAL as a noun and verb in academic prose is the exact opposite of that in fiction. A corpus restricted to either of these registers would give no clue at all for the language use patterns found in the other.

Furthermore, this example shows how overall generalizations about English are often misleading. The overall corpus frequencies in this case show that *deal/deals* is about equally common as a noun and a verb. This generalization merely blurs the opposite patterns of use that in fact occur in academic prose and fiction. This small example illustrates a finding that is common in corpus-based research: that overall generalizations of a language are often misleading, because they average out the important differences among registers. As a result, overall generalizations are often not accurate for any variety, instead describing a kind of language that doesn't actually exist at all.

Returning to the investigation of the word DEAL, we are now left with the question of what accounts for the different distributions of noun and verb forms across these two registers. To delve into this, we must consider the different senses of the word and the ways that they are used in each register.

2.5 Distribution of senses across registers

Section 2.2 illustrated how a corpus allows us to investigate the meanings of words through the use of concordance listings, which can provide an exhaustive list of all occurrences of a word in context. However, this thoroughness also presents a challenge: How are we to sort and analyze all the information from a concordance file? It presents a daunting enough task for DEAL, with around 2,000 occurrences in a 10-million-word corpus. For a more common word we could be faced with sorting and classifying tens of thousands of entries, trying to find the patterns of meaning in them. Such a task is simply not feasible.

Rather than manually sorting through concordances, one way to begin investigating the senses of words is to look at their "collocates": the words that a target word commonly co-occurs with. There is a strong tendency for each collocate of a word to be

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associated with a single sense or meaning (although more than one collocate can be associated with that same sense). Thus, identifying the most common collocates of a word provides an efficient and effective means to begin analyzing senses.

In this section we first look at the collocates and senses of DEAL as a noun, comparing the corpus findings with dictionary definitions. This illustration is then supplemented by a short discussion of the collocates and senses of DEAL as a verb.

2.5.1 DEAL *as a noun*

Table 2.3 displays the collocates for DEAL as a noun in two registers from the Longman–Lancaster Corpus (a 2.7-million-word sample from academic prose, and a 3-million-word sample from fiction). Similar displays, showing a list of collocates ordered according to frequency, can be produced using most commercially available concordancing packages (such as TACT, CorpusBench, etc.).

Left collocates are those co-occurring words immediately preceding the target word; for example, Table 2.3 shows that *good* is a common left collocate of *deal*. Right collocates are the co-occurring words immediately following the target word (e.g., *deal* + *of*). The lists in this table present only the first word immediately to the right or to the left of *deal*; the same techniques could be used to investigate collocations at a distance (e.g., two or three words removed from the target word; see Section 2.6.3). The table lists all collocates that occur at least three times per million words in each register.

Table 2.3 shows that in academic prose, the most common left collocate is *great* (45 times per million words), followed by *good* (23 times per million words). These are, in fact, the only collocates with substantial frequencies. They account for 185 of the 196 instances of DEAL as a noun in that register. The next most common collocates (*package* and *that*) occur only twice, or 0.7 times per million words.

What does this pattern tell us about the sense of DEAL that is most commonly used in academic prose? It seems most likely that *good/great deal* will refer to either an amount of something or to a

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Table 2.3 *Common collocates of* deal *as a noun in a 5.7-million-word sample from the Longman–Lancaster Corpus*

	Raw counts	Normed per million words
Academic prose		
(2.7-million-word sample)		
Left collocates		
great	122	45
good	63	23
Right collocates		
of	106	39
more	18	7
in	8	3
to	8	3
Fiction		
(3-million-word sample)		
Left collocates		
great	122	40
good	84	28
the	24	8
big	10	3
Right collocates		
of	84	28
to	22	7
about	15	5
more	10	3
with	9	3

business transaction. In fact, when we consider the right collocates, it becomes clear that the meaning referring to an amount is the most common use of *deal*. Again the pattern of collocations is very striking: the most common right collocate is *of*, which occurs 39 times per million words. There is then a very large drop to the next most common right collocate: *more*, with 7 occurrences per million words. The next two, *in* and *to*, occur only 3 times per million words. By far the greatest use of *deal* as a noun, then, appears to be in the sense of amount, as in *a good/great deal of*.

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We can verify our analysis of word senses by looking at concordance listings which show these collocations. For example, common uses of *good/great deal of* include *a good deal of work* and *a good deal of attention*. In addition, the other most common right collocates also refer to amount. The collocate *more*, for instance, is used in *a great deal more tolerance* and *a good deal more inhibited. In* and *to* are also used with *deal* referring to amount, as in *a great deal in common, differ a great deal in their understanding, a great deal to be desired,* and *a great deal to offer*. By far, then, the overwhelming majority of the occurrences of DEAL as a noun in academic prose are used to show a large amount.

Table 2.3 shows that the collocations for fiction have both interesting similarities and differences with academic prose. Again, by far, the most common left collocates are *great* and *good*. In fact, the combined occurrence of *good/great deal* is very similar in the two registers (both approximately 68 times per million words). However, in fiction, there are still a substantial number of occurrences (96 instances) not accounted for by *good/great + deal*. The collocates for these occurrences are distributed more broadly: *the* occurs 8 times per million words and *big* occurs 3 times per million words, while there are 37 other collocate pairs which occur only once or twice.

This collocational pattern suggests that DEAL in the sense of amount is the most common meaning in both academic prose and fiction, but that other uses are relatively common in fiction (but not in academic prose). For example, the left collocate *the* is used when DEAL has the sense of an agreement, as in *part of the deal is . . .* and *Isn't that the deal?* The collocate *big* represents a different sense. It is used to convey lack of importance in expressions such as *no big deal* and *what's the big deal?*

In addition, many collocates that are not frequent individually (and therefore not included in Table 2.3) are associated with yet another meaning relating to a business transaction, as in *property deal, record deal, cash deal, land deal, mining deal*. Although none of these words is among the top five collocates of DEAL, together they show another important sense.

The right collocates of *deal* in fiction are very similar to those in academic prose. *About* and *with* are the only notable right colloc-

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ates that did not appear in academic prose. *About* is similar to *of* in being used with the sense of amount:

(1) a. I also knew a *great deal about* love.
b. We both laughed a *great deal about* this.

In contrast, the right collocate *with* is used with DEAL to refer to an agreement:

(2) a. I made a *deal with* the doctors.
b. I'll cut a *deal with* you.

The collocations of *deal* in fiction also uncover a sense not found at all in academic prose. Farther down the list of right collocates, we find 4 occurrences of *table* and 1 occurrence of *box*. These collocates are used with *deal* to refer to a type of wood. While these are not common collocates, they point out another use of the word in fiction, and they come from five different texts, showing that this use is relatively widespread.

In general, then, DEAL as a noun is used most commonly to refer to an amount. However, fiction has a higher frequency of DEAL as a noun, and correspondingly it uses this word with more different senses, referring to agreements, a type of wood, and an expression of unimportance, in addition to amounts.

How do these frequencies in the corpus relate to dictionary definitions? A survey of dictionaries shows a bewildering array of meanings for the entry DEAL. Some dictionaries have only one main entry while others have as many as four. Within the entries the number of definitions varies from two or three to twenty or thirty. With this variability in presentation, it is difficult for a user to guess at the most common uses of DEAL.

Table 2.4 displays the seven senses of DEAL as a noun that are most commonly repeated in a sample of five dictionaries: 1. a large but indefinite amount; 2. an agreement or arrangement; 3. the distribution of cards in a game; 4. treatment received; 5. the act of distributing; 6. wood of fir or pine trees; and 7. the act of buying or selling, or a business transaction. Even this sub-sample of definitions shows great variation. Most of the dictionaries cover all seven definitions, but two do not mention the "act of distributing,"

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Table 2.4 *Common dictionary definitions of* DEAL *as a noun*

Definition	*Webster's Encyclo.* 1989	*Webster's Third* 1981	*Chambers* 1993	*Random House* 1993	*Longman Lang. & Culture* 1992
large but indefinite amount	entry 1 sense 16	entry 1 sense 2	entry 1 sense 3	entry 1 sense 21	entry 2 sense 3
agreement/ arrangement	entry 1 sense 13	entry 3 sense 3	—	entry 1 sense 18	entry 2 sense 1
distribution of cards in a game	entry 1 sense 18	entry 1 sense 3	entry 1 sense 4	entry 1 sense 21	entry 2 sense 4
treatment received	entry 1 sense 15	entry 3 sense 2	entry 1 sense 6	entry 1 sense 6	entry 2 sense 2
act of distributing	entry 1 sense 17	—	—	entry 1 sense 23	—
pine or fir wood	entry 2 3 senses	entry 4 2 senses	entry 2 sense 1	entry 2 3 senses	entry 3 sense 1
act of buying or selling/ a business transaction	entry 1 sense 13	entry 3 sense 2	entry 1 sense 5	entry 1 sense 17	entry 2 sense 1

and one does not mention "agreement/arrangement" except in the sense of a business transaction. The ordering of definitions also varies widely. For example, "a large but indefinite amount" is the second sense given in the first entry in *Webster's Third Dictionary*, but it is the twenty-first sense in the *Random House Dictionary*.

Comparing these definitions to our corpus-based investigation of DEAL raises several points. The use of DEAL marking an amount is by far the most common sense in both registers of the corpus, yet this sense is not covered until very late in some dictionaries – the sixteenth and twenty-first sense in two cases. In addition, the analysis of collocates found a relatively common sense not covered by these dictionaries: the use of *big deal* to mean unimportance. Finally, register differences are disregarded by all these dictionar-

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ies, although more recent corpus-based dictionaries are beginning to take account of important register patterns.

The flip side is that all five dictionaries identify a sense not found in our corpus, referring to the distribution of cards in a game. Although this is one of the first meanings that speakers associate with the word DEAL, it is in fact not common in actual use (apart from card games!). This omission highlights the importance of large representative corpora for lexicographic work. It also shows how our intuitions should complement corpus-based analysis. That is, to be complete, a dictionary should include the card game sense of DEAL, even if that sense does not appear in our corpus – every native speaker of English recognizes this as a distinct sense of DEAL. However, we can rely on our corpus analysis to tell us that this is a relatively rare use of DEAL, one that language learners are not likely to encounter outside of restricted domains. Lexicographic work should incorporate both perspectives, being complete in identifying the range of meanings but useful in marking senses that are most common or important.

2.5.2 DEAL as a verb

The senses of DEAL as a verb can be investigated with collocations using these same techniques. Here we will concentrate on the collocation *deal with* to illustrate the case where a collocate pair is associated with several different senses.

The collocate pair *deal with* is much more common than any other collocation in both academic prose and fiction. In the Longman–Lancaster Corpus, it occurs approximately 157 times per million words in academic prose and 58 times per million words in fiction. In comparison, the next most common right collocate in academic prose is *only*, at 2.6 occurrences per million words. In fiction, the next most common collocate pair is *deal in*, with 4.6 occurrences per million words.

When we look at the concordance listings for *deal with*, it immediately becomes clear that several different senses are represented by this collocate pair. The most common of these is stating the subject matter or focus of a book, article, study, etc. Looking over the full concordance listing shows that this sense occurs far more

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often in academic prose, and it thus accounts for the greater frequency of *deal with* in that register. For example,

(3) a. The second controversy *dealt with* the source of nitrogen in plants.
b. An important point to note is that the preceding discussion has *dealt with* thermodynamic acidity . . .
c. Other environmental effects are *dealt with* in other chapters.
d. The interpretation and use of this information through soil survey, soil classification and land evaluation is *dealt with* in Chapter 24.

Another sense of *deal with* means to solve a problem, as illustrated in the following examples:

(4) a. When they had *dealt with* the fire another crisis arose.
b. Moreover many losses are due to chilling and crushing, both factors that can be *dealt with* by good environmental control and housing.

There also appears to be a sense of coping, or just handling a situation in some way without actually solving a problem. This sense is found primarily in fiction:

(5) a. He didn't have the right temperament to *deal with* the Hennigs of this world.
b. She would have rather there been a fight, anger – or even tears and pleadings. These she could *deal with*, not this deadly coldness exhibited by Alice.
c. But suffering is also a fact. It has to be *dealt with*.

Finally, *deal with* includes a sense denoting interactions with a person, especially a person you do business with:

(6) a. Hansie De Beer runs the farm, he's the one Mehring usually *deals with* . . .
b. The son handed me a small suitcase with the distant eyes of a man *dealing with* a chauffeur.

Clearly, these senses of *deal with* deserve more detailed analysis. The corpus could be analyzed to look at the words occurring after *deal with*, coding them for different semantic categories (subject matter, problems, etc.). In this way, extended collocational frames

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could be used to help distinguish among these senses. The main point of the present section, however, has been to show how collocational pairs do not necessarily have a one-to-one mapping with meanings – rather, in some cases, a single collocation can be used with a range of senses, depending on the larger context.

2.6 Analyzing the use of seemingly synonymous words

Languages have many words that are considered synonymous, in that dictionaries and thesauruses often characterize such words as identical in meaning. The present section shows how these equivalencies can be very misleading, because "synonymous" words are typically used in very different ways. Corpus-based lexicographic analyses are particularly well suited to uncovering such systematic differences in the patterns of use.

In this section we consider the words *big*, *large*, and *great*. Thesauruses often list these three words as synonyms of size. We saw in the last section that *great deal* is a common way to refer to a large amount of something. This section will analyze additional collocations to investigate the various uses of *great* relative to *big* and *large*.

2.6.1 *Frequency distributions of* big, large, *and* great

Table 2.5 presents the frequency distributions of *big*, *large*, and *great* in a 5.7-million-word sample from the Longman–Lancaster Corpus. Similar to the frequency distributions considered earlier in this chapter, this table shows that when registers are combined, we get overall frequency counts that do not accurately reflect any register. In the present case, the combined corpus shows *large* to be the most common of the three adjectives, followed by *great*, and then *big* (with normed frequencies of approximately 408, 393, and 230, respectively).

Academic prose actually has this same ordering of the three adjectives, but the frequency difference between *large* and *big* is much different: *large* is particularly common, occurring about 605 times per million words, while *big* is very rare, occurring only

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Table 2.5 *Frequency of* big, large, *and* great *in a 5.7-million-word sample from the Longman–Lancaster Corpus*

	Raw counts	Normed per million words
Total sample (5.7 million words)		
big	1,319	230
large	2,342	408
great	2,254	393
Academic prose (2.7 million words)		
big	84	31
large	1,641	605
great	772	284
Fiction (3 million words)		
big	1,235	408
large	701	232
great	1,482	490

Graphic representation of normed frequencies (each * represents 100 occurrences per million words)

	Combined	Academic	Fiction
big	**	—	****
large	****	******	**
great	****	***	*****

31 times per million words. Such extremes are not at all predictable from the combined counts. The patterns in fiction are even less expected from the overall counts, since they are almost the opposite from that found in academic prose: *great* and *big* are both notably common (with frequencies of 490 and 408 occurrences per million words), while *large* has a much lower frequency (232 occurrences per million words).

It is clear from the table that there is great variation in the use of these seemingly synonymous words between the two registers. *Big* is over ten times more common in fiction than it is in academic prose, and *great* is over one-and-a-half times more common

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in fiction. *Large*, on the other hand, is three times more common in academic prose than in fiction. To understand why these differences exist, it is useful to compare the common collocations of the three adjectives.

2.6.2 Immediate right collocates of big, large, *and* great

Table 2.6 shows the most frequently recurring right collocates of *big*, *large*, and *great* in the academic prose and fiction samples from the Longman–Lancaster Corpus. A fuller analysis would require looking at the complete list of collocates; here we focus only on the top ten collocates in each register, excluding collocates that occur less than once per million words.

In both registers, the collocates of *big* show that it is most commonly used to refer to physical size. In academic prose, there are only two common collocates of *big*. The collocate pair *big enough* occurs 6 times (2.2 per million words), usually referring to physical size, as in

(7) small trees which are not *big enough* to be converted into timber

The second most common collocate pair in academic prose is *big traders*, which occurs only 3 times – all in the same text about economic development in West Africa, which contrasts big and small traders.

In fiction, there are many more common collocations with *big*. The vast majority show that big is used to describe the size of physical objects, such as *man*, *house*, *toe*, *boy*, *room*, and the indefinite pronoun *one*. For example,

(8) a. The *big man* gave me a raking glance and grinned.
b. . . . she was overawed by the *big house*.
c. The sitting room was a *big room*.

In addition, *big* is used with other descriptive adjectives, such as *black*, *old*, and *red*. Consideration of the concordance listings for these pairings shows that they are also being used to refer to the physical size of objects. For example:

(9) a. "The beast had teeth," said Ralph, "and *big black eyes*."
b. his *big black mongrel dog*
c. the *big black saucepan*

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Table 2.6 *Ten most common right collocates of* big, large, *and* great *in two registers from the Longman–Lancaster Corpus (Frequency counts normed per million words; collocates with frequencies less than 1 per million words excluded)*

Academic prose (2.7 million words)

Big		Large		Great	
right collocate	freq. per million	right collocate	freq. per million	right collocate	freq. per million
enough	2.2	number	48.3	deal	44.6
traders	1.1	numbers	31.3	importance	12.5
		scale	29.4	number	8.9
		and	28.0	majority	8.1
		enough	15.9	variety	7.0
		proportion	11.8	extent	7.0
		amounts	10.7	part	4.1
		quantities	10.3	care	3.3
		part	10.0	advantage	2.6
		extent	8.9	detail	2.6
				interest	2.6

Fiction (3 million words)

Big		Large		Great	
right collocate	freq. per million	right collocate	freq. per million	right collocate	freq. per million
man	9.6	and	15.2	deal	40.4
enough	8.9	black	4.3	man	6.6
and	8.3	enough	3.6	burrow	5.6
black	8.3	house	3.0	big	4.6
house	7.6	room	2.7	aunt	4.3
one	7.0	white	2.7	care	4.0
toe	5.0	number	2.3	pleasure	4.0
old	4.6	for	2.3	and	3.0
red	4.3	man	2.0	relief	3.0
boy	3.6	one	2.0	black	2.7
room	3.6	in	2.0	to	2.7

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Similarly, the collocate pair *big enough* in fiction is used to show physical size:

(10) a. The cart was not really *big enough*, he realized
b. The revolver, which looked *big enough* to stop a florist's van, was supposed to serve as a deterrent . . .
c. "These idiotic beds aren't *big enough* for one person, let alone two."

Finally, the coordinator *and* is a common collocate of *big* in fiction. As discussed in Section 2.3, function words are the most common words in a corpus. In the present case, *and* is a common collocate for all three adjectives under consideration. However, the coordinator *and* is extremely frequent overall, making it hard to assess the importance of this collocational association. Methodology Box 7 addresses this problem and introduces a number of statistical tools that measure the strength of association between members of a collocate pair relative to the overall frequency of each word in the pair.

In contrast to *big*, the collocates of *large* in academic prose show that it is most often used to refer to a quantity or amount of something. Seven of the top ten collocates clearly have this meaning: *large + number(s)*, *proportion*, *amount(s)*, *quantities*, *part*, *extent*. For example:

(11) a. [. . . glutinous rices] are widely grown in Asia where a *large number* of varieties are recognized.
b. There are a *large number* of processes grouped together under the general term weathering.
c. A *large proportion* of his dependent clauses are in fact noun clauses.

Large + enough is also used commonly (29 of its 43 occurrences) to show quantities or proportions, as in:

(12) a. The ratio is *large enough*, however, to allow . . .
b. . . . which will always tell in a finite number of steps (which may easily be *large enough* to require a computer) whether or not a given element of Q[x] is irreducible.

The other occurrences of *large enough* refer to physical size, as in

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(13) a. The pore size of the agar gel and cellulose acetate is *large enough* that the protein molecules are able to move freely . . .

Finally, the collocate pair *large scale* is common in academic prose. This pairing is used as an adjective chunk, referring to the magnitude of various processes, for example,

(14) a. This is the type of industrial organization, according to Marx, which is most compatible with *large-scale* centralisation . . .
b. [these] can then be treated in the context of *large-scale* motions of lithospheric plates.

In fiction, the most common use of *large* is in the sense of physical size, co-occurring with the noun and adjective collocates *house*, *room*, *man*, *black*, and *white*. For example,

(15) a. a *large black* saucepan
b. the *large black* barrels of a sawn-off shotgun
c. a *large white* bird

(16) a. Apparently Philbrick has a *large house* in Carlton House Terrace.
b. It was a *large room*, totally silent save for the voice of one sister. . . .

This use of *large* is the same as the most common use of *big* in fiction, and many words occur as right collocates to both adjectives. However, as the overall frequencies show, *large* is not used as commonly as *big* in fiction, and therefore these collocates are not as common with *large* as with *big*. For example, *big man* occurs 9.6 times per million, while *large man* occurs only twice per million; similarly, *big house* occurs 7.6 times per million, and *large house* occurs only 3 times per million.

In fiction, *large* is also used to refer to quantities or amounts. The collocate pair *large number* is relatively common, and other right collocates such as *amount*, *proportion*, and *sum* are rare individually, but together compose a class of collocates referring to amount:

(17) a. A *large number* of people sat round a table . . .
b. . . . Ichiro was fascinated by the *large amount* of space in our house . . . .

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The third adjective, *great*, has yet another pattern of collocations. In academic prose, its most common use is to show amount or quantity. This occurs by far most commonly as the collocate pair *great deal*, discussed in the last section. However, *great number*, *great majority*, *great variety*, *great extent*, and *great part* also recur. For example:

(18) a. There is not a *great deal* of information on the minimum size of pore into which a root can grow . . .
b. The *great majority* of mechanical problems give rise to matrices having distinct eigenvalues . . .

This use of *great* in academic prose is similar to the use of *large*, although *large* never co-occurs with *deal*. However, *great* also has a very different sense, one that shows intensity. In academic prose, this sense occurs with the right collocates *importance*, *care*, *advantage*, *detail*, and *interest*:

(19) a. It is of *great importance* to control ectoparasites. . . .
b. The figures have to be interpreted with *great care*.
c. . . . the structure of some is now known in *great detail* . . .

In fiction, *great* is used most commonly to show amount, with the collocate pair *great deal*:

(20) a. He stood and drank a *great deal* of apple juice.
b. . . . Sandy was a bright young woman who seemed to know a *great deal* about life in Alaska.

However, the collocates of *great* in fiction show that it has a much wider range of senses. For example, *great man* occurs 6.6 times per million words, referring to a sense of very important or very good:

(21) a. We approach a *great man* through his servants.
b. ". . . He was a *great man*, and we all feel as though we've been orphaned."

Certain occurrences of *great care* also seem to refer to a meaning of "very good":

(22) . . . I promise you we will take *great care* of him.

In addition, *great* is sometimes used to refer to a large physical size, as in:

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(23) a. a *great, black* bird
b. his *great black* moustache

The collocate pair *great burrow* also refers to physical size. This pair again shows the influence that a single text can have; all occurrences of this collocation come from the book *Watership Down*, where the rabbits meet together in a large area called "the great burrow":

(24) In the *great burrow*, however, things happened differently.

In addition to the above, *great* in fiction is used as an intensifier in the collocation *great big*:

(25) a. It's a *great big* country with a continent of promise.
b. . . . there're *great big* gaps where they cut the wire and come out at night.
c. . . . All those ones who live in those *great big* piecrust mock-two-door houses with His and Her Caddies parked out by the hydrangea bushes.

It is also used in the sense of intensity that was seen in academic prose, as in *great care*, *great pleasure*, and *great relief*:

(26) a. . . . Jimmy found *great pleasure* in the society of one who had seen so much of the world.
b. He laid the conch with *great care* in the grass at his feet.
c. . . . his fingers were numb, and he had *great difficulty* in undoing his collar.

Finally, one collocation has a specialized meaning, showing the sense of family relationship: *great aunt*:

(27) a. He was almost as old as her *great aunt* had been.
b. . . . my *great-aunt* has appeared unexpectedly and is carrying me off to her home in Surrey.

Even this brief analysis shows that, despite variation, there are certain distinct patterns discernable in the use of these seemingly synonymous words. *Big* is most often used to refer to physical size, while *large* is used more to refer to quantities and amounts. *Great* is also used for amounts, especially in the collocation *great deal*, but this adjective has a wider range of meanings, ranging from intensity to size to familial relationships.

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These preferred senses of the three adjectives help to explain their frequency distributions across the two registers. Fiction texts contain frequent physical descriptions, referring to the size of objects, people, and places. In contrast, when academic prose deals with size, it is more likely to use specific measurements. Thus, it makes sense that fiction has a much higher occurrence of *big* than academic prose, given the preferred use of that adjective to refer to physical size. Academic prose, on the other hand, is more concerned with amounts and quantities, and thus it has a higher occurrence of *large*. Both registers make use of *great* to mark amounts with the collocate pair *great deal*. However, fiction uses many more senses of *great*, which corresponds to its more varied descriptions and concerns as compared with academic prose.

In sum, the present section shows that supposedly synonymous adjectives are not at all equivalent in their meaning, when their actual patterns of use are analyzed empirically. Rather, corpus-based analysis can be used to show how each adjective has its own preferred collocates, different preferred senses, and different distributions across registers.

2.6.3 *Collocates of* large *at a distance*

Words do not necessarily need to be adjacent to be associated with each other. That is, two words may tend to co-occur even with a few other words between them. To illustrate the usefulness of looking at collocations at a distance, the present section describes collocational patterns in the second word to the right of *large*, e.g. *of* in the sequence *large X of*.

Table 2.7 displays the common collocates that occur two words to the right of *large*. The most common of these by far is the collocation *large* X *of* in academic prose (and to a lesser extent in fiction). In fact, this collocational association is so common that it can be considered as a kind of frame, allowing substitution of a range of intervening words, such as *large amount of*, *large proportion of*, and *large group of*.

In the last section, we found that *large* is often used with nouns of amount. These nouns are commonly used in the frame *large* + N + *of*. Examination of this extended frame shows that these

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Table 2.7 *Most common collocates of* large *occurring two words to the right:* large *X — (based on analysis of a sample from the Longman–Lancaster Corpus; frequency counts normed per million words)*

Academic prose (2.7-million-word sample)		Fiction (3-million-word sample)	
collocate	freq. per million	collocate	freq. per million
of	167.4	of	31.1
in	19.9	and	7.9
and	16.6	in	5.6
to	14.4	eyes	4.3
the	7.7	on	3.3
are	7.0	which	3.0
with	6.6	with	2.7
on	5.5	for	2.0
is	4.1	room	2.0
open	4.1	the	2.0
small	4.1	to	2.0
that	4.1	was	2.0

collocational sequences serve two major functions: 1. marking an amount or measurement, or 2. marking some part of a larger entity. Almost all of the common noun collocates of *large* in academic prose fit into this frame:

– Marking an amount or measurement:
a large number of, large numbers of, large amounts of, large quantities of, a large volume of, large bundles of, large masses of, a large batch of, large areas of
– Marking some part of a larger entity:
a large proportion of, a large part of, a large sample of, a large fraction of, a large segment of

Thus, in this case the strongest collocational association for *large* occurs with the function word *of* occurring at a distance. However, examination of the intervening nouns shows that this frame is typically used to mark an amount or a part of a larger whole.

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Table 2.7 identifies many other collocational associations of interest. Here we focus on only one of those: the pattern *large* X *eyes* in fiction. This association is particularly interesting because it is much more common than the immediate collocate pair *large eyes*; *large* X *eyes* occurs 4.3 times per million words, while *large eyes* occurs only about 1.6 times per million words. The reason for the more common association at a distance is that the noun *eyes* is commonly modified by an adjective of color or quality, in addition to the descriptor *large*. As this collocational pattern shows, these modifiers consistently occur in the order *large* + color/quality-adjective + *eyes*, as in:

his large hazel eyes
large brown eyes
large black eyes
very large dark eyes
large watery eyes

If space permitted, we could consider numerous other collocational patterns at a distance, identifying other patterns of use for *large*, and identifying additional differences from the adjectives *great* and *big*. Our main purpose in the present section has been to simply illustrate how such patterns can be an extremely productive area of investigation in describing the use and meaning of a word.

2.7 Conclusion

In the present chapter, we have illustrated many different kinds of corpus-based lexical investigations. As noted in the introduction, this is one of the earliest uses of a corpus-based approach. In part, this is due to the central importance of word use to language learners. That is, learners need to understand how words are actually used, in addition to simple meanings or lists of supposedly "synonymous" terms. As we have seen, these patterns of use are both highly systematic and complex. Through corpus-based analysis, we can document such patterns, providing the information needed for more informative and helpful language instruction and materials development.

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The need for empirical evidence concerning the actual patterns of use is perhaps most obvious for lexicographic questions. However, similar questions arise across all areas of grammar. In the following chapters, we show how corpus-based research can be used to address language use patterns for grammatical and discourse features. In addition, we will show that the patterns of language use are not autonomous across the different levels of grammar. In particular, we will return to lexical research questions in Chapter 4 and show how word use is often intimately tied to variation in grammatical characteristics.

Further reading

More comprehensive discussions of corpus-based approaches to lexicography are found in Francis (1992), Sinclair (1985, 1987, 1991), Stubbs (1995), and Zernik (1991). In addition, there have been numerous corpus-based publications to investigate specific lexical issues, including studies:

- to disambiguate the functions of multifunctional words (e.g., Aijmer 1986 on *actually*; Altenberg 1994 on *such*; Stenström 1986 on *really*);
- to investigate the distribution and use of closely related words (see Biber, Conrad, and Reppen 1994 on certainty adjectives; Kennedy 1991 on *between* and *through*; Coates 1983, and Collins 1991a on modal verbs);
- to identify and characterize the use of relatively fixed lexical expressions (e.g., Altenberg 1991, 1993; Kjellmer 1991; Nattinger and DeCarrico 1992; Renouf and Sinclair 1991).

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3 Grammar

3.1 The study of grammar

The study of grammar is concerned with understanding the structure of a language, including: morphology – the structure of words; syntax – the way words are arranged into sentences; and other properties of words, such as their grammatical classes (e.g., nouns, verbs, adjectives).

Unlike lexicography, grammar does not have a long tradition of empirical study. Until recently, little attention was given to studying how native speakers actually exploit the grammatical resources in their language. Instead, interest in grammar typically has had one of three major concerns.

First, the study of grammar is often seen as a prescriptive enterprise, itemizing the grammatical rules required for "correct" speech. Rules such as "Don't end a sentence with a preposition" and "Be consistent in the use of tense throughout a sentence" are prescriptive rules that many students associate with English grammar.

Within linguistics, though, grammatical studies are almost always descriptive rather than prescriptive. Traditionally, grammatical studies had the goal of providing a relatively complete catalog of the forms in a language and a description of the rules for combining forms. A grammar of this type would typically describe the full range of morphological forms, including inflectional morphology (e.g., marking person or tense for verbs, or marking number or case relations for nouns) and derivational morphology (e.g., the morphological processes for turning verbs into nouns). Descriptive grammars would also describe the ways that words

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are combined into clauses and sentences, focusing on word order and various kinds of subordination.[1]

More recently, many linguists studying grammar have had more theoretical goals, focusing on identification and analysis of the basic, underlying principles controlling the syntax of a language. This sort of work is concerned with showing both how languages are structured and how human minds are working as they produce or process language.

Although grammatical research within linguistics is almost exclusively descriptive rather than prescriptive, it generally has not used empirical methods to study language use. Descriptive grammarians typically use "field methods" to identify the various paradigms in a language, while theoretical grammarians typically rely on their own intuitions about language, sometimes supplemented by asking native speakers to judge whether made-up sentences are grammatical or not. Neither approach analyzes the patterned use of grammatical features in texts. Further, none of the above approaches focus on variation in language use. In prescriptive approaches, deviations from the norm are generally considered to be errors; in contrast, descriptive approaches are likely to note the existence of grammatically correct alternatives (such as active versus passive voice) but often give little attention to functional reasons for choosing between the alternatives.

The areas that traditional studies have neglected turn out to be the strengths of corpus-based studies of grammar. The availability of large corpora and computer tools make it possible to study the patterned ways in which speakers use the grammatical resources of a language – by investigating the frequency distribution of various constructions, the association patterns between grammatical structures and other linguistic and non-linguistic factors, and the factors that affect choices between structural variants.

Corpus-based research can be applied to grammar on the word level, sentence level, and discourse level. Chapter 5 investigates questions of discourse. In this chapter we illustrate research questions related to morphology, grammatical categories, and syntactic constructions. Specifically, we investigate four research questions, as follows:

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1. How can the use and function of morphological characteristics be better understood by analyzing their distribution across registers?

   Using a corpus, we can relate the distribution of a morphological characteristic to its contexts of use and better understand the functions that it serves. In Section 3.2, this type of question is illustrated by investigating the distribution of nominalizations across three registers.

2. How can the use and function of grammatical classes be better understood by analyzing their distribution across registers?

   Investigating this question can help us to understand how different varieties exploit the grammatical categories of words available to them. For example, previous studies have characterized styles as "nominal" or "verbal," describing how certain authors make greater use of nouns or verbs. Section 3.3 illustrates a corpus-based investigation of this issue by quantitatively studying the ratio of nouns to verbs in three registers. Using the corpus, we can analyze the distribution and function of these different categories of words and study the part that they play in fulfilling the communicative function of different registers.

3. How can the function of syntactic constructions be better understood by analyzing their distribution and linguistic associations across registers?

   As discussed above, previous grammatical studies often neglected questions of variation and discourse function. Here we directly address these issues by examining the complementation of verbs and adjectives with *that*-clauses versus *to*-clauses (e.g., "I think *that I'll go*" versus "I want *to go*"). We show how these two types of complement clause have very different distributions across registers, indicating different functional associations. These differences can be explained in part by considering the lexical and grammatical associations of each type of complement clause. Grammatical associations are considered in Section 3.4, while the lexical-grammatical association patterns are taken up in Chapter 4.

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4. What linguistic and non-linguistic features are associated with the choice between seemingly synonymous structural variants?

Similar to synonymy with words, languages often contain grammatical variants that seem to mean the same thing. These variants are typically considered to be interchangeable. In our investigation of this research question, we show how the use of such structures depends on various linguistic and non-linguistic factors. Specifically, we continue our analysis of *that*-complement clauses, examining the factors associated with the choice between subject clauses versus extraposed clauses (e.g., *That we mustn't worry too much is clear* versus *It is clear that we mustn't worry too much*).

Throughout the sample analyses in this chapter, we show that there are important and systematic patterns of use associated with grammatical features at all levels. Understanding these patterns is crucial for a complete understanding of grammar. That is, grammatical variability is inherent in all human languages – and empirical studies of language use allow us to begin to understand the functional underpinnings of these structural alternatives. Such research concerns are obviously central for language instruction as well as purely academic pursuits. However, our intuitions are often not reliable in predicting when one variant should be used over another. Rather, corpus-based analyses of actual texts are required to uncover these patterns. To illustrate this point, the conclusion of the present chapter contrasts the treatment of subject *that*-clauses in traditional ESL textbooks with the actual patterns of use uncovered through corpus-based analysis.

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3.2 Investigating a morphological characteristic

Studying a morphological characteristic in a corpus can teach us both about the frequency and distribution of the characteristic and about the differing functions of particular variants. In comparison to other grammatical characteristics, corpus-based analyses of morphological characteristics are relatively straightforward. In fact, morphological characteristics can often be identified using the search function in a concordancing package, analyzing an untagged corpus. That is, most concordancing packages

allow the user to search for particular prefixes or suffixes, such as *un+* or *+ment*. For this reason, morphological analysis provides a good introduction to corpus-based investigations of grammar. In this section, the investigation of a morphological characteristic is illustrated by looking at the distribution and function of nominalizations.

3.2.1 The distribution and function of nominalizations

Nominalizations are nouns that are related to verbs or adjectives morphologically; the nominalization is said to be "derived" from the verb or adjective. For example, *civilization* is a nominalization derived from *civilize*, and *kindness* is a nominalization derived from *kind*.

In the present section, we focus on only four common derivational endings: *-tion/-sion*, *-ness*, *-ment*, and *-ity* (and their plural forms). These are very productive endings in English, and a concordancing package can be used to identify nominalizations ending with these suffixes with a high degree of accuracy. However, hand-editing of an automatic search is necessary, since some items will match the search template but not be nominalizations (such as *nation* and *city*). The amount of hand-editing can be limited by making prudent choices in the search template for automatic analysis. For example, a search for all words ending in *-ion* would locate many words that were not nominalizations (e.g., *cushion*, *dandelion*). In contrast, searching for *-sion* provides a much more accurate identification of nominalizations (e.g., *decision*, *division*, *discussion*, *expansion*, *extension*, *submission*), although a few inaccurate items will still be included (e.g., *mansion*).

Although most concordancing programs can search for all words with a particular ending, we have written our own computer programs for the sample analyses in this section. Once students have developed proficient programming skills, they will find that it is often easier to write their own programs specifically for a given analysis. Such programs can be designed to achieve higher accuracy in the search, the output can be tailored exactly for individual preferences, and there is no limit to the size of the corpus that can be analyzed. Methodology Box 3 further explores

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Table 3.1 *Frequency distribution of nominalizations across three registers (academic prose and fiction from the Longman–Lancaster Corpus; speech from the London–Lund Corpus)*

	Academic prose (2.7 mill. wds)	Fiction (3 mill. wds)	Speech (.5 mill. wds)
Nominalizations per million words	44,000	11,200	11,300

the advantages and disadvantages of concordancing packages in comparison to individually designed computer programs.

The analysis of nominalizations was conducted on three registers. The first two – academic prose and fiction – are the same as those used in Chapter 2, with approximately 3 million words of each register from the Longman–Lancaster Corpus. A third register – spoken English – is added in this chapter, to provide a contrast with the characteristics of written registers. We use the entire London–Lund Corpus, a collection of about 500,000 words from spoken monologue and dialogue (see Chapter 1 for more details). All frequencies reported in this section are normed to a basis of a million words of text (see Methodology Box 6).

Table 3.1 shows the frequency distributions for nominalizations across the three registers. The table shows that, while fiction and speech have similar frequencies, academic prose has a frequency almost four times greater.

We can gain some insight into why these registers have such different distributions by studying the more common forms in context. Here we can illustrate one of the advantages of writing your own computer program for corpus analyses: our program is designed so that it counts each individual nominalization and writes out concordance listings for each of them, at the same time that it calculates an overall count for all nominalizations in each register. It is therefore easy to further investigate each type of nominalization in its contexts (see also Methodology Box 3).

The specific nominalizations that are common in a register depend, of course, on the topics covered in the texts of the corpus.

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As we saw in Chapter 2, a very large corpus is necessary to get a representative sample for lexicographic purposes; the London-Lund Corpus is too small for much lexicographic work. Here, however, we want to look not at the use of specific words, but at the role played generally by nominalizations.

In addition to overall frequency of nominalizations, another perspective can be provided by considering the number of different nominalizations occurring in each register. Academic prose is again strikingly different from the other two registers. In fact, six different nominalized words are notably common in academic prose, with frequencies over 500 per million words: *movement* (with almost 900 occurrences per million words), *activity*, *information*, *development*, *relation*, and *equation*. In contrast, no nominalizations in fiction or speech occur this frequently.[2] For example, *movement* occurs around 100 times per million words in fiction and about 60 times per million words in the spoken corpus. *Development* occurs only about 10 times per million words in fiction and is virtually non-existent in the spoken corpus.

When we look at concordance listings for these words, we find that academic prose uses nominalizations to treat actions and processes as abstract objects separated from human participants. We see this pattern with the use of *movement*:

(1) a. If *movement* has occurred recently, the effect on topography, drainage patterns, vegetation and human artifacts can often be recognized.
b. The legs and hips, or arms and shoulders, may be used to initiate *movement* in any direction . . .

Here, *movement* is a process which has been nominalized so that it can be used as the subject or object in clauses. These texts are discussing the generalized action of moving, rather than a particular person moving.

Fiction and spoken discourse, on the other hand, are more often concerned with people and use verbs and adjectives to describe how they are behaving. Thus, these registers more often have particular people as agents and more commonly use *move* as a verb:

(2) a. Garth whistled breathily to himself and *moved* his hand crabwise along the table. (fiction)

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b. He . . . gave a smile which barely *moved* his lips. (fiction)
c. When we *moved* into the new house and . . . (spoken)
d. It's how much they *move* it that counts. (spoken)

This same pattern is seen in the use of nominalizations such as *activity*, *development*, and *information*. In academic prose, these nominalizations are used to refer to generalized processes:

(3) a. The third important aspect of *information* is speed.
b. Sometimes algae can stop the *development* and growth of these plants.
c. The experimental results can be described quantitatively by defining the size and *activity* of the shoot and root systems . . .

In contrast, fiction and the spoken corpus more often refer to similar processes and actions with verbs or adjectives, describing what specific people are doing:

(4) a. I do hope you know that never in this country do we *develop* the sort of mob war that makes a protest against something however unjust *develop* into an organized riot. (spoken)
b. I've *informed* the Soviet government of that visit. (spoken)
c. "Aye, the big fellow is *active* again you'll be pleased to know." (fiction)

These generalizations are not to say that fiction and speech do not use nominalizations, or that academic prose does not refer to processes with verbs. However, there is a pattern of association between the registers and the distribution of nominalizations. Academic prose more often refers to a process with a stative nominalization, where fiction and the spoken corpus describe a specific person's action with a verb or adjective. That is, the focus of academic prose is usually on generalized processes, states, or relationships, whereas the other two registers commonly focus on a specific person doing some activity.

3.2.2 The distribution and function of nominalization endings

In the previous section we looked at the distribution of nominalizations as a group. Another perspective on nominalizations

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Table 3.2 *Proportion of nominalizations formed with each suffix*

	Academic prose	Fiction	Speech
-tion/-sion	68%	51%	56%
-ment	15%	21%	24%
-ness	2%	13%	5%
-ity	15%	15%	15%

can be gained by looking at the distribution of each suffix separately. From this perspective we can learn more about the functions of the different kinds of nominalizations and the role they play in the registers.

Table 3.2 shows, for each register, the percentage of all nominalizations that are accounted for by each of the four suffixes considered here. Rather than showing actual frequencies, the table shows the relative proportion of nominalizations with each ending in each register.

Several interesting patterns are evident in Table 3.2. Here we will briefly discuss three of these:

1. Although the *-tion/-sion* suffix accounts for the majority of instances in all three registers, the proportion is by far highest in academic prose (68 percent).
2. The *-ment* suffix accounts for a greater percentage of the nominalizations in fiction and the spoken corpus than in academic prose.
3. The *-ness* ending is more important in fiction than in either of the other two registers.

Although there are no absolute rules governing the choice among nominalizing suffixes, a detailed consideration of each type reveals certain systematic differences in meaning, providing some insight into the distribution patterns shown in Table 3.2. For example, the *-tion/-sion* ending is used to convert an action expressed by a verb into a noun, usually referring to a generalized process or state. Typical examples are *relate/relation* and *educate/education*. In the last section, we saw that academic prose has the

greatest concern for explaining such states and processes. It thus makes sense that this register would have the highest percentage of its nominalizations with this ending. Other examples include:

(5) a. the *conclusion* is incorrect . . .
b. it must be noted that the natural division of *expansion* between a compressor turbine and power turbine may not . . .
c. designers can make a choice of cycle for a particular *application.*
d. it provides a direct *indication* of fuel *consumption* . . .

Like *-tion/-sion*, the *-ment* ending is used to convert a verb to a noun. Count nouns formed with *-ment* frequently refer to a process making or doing something. Many of these nominalizations occur in all three registers; for example, *movement, government, achievement, agreement, argument.*

However, many nominalizations formed with *-ment* are non-count nouns, which usually describe mental states; for example: *amazement, agreement, astonishment, disappointment, embarrassment, excitement.* These kinds of nominalizations are rare in both academic prose and the spoken corpus. In fiction, though, they are relatively common, where they are used to describe the mental states of characters:

(6) a. Patrick shrugged in *embarrassment.*
b. The assembly cried out savagely and Ralph stood up in *amazement.*
c. So they're gone, she thought, sighing with relief and *disappointment.*

In the spoken corpus, these same nominalizations are occasionally used for similar purposes:

(7) a. . . . I can quite see there's cause for *disappointment.*
b. . . . the *excitement* of the recurrence of the seasons . . .

However, it is more common for the spoken texts to refer to mental states as verbs and adjectives:

(8) a. you'll be *amazed.*
b. are you *disappointed* by not getting honors yourself?
c. It would have been *embarrassing* to have had to tell him later that he wouldn't have got it.

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Finally, we turn to the high proportion of *-ness* in fiction. Even though fiction has the lowest frequency of nominalizations overall, the normalized counts for *-ness* nominalizations are highest in that register (c. 1,430 per million words, versus 890 per million in academic prose and 480 per million in the spoken corpus). The *-ness* ending generally converts adjectives into nouns that often describe personal qualities. Fiction uses a number of these *-ness* nouns that are rarely found in the other registers: *awareness*, *bitterness*, *darkness*, *goodness*, *happiness*, *politeness*, *weakness*. Again, these words are important for the detailed description that occurs in fiction:

(9) a. The *bitterness* in his heart was now mixed with a kind of childlike excitement.
b. He could see Phyllis's face in profile, and it radiated energy and *happiness*.
c. They spoke to each other with the wary *politeness* of uncongenial strangers stranded in an airport lounge.

Describing personal feelings and qualities is also important in the spoken corpus. However, in that register these descriptions are more commonly packaged as adjectives, often directly attributed to the speaker or hearer:

(10) a. We feel frustrated and *bitter* and annoyed.
b. I'm not too *happy* about this lauding of language . . .

Much more could be said about the use of nominalizations. We could look, for example, at the number of different words occurring with each ending to see which is the most productive; or we could undertake a historical study to explore the development and use of nominalizations through time. However, even this brief example illustrates the strengths of corpus-based work for morphological studies. In particular, this section has shown how nominalizations as a group, as well as particular derivational suffixes, have strong association patterns with different registers, reflecting the primary communicative functions of the registers.

3.3 The distribution of grammatical categories

In the last section we saw that academic prose had more nominalizations than fiction or speech, suggesting a greater focus

on abstract states, processes, and objects. Fiction and speech, on the other hand, were characterized as including more personal descriptions and actions attributed to specific people. If these characterizations are accurate, we should also see a difference in the overall frequencies of nouns and verbs across these three registers. In fact, stylistic studies have sometimes used comparative counts of nouns and verbs to characterize the styles of particular authors or texts. Such counts can also be used to describe differences among registers, as illustrated in the present section.

3.3.1 Issues in counting grammatical categories

With a grammatically tagged corpus, counting nouns and verbs is a relatively easy matter. Every noun will have a tag beginning with "n," and every verb will have a tag beginning with "v." It is thus easy to write a program or use a commercially available program to count every word marked with one of these "tags." (See Methodology Box 4 for a discussion of tagged corpora and Methodology Box 5 for an explanation of the process of tagging.)

A more complicated question, however, is what precisely to include as a noun or as a verb. There is no single correct answer to this question. Rather, the important thing is to make a principled decision and then be consistent in making counts.

Several issues are problematic in deciding how to count nouns and verbs. The first is whether to count nouns that function as adjectives, modifying a following noun. For example, should *grasshopper ecology* and *animal groups* each be counted as one noun or two? On the one hand, *grasshopper* and *animal* serve to modify the following nouns, and in that way they are similar to adjectives in phrases such as *general ecology* or *small groups*. On the other hand, a phrase such as *grasshopper ecology* does contain two distinct referents – grasshoppers and ecology – and in that way it differs from phrases with an adjective as pre-modifier (such as *general ecology*). Given that our goal here is to count the extent to which nominal versus verbal references are used, it seems appropriate to include nouns used to modify other nouns.

A question also arises with pronouns. Since pronouns take the place of nouns, in a sense they refer to a nominal entity or

abstraction. However, they differ from nouns in that they do not refer to anything when used in isolation. For example, the referent for *he* cannot be identified apart from a specific context, making it different from *grasshopper*. If we want a count of the words that directly refer to things, then it seems most appropriate to omit pronouns.

In counting verbs, similar kinds of questions arise. One issue is whether auxiliary verbs should be included in the overall verb count. For example, the following sentences from fiction each include an auxiliary verb (italicized) in addition to a lexical main verb:

(11) a. He *had* left home a little before eight.
b. Joanne and her mother *were* talking.
c. He *did*n't stop.

These auxiliary verbs do not provide any lexical content. Instead, they serve only to mark aspectual meaning (perfect or progressive aspect), or they are required for negation (in the case of *do*). It thus makes sense to exclude auxiliary verbs from an overall count which shows reliance on verbs rather than nouns.

As stated above, there is no single correct answer to the question of what to count as nouns and verbs. It is most important to consider the research questions that are being investigated and make principled decisions. It is also important to determine how counts were made in other studies before making comparisons across studies. As we shall see, different ways of counting give different views of just how "nouny" a register is.

Table 3.3 displays the noun–verb ratios for three different ways of counting nouns and verbs. Line A includes all nouns and verbs, line B excludes auxiliary verbs, and line C excludes nouns acting as premodifiers of other nouns as well as auxiliary verbs. (Pronouns are excluded from all noun counts.) For the calculation in each line, the total count of nouns was divided by the total count of verbs, thus showing how many nouns occur relative to each verb.

Before we focus in on one way of counting, notice that the same relative relationships hold in all three counts. Fiction and the spoken corpus have very similar noun–verb ratios, while

Table 3.3 *Noun-to-verb ratios in three registers*

	Academic prose	Fiction	Speech
A. all nouns and verbs	2.2:1	1.2:1	1.2:1
B. all nouns and verbs excluding auxiliaries	2.9:1	1.5:1	1.6:1
C. nouns excluding premodifiers of other nouns and verbs excluding auxiliaries	2.5:1	1.3:1	1.3:1

academic prose is close to double that ratio. With any method of counting, then, we find similar patterns among the registers.

On the other hand, the exact ratios vary and give different impressions of the nouniness of the registers. When all nouns are counted and auxiliary verbs are excluded, academic prose has a ratio of almost 3 to 1. When all nouns and verbs are counted, however, the ratio in academic prose is just over 2 to 1. As noted above, these differences are especially important to note for comparisons with different studies.

3.3.2 Comparison of noun–verb ratios across registers

For our purposes here, the most appropriate way of counting nouns and verbs is to exclude pronouns from the noun count, and to exclude auxiliary verbs from the verb count (i.e. Line B in Table 3.3). The finding that academic prose has a much higher noun-to-verb ratio than the other two registers (2.9:1 versus about 1.5:1) can be interpreted in a similar way to our interpretation of the findings about nominalizations in Section 3.2. For example, consider the following excerpt from an academic prose text, which has 10 nouns (italicized) and only 1 main verb (bold):

(12) In *planning* a *livestock building* or *conversion*, the psychological and *health requirements* of the *livestock* should undoubtedly be **given** absolute *priority* together with the basic *needs* of the *stockman*.

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In this passage, we can see the emphasis in academic prose on objects, states, and processes – all referred to with nouns – rather than human agents and their actions (described with verbs). For example, instead of describing how a person planned to construct or convert a building, we get nominal descriptions of general processes: *planning*, *conversion*.

In contrast, typical passages from fiction or the spoken corpus contain much more action. The following fiction passage has 7 nouns and 5 verbs, used to describe the actions of a specific person:

(13) He **emerged** and **locked** the *door*. He **unsnapped** the protective *strap* on his *holster* and **scanned** the parking *lot*. He **walked** quickly to the *glass door* of the *bank*.

The following short passage from a conversation contains even more verbs than nouns – 14 main verbs and 4 nouns:

(14) A: Oh yeah, it's **called washing** your *hair*. Don't you **know** how to **wash** your *hair*?
B: Might **be**.
C: I **know**, I **know** how to **have** a *bath*.
B: **Go** away, I'm **cooking**. . . . **Excuse** me please, I'm **trying** to **cook**. I haven't **got** enough *potatoes*.

In the fiction and spoken passages, it is also apparent that pronouns take the place of many nouns, and this reduces the noun-to-verb ratio. For example, in the conversation sample there are more pronouns than nouns: *I*, *me*, *you*, and *it* occur eight times. In the academic prose passage there are no pronouns. Thus, looking simply at noun-to-verb ratios does not provide adequate overall characterizations of these registers. Even these short passages suggest that other grammatical features co-occur with nouns and verbs. We will return to the distribution of nouns and verbs in Chapter 6, showing how these grammatical classes co-occur with a wide range of other linguistic features across registers.

3.4 Distribution and function of a syntactic construction

To illustrate a corpus-based investigation of a syntactic construction, we briefly describe certain aspects of the grammar of

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complement clauses in English. Complement clauses complete the meaning of the verb or adjective. For example, in

(15) Convention requires *that precedents be followed . . .*

the italicized *that*-clause completes the meaning of the verb *requires*, by explaining what is required. In

(16) I'm sure *that you know he would be awfully grateful . . .*

the italicized clause specifies what the speaker is sure about, completing the meaning of an adjectival predicate (*be sure*).

The two most common types of complement clause are *that*-clauses and *to*-clauses. In some contexts, these two are similar in meaning. Thus compare:

(17) a. I hope that I can go.
b. I hope to go.

However, corpus-based study shows that the actual use of these two structures is quite different.

In the following sections, we first consider several methodological issues relating to the identification and coding of *that*- and *to*-complement clauses (Section 3.4.1). Then, in Section 3.4.2 we compare the distribution of the two clause types across registers, and we briefly consider associations with various structural factors. We note in passing that lexical associations are very important in accounting for the differing functions and distributions of the two complement clause types, but we defer detailed discussion of those factors until Chapter 4. Finally, in Section 3.5 we turn to a more detailed discussion of two structural variants that have nearly equivalent meaning: *that*-clauses in subject position versus extraposed constructions (explained below). In all of these cases, we see how corpus-based analyses can be used to uncover association patterns that would go unnoticed otherwise.

3.4.1 Identifying and coding that- *and* to-*complement clauses*

As with most syntactic constructions, it is very difficult to use a concordancing package to identify *that*- or *to*-complement clauses

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in an untagged corpus. It would require looking at every occurrence of *that* and *to*, to determine whether or not it was a complement clause. What makes this task so difficult is the fact that these words are multiply ambiguous. For example, the word *that* has numerous grammatical functions, including a demonstrative determiner (*that horse*), a demonstrative pronoun (*that is ridiculous*), a relative pronoun (*the person that I like*), and part of a complex clause connector (*such that* and *so that*). The word *to* is similarly multifunctional (e.g., marking adverbial clauses, relative clauses, and prepositional phrases).

In addition, the complementizer *that* can be optionally omitted from a *that*-clause, as in

(18) a. I thought *you were going to go and pick them up.*
b. I don't think *we will see any very dramatic changes . . .*
c. I'm sure *you'll do very well . . .*

To identify clauses with omitted *that* using a concordancer, we would need to examine every occurrence of every verb and adjective that can take a *that*-clause, to check whether they are followed by a complement clause with the *that* deleted. Such work would be inordinately time-consuming and unfeasible with a large corpus.

With a tagged corpus and computational techniques, the analysis can proceed much more quickly. Using on-line dictionaries of the verbs and adjectives that take complement clauses, grammatical taggers can identify *that*-clauses and *to*-clauses with a high degree of accuracy. This same approach can be used to search for *that*-clauses with the *that* omitted.

However, identifying *that*-clauses and *to*-clauses in conversation can still be a tricky business, and hand-editing is necessary. Repetitions, false starts, and the transcribing conventions of a corpus can make automatic coding difficult. For example, consider this passage as it looks in the London–Lund Corpus:

(19) B: Oh no . I have never said that – in fact I went to great pains I will be perfectly frank with you . I went to great pains to put it . about – quite publicly that you were the one who in fact was supporting me in the interview.

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In this transcript, the periods stand for short pauses, and the dashes mark longer pauses. Thus, these punctuation marks in spoken transcription are no aid in identifying syntactic units. As a result, there are often structures that are unclear or ambiguous for an automatic tagger. For example, the first line in the above sample could be interpreted in two different ways:

1. "I have never said that I went to great pains"
   where the *that* begins a verb complement clause, and
2. "I have never said that. I went to great pains to . . . put it about"
   where the *that* is a demonstrative pronoun.

By looking at the extended context, a human analyst can tell that the second analysis is the correct one. That is, speaker B is responding to the following statement by speaker A:

(20) A: I've heard from a number of sources that you [. . .] think you did not get the job here because of me –

Thus, the accurate interpretation of B's statement seems to be "Oh no. I have never said that. In fact, I went to great pains to say you supported me." The *that* in this case is thus a demonstrative pronoun rather than a *that*-complement clause. In cases like this, however, only a human can analyze the sentence accurately. Thus, the codes assigned for complex constructions by automatic programs must be edited by hand, especially when dealing with spoken texts.

In addition, there are some ambiguous constructions that even a person cannot analyze with certainty. Expressions such as *I mean*, *you know*, and *you see* in spoken texts are particularly problematic with *that*-clauses. For example, in the following utterance, it is impossible to be certain how *I mean* is being used:

(21) oh no I don't guarantee them I mean you can't do that

Is this speaker literally clarifying what she means, as in "I mean that you can't do that"; or is she using *I mean* as a discourse marker inserted at this clause boundary? Similarly, the phrase *you know* can sometimes have the meaning of "you know that", as in:

(22) well you know I don't know very much about what he does

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That is, this utterance could be interpreted as a literal statement of what the hearer knows – "you know that I don't know very much . . ." On the other hand, the phrase *you know* frequently functions as a filler or discourse marker, such as:

(23) you know I'm just – I literally I was there this morning . . .

Here it seems unlikely that the speaker is making a statement about what the hearer knows.

In many cases, there is simply no straightforward way of deciding which interpretation is correct – or whether there is a single "correct" interpretation! It is therefore best to make a consistent judgment about how to handle such expressions; in the case of our sample analysis, ambiguous uses of *you know*, *I mean*, and *you see* have been omitted from the analysis.

For the reasons illustrated above, many complex grammatical constructions are difficult to identify consistently using automatic techniques. Therefore, several analytical steps are necessary before counts can be compiled of such constructions. For our example here, the first step involved using a computer program to automatically identify constructions that were likely to be *that*- or *to*-complement clauses. Each automatically identified construction was then checked using an interactive program. Although this is time-consuming, the interactive checking program is much faster than simply reading through texts (see Chapter 5 for more details about interactive programs). Finally, after the codes had been edited, another program was used to calculate frequency counts of each structural variant.

3.4.2 The distribution and function of that- *and* to- *complement clauses across registers*

To compare the distribution and function of *that*-clauses and *to*-clauses, we analyze their use in two registers: academic prose, based on a sample of approximately 4 million words from the Longman–Lancaster Corpus, and conversation, based on a sample of approximately 5 million words from the British National Corpus. Even the overall frequencies of the two complement clause types, plotted in Figure 3.1, show striking differences. *That*-clauses

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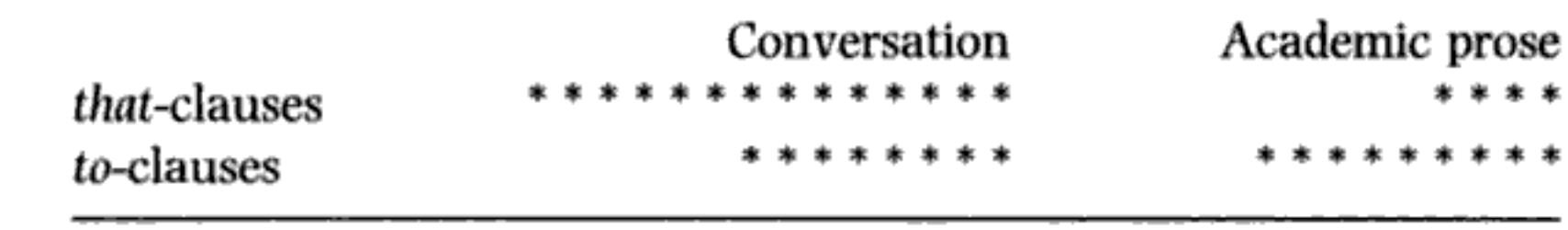

Figure 3.1 Overall distribution of *that*-clauses and *to*-clauses in conversation and academic prose (each * represents 500 occurrences per million words)

are very common in conversation but not so common in academic prose. In contrast, *to*-clauses are moderately common in both conversation and academic prose.

This difference in overall distribution can be related in part to the differing lexical associations of the two types of complement clause. In fact, there are only a few verbs that can control both *that*-clauses and *to*-clauses, such as *hope*, *decide*, and *wish*. Instead, most verbs control only one or the other type of complement clause. For example, the verbs *imagine*, *mention*, *suggest*, *conclude*, *guess*, and *argue* can control a *that*-clause but not a *to*-clause; the verbs *begin*, *start*, *like*, *love*, *try*, and *want* can control a *to*-clause but not a *that*-clause. Lexico-grammatical association patterns – between grammatical constructions and sets of words – are so important that we have devoted a separate chapter to them (Chapter 4), and we return to the differences between the lexical associations of *that*-clauses and *to*-clauses in that chapter.

However, the differing distributions of *that*-clauses and *to*-clauses can also be related to their grammatical associations. One reflection of this difference is their use in "extraposed" versus "non-extraposed" constructions. Extraposed constructions have a complement clause occurring after the verb phrase, with a dummy *it* as the subject of the sentence.

(24) *That*-clauses
   a. Non-extraposed:
      I think that you might be wrong.
   b. Extraposed:
      It's possible that it'll happen again.

(25) *To*-clauses
   a. Non-extraposed:
      I want to sleep here.
   b. Extraposed:
      It's possible to adjust the limit upwards.

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	Conversation	Academic prose
Extraposed *that*-clauses	* *	* * * * * *
Extraposed *to*-clauses	* *	* * * * * * * * * * * * * * * *

Figure 3.2 Use of *that*-clauses and *to*-clauses in extraposed constructions (each * represents 100 occurrences per million words)

	Conversation	Academic prose
Extraposed *to*-clauses with verb predicates	*	*
Extraposed *to*-clauses with adjective predicates	*	* * * * * * * * * * * * * * * *

Figure 3.3 Use of *to*-clauses in extraposed constructions controlled by verbs versus adjectives (each * represents 100 occurrences per million words)

Figure 3.2 shows that *to*-clauses are in fact used much more commonly in extraposed constructions than *that*-clauses, especially in academic prose.

It turns out that the association of extraposition with *to*-clauses in academic prose is even more specialized than Figure 3.2 indicates. That is, as we mentioned above, complement clauses can be used to complete the meaning of either a verb predicate or an adjectival predicate. However, Figure 3.3 shows that extraposed *to*-clauses have a very strong association with adjectival predicates, especially in academic prose. These constructions are especially common with adjectives such as *possible*, *impossible*, *difficult*, and *hard*; for example:

(26) a. It is impossible to know how these levels will be viewed.
b. It is often difficult to find a suitable donor.
c. It is hard to resist the temptation . . .

These grammatical associations – of *to*-clauses associated with extraposition, which is in turn associated with adjectival predicates – help to explain the general preference for *to*-clauses over *that*-clauses in academic prose. In fact, these patterns fit well with the general patterns of use illustrated in the earlier sections of this chapter. Thus, we saw earlier that academic prose has a preference for static rather than dynamic packaging of information, reflected in more frequent nominalizations and a much higher

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noun-to-verb ratio. Similarly, we saw the preference for generalized states and processes in academic prose, in contrast to the descriptions of specific people performing actions in fiction and conversation. Extraposed constructions with adjectival predicates serve both of these functions: they present a proposition that cannot be attributed to any particular person, and they frame the proposition in terms of a static condition (with an adjectival predicate such as *possible* or *difficult*) rather than in terms of a dynamic action or process (such as *think* or *feel*). Thus, the typical grammatical associations of *to*-clauses fit well with the typical communicative priorities of academic prose, resulting in a greater reliance on *to*-clauses in that register.

A fuller comparison of *that*-clauses and *to*-clauses would require a similarly detailed interpretation of the patterns for *that*-clauses, allowing us to account for the overall preference for those constructions in conversation. (We return to this question in Chapter 4, focusing on lexical associations.) The main point here is simply to illustrate the systematicity and importance of grammatical association patterns in accounting for the patterns of use for grammatical features across registers.

3.5 Factors associated with the choice between synonymous structural variants

The grammar of any language contains numerous structural variants that are nearly equivalent in meaning. For example, active and passive constructions are often very close in meaning, as in:

(27) a. Active:
A split exhaust pipe had caused Brundle's trouble.
b. Passive:
Brundle's trouble had been caused by a split exhaust pipe.

Similarly, clauses with two objects can often be alternatively expressed using a direct object and an indirect object, with little difference in meaning; for example:

(28) a. Verb + noun phrase + noun phrase:
The strain of unrelenting analytical effort gave Mr. Sammler a headache.

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b. Verb + noun phrase + prepositional phrase:
The strain of unrelenting analytical effort gave a headache to Mr. Sammler.

That-clauses also have two structural variants that are nearly equivalent in meaning: subject position versus extraposed constructions. In addition to the structural options described in the last section, *that*-clauses can occur in subject position:

(29) *That these identifications are circular* is unimportant at the moment.

Extraposed constructions (introduced in the last section) provide a structural alternative to *that*-clauses in subject position:

(30) It is unimportant at the moment *that these identifications are circular*.

An obvious question from a use perspective is why a language should have structural alternatives with similar or equivalent meaning. That is, what different purposes do they serve, and how does a speaker know when to use each option? These questions are similar to the questions surrounding the choice among "synonymous" words, addressed in Chapter 2. In the present section, we illustrate a corpus-based approach to these grammatical issues by comparing the use of subject versus extraposed *that*-clauses. We first consider various discourse factors influencing this choice, and then compare these findings with the treatment of this structural option in selected ESL textbooks.

3.5.1 Factors associated with the choice between subject and extraposed that*-clauses*

That-clauses in subject position are very rare in all registers. In academic prose, subject *that*-clauses occur only about 5–10 times per million words, making extraposed *that*-clauses over twenty times more common. Subject *that*-clauses are virtually non-existent in the spoken corpus.

The preference for extraposed constructions can be explained in part by a general preference for putting heavier constituents at the ends of sentences. From a processing perspective, this preference allows readers to find the main verb of the sentence sooner. With a subject *that*-clause, readers often must process a long,

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complicated structure in subject position before coming to the main verb of the sentence, which can cause difficulty in understanding. Thus compare:

(31) a. Extraposed:
It follows *that frequentist probability is conceptually inadequate for the design or licensing of hazardous facilities.*
b. Subject *that*-clause:
That frequentist probability is conceptually inadequate for the design or licensing of hazardous facilities follows.

Given this overall preference, though, we might wonder why speakers use subject *that*-clauses at all. What factors are associated with their use? These are the sorts of questions that could be addressed through detailed corpus-based investigation. In the present case, we generated a concordance listing with every occurrence of a subject *that*-clause in the Longman-Lancaster Corpus. Since this is a rare construction, it is easy to go through each occurrence looking for systematic patterns of use. Note that this is a different approach from that illustrated in most previous sections. On the one hand, we examined every occurrence of this construction in context (rather than using computational techniques to uncover general patterns over hundreds of occurrences). On the other hand, we do not use quantitative techniques to describe the patterns of use; with so few tokens, quantitative patterns are less likely to be replicable in other corpora.

However, the use of a detailed qualitative approach does not preclude the identification of strong patterns of use. In the present case, two patterns emerge. Subject *that*-clauses are used:

1. when the predicate of the sentence (occurring after the main verb) is a long or complex structure, often another *that*-clause, and
2. when the subject *that*-clause states "given" information – either information previously stated or taken to be a known fact.

For example, in the following text, the italicized sentence has two *that*-clauses, one as subject and one as predicate. Furthermore, the subject *that*-clause (marked in square brackets) restates the idea that has just been established in the previous sentence (i.e., that labor power is a commodity):

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(32) Under the capitalist process of production, labour power itself becomes a commodity belonging to the capitalist and "the right to use that power for a day belongs to him, just as much as he has a right to use any other commodity . . ." *[That labor power is a commodity] means that the worker functions as a thing . . .*

If this sentence is converted into an extraposed construction, it becomes extremely difficult to process, because it has two complex structures coming after the main verb (i.e., the two *that*-clauses):

(33) *It means that the worker functions as a thing that labor power is a commodity.*

The sentence with a subject *that*-clause in the following excerpt also has a long, complex structure occurring after the main verb:

(34) The location and other characteristics of the schools associated with the project suggested that the sample of low attainers drawn from them would be representative of the lowest 40 per cent of attainers nationally. *[That this was so] was demonstrated by using a reference test of APU items which had parameters established in national surveys.*

Here, the given information in the subject *that*-clause (marked in square brackets) is referred to simply with the demonstrative pronoun *this*, which stands for the idea that "the sample was representative." In this case, the structure occurring after the main verb (*was demonstrated*) is extremely complex, consisting of a *by* prepositional phrase, which includes another prepositional phrase (*of APU items*) plus a relative clause (*which had parameters . . .*), which in turn includes yet another prepositional phrase (*in national surveys*).

Finally, in some cases the information in the subject clause may be "given" in the sense that it reports an established fact. The following passage discusses the evolution of winged insects. Instead of a simple subject *that*-clause, this example actually uses a subject noun phrase beginning with *the fact that*:

(35) The remains of the earliest winged insect from the Upper Carboniferous give no indications of how and where the Pterygota arose. *[The fact that in many insect groups the newly emerged adults show a slow gliding type of flight linked to dispersal] suggests,*

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> *however, that the earliest winged insects may have evolved in temporary habitats of small, erect plants . . .*

In this case, the proposition being reported – that newly emerged adult insects have a slow gliding type of flight – is not introduced in the previous discourse; but it is presented as an established fact through the use of this construction. (Note again that this example also has a *that*-clause occurring after the main verb, making the alternative with an extraposed construction very difficult to process.)

One or both of these two discourse factors are found in nearly all the examples of subject *that*-clauses examined from the Longman–Lancaster Corpus. If we analyzed a very large corpus (100 million words or larger), we would have sufficient tokens to give more precise quantitative measures of the strength of each factor. However, the detailed discourse analyses summarized here are sufficient to identify these factors as important influences on the choice of a subject *that*-clause over an extraposed alternative.

3.5.2 Comparison with ESL textbooks

In the last chapter, we saw that current dictionaries are not equally effective in presenting the actual patterns of use most commonly found in spoken and written texts. In a similar way, the findings of corpus-based investigations can be used to inform the presentations in textbooks for ESL students. Findings about grammatical constructions are particularly important in this case.

For example, the previous section showed that the use of subject *that*-clauses are associated with both non-linguistic and linguistic characteristics. First, they occurred primarily in expository written registers and were virtually non-existent in conversation. Secondly, their use in technical writing was associated with two discourse factors: complex predicates and the expression of given information. Patterns such as this can be very useful to ESL students, since knowing when to use structures appropriately is an essential part of developing communicative competence in a language.

However, ESL textbooks often do not provide reliable information about language use. For example, Table 3.4 briefly summar-

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Table 3.4 *Coverage of subject position* that*-clauses in four ESL textbooks*

type of book	Examples + description	Number of exercises	Explanation of variants
grammar (Azar 1989)	yes	2 oral, 1 written	none
academic grammar (Danielson & Porter 1990)	yes	0	none
college composition (Jenkins 1986)	yes	1 written	none
academic composition (Spack 1990)	yes	0	none

izes the way that four ESL textbooks present subject *that*-clauses. Two of these books focus on grammar, and two on academic/college writing. All four of the books give examples or descriptions of subject *that*-clauses. In addition, two of the books include exercises for practicing subject *that*-clauses (one of these includes oral exercises). However, none of the books gives any explanation of when to use subject *that*-clauses. Students are left knowing that this construction exists, but without knowing what its function is or when it is appropriate to use it.

Using the findings of corpus-based studies could improve such textbooks in two ways. First, books could emphasize those constructions most commonly found in the target register. Students typically study English for particular purposes – for example, conversational English for fluency in everyday interactions, or academic English for proficiency in educational reading and writing tasks. Textbooks can build on those goals to teach the grammatical constructions that students are most likely to encounter, given their communicative goals. In the present case, there is little

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to be gained by having students practice subject *that*-clauses orally, given that these constructions are almost never used by native speakers in spoken discourse.

Second, students can be given explanations of when to use different structural variants. In the present case, the associations between given information and complex predicates in relation to subject *that*-clauses could be explained, and examples and exercises consonant with these patterns could be included. This approach would help students understand the informational implications of subject *that*-clauses when they encounter them in reading, and eventually help them to produce such structures appropriately in writing.

In sum, corpus-based research is not only useful for our academic understanding of language function and use. It is also helpful for language teaching. By basing instruction on the actual patterns of use, we are able to equip students with the tools that they need for producing and understanding language appropriately across different contexts.

3.6 Conclusion

The present chapter has focused on the empirical study of grammar. We have illustrated how corpus-based techniques can be used to study grammar at many different levels, including morphological features, grammatical classes, and syntactic constructions. We have shown that grammatical features are distributed in systematic ways across registers, illustrating our recurring theme of the importance of register differences. Once again, any patterns generalized for all of English are not likely to be valid for any actual text or register – rather, generalized patterns would merely level the important patterns of use found across registers. Further, we have illustrated the way in which these register patterns can be interpreted functionally, in terms of the differing communicative goals and characteristics of each register.

The following two chapters continue our exploration of linguistic characteristics. In Chapter 4, we investigate patterns of use at the interface between lexicography and grammar, showing how there are systematic association patterns between grammat-

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ical features and classes of words. In Chapter 5, then, we move on to a higher level of structure, looking at the ways that discourse features can be studied using corpus-based techniques.

Notes

1. Such grammatical descriptions were sometimes used as an aid to language learners. This was the case in schools as early as the Middle Ages and Renaissance, when scholars learned Greek and Latin by studying the grammar of those languages. Even today, some people equate knowing a second language with being able to recite grammatical rules.
2. Several of the more common words identified by an automatic search for derivational suffixes should not be considered as productive nominalizations. These include: *condition, position, question, solution, university, department, apartment,* and *parliament.*

Further reading

Book-length treatments reporting corpus-based grammatical investigations include: Coates (1983) on modal verbs, Collins (1991b) on clefts, Granger (1983) on passives, Mair (1990) on infinitival complement clauses, Meyer (1992) on apposition, Tottie (1991) on negation. Several books and articles focus on nominal structures: de Haan (1989) with an overview of the structural options for postmodifying clauses; Fox and Thompson (1990) on relative clauses in conversation; Geisler (1995) on relative infinitives; Hudson (1994) discussing the constant proportion of nouns across registers; Johansson (1995) on the relativizers *whose* and *of which*; Varantola (1984) on noun phrase structures in technical English. In addition, there have been numerous research papers published using corpus-based techniques to study English grammar (see many of the papers in collected volumes such as Aijmer and Altenberg 1991; Johansson and Stenström 1991; Aarts and Meyer 1995).

The Longman Grammar of Spoken and Written English (Biber, Johansson, Leech, Conrad, and Finegan, to appear) provides a comprehensive survey of English grammar from a corpus-based perspective, including many analyses similar to those found in this chapter.

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4 Lexico-grammar

4.1 Investigating lexico-grammatical issues

In Chapters 2 and 3, particular lexical and grammatical features were explored from a corpus-based perspective. Specifically, in Chapter 2 we discussed how words have strong association patterns with other words – their "collocations." Then, in Chapter 3 we discussed how grammatical structures can similarly have strong association patterns with other grammatical structures. In addition to those kinds of patterns, however, there may also be associations between words and grammatical structures – i.e., lexico-grammatical associations. In the present chapter, we focus on these relationships: between words and their grammatical environments, or between grammatical structures and their lexical environments.

The investigation of lexico-grammatical associations is particularly useful when we are attempting to distinguish between words or structures that are nearly synonymous in meaning. It turns out that lexico-grammatical associations can be considered from either end of the relationship. That is, we can distinguish between two nearly synonymous words by considering their associations with different grammatical constructions; and we can distinguish between two nearly synonymous grammatical constructions by considering their associations with different classes of words.

In the present chapter, we illustrate both kinds of research question:

1. How can nearly synonymous words, with the same grammatical potential, be distinguished in terms of use patterns relating to their grammatical associations?

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We consider two different examples of this type. First, in Section 4.2, we examine the grammatical associations of two nearly synonymous adjectives: *little* and *small*. These two words are very similar in meaning, and both can occur in either attributive or predicative position, supporting the view that they are nearly synonymous. However, corpus-based study shows that the two words have very different preferences for attributive or predicative positions, and that these preferences are strongly associated with their distribution across registers.

In our second example (Section 4.3), we investigate the grammatical associations of two nearly synonymous verbs: *begin* and *start*. These two verbs are grammatical in the same range of valency patterns; that is, they are grammatical with the same variety of clause elements (e.g., both can occur as an intransitive verb or as a transitive verb with a direct object). However, our corpus-based analysis shows that the two words are in fact very different in their preferred grammatical associations.

2. How can the use of nearly synonymous grammatical constructions be understood in terms of their differing lexical associations?

This question continues our exploration of *that*- and *to*-complement clauses begun in the last chapter. In Section 4.4, we show how differences in the overall distribution of these clause types can be related to their differing lexical associations. This analysis shows that each type of complement clause has a distinct set of verbs that it commonly occurs with; and in those cases where a verb has the potential to occur with both clause types, the analysis shows that in actual use there is a very strong preference for one or the other type.

4.2 The grammatical associations of nearly synonymous words: *little* versus *small*

Chapter 2 described how nearly synonymous words can be distinguished in terms of their lexical association patterns, or their "collocations." Thus, we showed how the adjectives *big*, *large*, and *great* tend to co-occur with different sets of nouns. It turns out that related words are also distinguished by having different grammatical

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associations. In the present section, we compare the adjectives *little* and *small*, investigating their associations with attributive and predicative functions.

Attributive adjectives precede nouns and provide information about that noun. For example,

(1) a. The *little* girl next door pulled him through the fence.
b. But I'm not a *small* person.

Adjectives in predicative position, on the other hand, occur after a copula, although they are similar to attributive adjectives in providing information about a noun (i.e., the subject of the sentence). For example,

(2) a. When she was *little*, she couldn't say Jessica.
b. Did you think it would be too *small*?

The use of attributive adjectives results in a more integrated structure than predicative adjectives. That is, predicative adjectives require an entire clause to present the same information contained in a noun phrase with an attributive adjective:

Predicative:
Noun phrase BE adjective
Attributive:
Adjective + noun

In considering factors that might distinguish between *little* and *small*, we asked ourselves whether they might be used to differing exents in attributive versus predicative positions. To investigate this possibility, we analyzed their distribution in a corpus of conversation and academic prose: c. 5 million words of conversation from the British National Corpus, and c. 5 million words of academic prose from the Longman–Lancaster Corpus. The texts in this corpus were automatically tagged with a tagging program (see Methodology Boxes 4 and 5) which identified the grammatical category of each word, including attributive and predicative functions of adjectives. However, the analysis of *little* and *small* provides a good example of problems that can occur with automatic techniques because of the complexity of real language use. Before the results for *little* and *small* are discussed in Section 4.2.2,

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therefore, Section 4.2.1 discusses methodological issues in analyzing potentially ambiguous structures and describes the integration of hand-editing techniques with automatic analyses.

4.2.1 Incorporating automatic and human analytical techniques

As described in previous chapters and Methodology Box 5, it is important to verify the accuracy of automatic analyses and to correct any inaccurate analyses that would affect results. For the present investigation, it was important to check the accuracy of the automatic identification of *little* and *small* as attributive or predicative adjectives.

A convenient way to check automatic analyses is with KWIC (Key Word in Context) files. In the present case, we wrote a computer program to cycle through the tagged texts in our corpus and produce KWIC files for each of the adjectives in each grammatical function category. In addition to attributive and predicative categories, it also turned out that some occurrences of *little* and *small* had not been tagged as either attributive or predicative, so a third category of "no tag" was made for each adjective.

A KWIC file is an efficient way to check the accuracy of the tagging because both the tag and the context are easy to see. For example, Figure 4.1 displays a sample of the KWIC file for *small* tagged as an attributive adjective; the "type" is identified as "Atrb" (i.e., the target word has been tagged as an attributive adjective), "file" gives the text identification number, and the target word *small* follows the arrow.

In Figure 4.1 all three of the occurrences are correctly tagged. In fact, almost all occurrences of these words identified as attributive were tagged correctly. However, problems arose with the other two categories (predicative and "no tag").

The first set of problems concerned the occurrences identified as predicative. Some of these were actually attributive adjectives. For example, in conversation, some adjectives were separated from their head nouns by another speaker's interruption, and the automatic analysis therefore failed to recognize them as attributive adjectives, as in:

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Type = Atrb
File = 00116 . TEC
worthless . In a single spikelet two to seven pistils
develop and are generally inherited as monogenic recessives . The
-----> small
misshapen grains add nothing to yield and break up
on milling . Plural embryos within a single ovule also

Type = Atrb
File = 00141 . TEC
generated in the soil , for it enters the plant
roots and affords extra nutrient over and above what the
-----> small
leaves are taking in . Hence a supply of
humus , which continuously yields carbonic acid , is advantageous

Type = Atrb
File = 00208 . TEC
cm in diameter (fig . 27) . On
section , at the centre of each lesion is a
-----> small
bronchus containing lungworms and mucopurulent exudate .
Microscopically , the epithelium is hyperplastic with an increase

Figure 4.1 KWIC file of *small* tagged as attributive adjective

(3) A: Cos they have little
B: We –
A: milk bottles at nursery school don't they?

Furthermore, in academic prose many occurrences of *little* were mistakenly identified as predicative adjectives when they were not adjectives at all. In these instances, *little* was actually a pro-form or an adverbial:

(4) Pro-form use of *little*:
There is *little* in the way of printed and audio-visual material that the clearinghouses do not attempt to acquire.

(5) Adverbial use of *little*:
The output for training patterns F'1 to F'3 is *little* affected by removal of X1 . . .

In the occurrences identified as predicative, mis-tagging of *little* appeared to be more common than mis-tagging of *small*; however,

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it was clear that a number of occurrences for both adjectives needed to be corrected.

The second problem area concerned the "no tag" category – occurrences that were not tagged as either attributive or predicative. Examination of these KWIC files showed that the occurrences were of four types. First, some occurrences were actually attributive adjectives. In these cases, automatic analysis was often impossible due to complex or ill-formed noun phrases. In conversation some noun phrases were interrupted by pauses (often transcribed with punctuation), fillers, or discourse markers (e.g., *you know*), and in academic prose there were long, hyphenated, or coordinated premodifiers:

(6) Interrupted noun phrases in conversation:
  a. Shut up you *little* . . . cow.
  b. It turns into one of those *little* you know turtles.

(7) Hyphenated or coordinated premodifiers in academic prose:
  a. The *small* rough-shelled larvated eggs are characteristic . . .
  b. Or it may have *little* or no practical effect.

The second type of occurrence in the "no tag" files were actually predicative adjectives. Punctuation and adverbials between the verb and predicative adjective caused many of the problematic cases for the automatic analysis, for example:

(8) a. Sterility is, however, *small* if the day temperature is high.
  b. . . . it is by no means *small* for a brachiopod.

The third type of occurrence concerned uses of *little* and *small* that were not actually adjectives. As with the occurrences mistakenly identified as predicative, these were more common in academic prose, and were either pro-forms or adverbials:

(9) a. Endosperm, this, the "kernal," is the part of the plant of greatest value but comparatively *little* is known of the genes involved in its characters.
  b. Rooting depth alone is rarely a good indicator of root growth and function so that radioisotopes are currently *little* used.

The fourth type of occurrence was also similar to mistakes in the predicative files: occurrences of the adjectives that could not

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be analysed because of the fragmented nature of conversation, for example:

(10) a. A: Cottage industries. What do you think I mean by cottage industries?
B: A small ???
C: *Small*. You know what a cottage is, it's a small house.
b. A: go on again then if you
B: Big and *small*.
C: Big and small well done.

In cases such as these, even analysis by a human cannot determine whether an attributive or predicative function was intended. In natural conversations, people often overlap utterances or repeat words. This does not cause problems for people when they are interacting, but it does present difficulties for linguists who are trying to analyze conversation! In this case these occurrences in conversation have not been mis-analyzed by the tagger; rather, they are grammatically indeterminate and need to be either a separate category or excluded from the quantitative analysis altogether.

Looking at the predicative and "no tag" KWIC files, it was clear that results based on the automatic analysis were likely skewed in several ways:

- the attributive count was artificially low because some attributive occurrences were mistakenly identified as predicative or left untagged;
- the predicative count was suspect – it might be artificially high because it included attributive and non-adjective occurrences; on the other hand, it might be low, because some predicative occurrences were left untagged;
- the total count of adjectives was artificially high because it included occurrences that were not adjectives or were grammatically indeterminate;
- many occurrences were not analyzed at all because they were included in the "no tag" category (up to 27 percent of all occurrences in the case of *little* in academic prose; see Table 4.1).

With problems of this type and the clear need for editing automatic counts, you might be questioning the usefulness of automatic analyses. Notice, however, that in this case over 6,000 of the 8,020 occurrences have been analyzed correctly. (That is, most

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Table 4.1 *Raw counts for* little *and* small *as adjectives, by grammatical function, based on automatic analysis*

	Attrib.	Predic.	"No tag" (Gram. function not tagged)	Total
Conversation (c. 5 mill. words)				
little	2,101	104	405 (16%)	2,610
small	399	72	158 (25%)	629
Academic prose (c. 5 mill. words)				
little	1,033	65	411 (27%)	1,509
small	2,557	316	399 (12%)	3,272
Total	6,090	557	1,373	8,020

of the 6,090 forms identified as attributive adjectives are correctly analyzed.) Furthermore, as we illustrate below, adjusting counts with some hand-editing is more efficient than analyzing occurrences entirely by hand.

When working with a large corpus, hand-editing can be extremely tedious and time-consuming; for example, in this sample study, there are almost 1400 occurrences in the "no tag" category alone. Fortunately, there is an alternative to checking every occurrence that is nearly as accurate and much more efficient. This alternative approach uses the proportions of occurrences in smaller samples to make adjustments to the total number of occurrences. In general, the steps are as follows: 1. extract a random sample of occurrences from the category that needs to be hand-edited; 2. analyze the grammatical feature by hand in that random sample; 3. compute the proportional use of each variant in the random sample; 4. multiply the total number of occurrences in the category by the proportions computed in step 3; 5. finally, adjust the original frequency counts by adding or subtracting the number computed in step 4. For a full study, more than one random sample should be taken from each category in order to make sure that the proportions are similar across samples. For simplicity, we illustrate this approach here using just one sample.

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In the present analysis, we had identified problems with the predicative and "no tag" occurrences of both *little* and *small*. Therefore, a random sample of 100 occurrences was analyzed by hand for each adjective in each register, for each of those categories – that is, 100 occurrences of predicative *little* in conversation, 100 occurrences of "no tag" *little* in conversation, 100 occurrences of predicative *small* in conversation, and 100 occurrences of "no tag" *small* in conversation, and the same for academic prose. (For categories that had fewer than or just over 100 occurrences, all occurrences were analyzed).

In the analysis, occurrences were classified into one of three groups: attributive, predicative, or not adjectival/grammatically indeterminate. The number of occurrences falling into each group was totaled. Then, using the percentage of each group in each of the categories, the automatic counts were adjusted.

To illustrate this procedure, consider the steps used for the adjustments of *little* in conversation. First, all 104 occurrences tagged as predicative were checked. Of these, 41 were accurately identified, 44 were actually attributive adjectives, and 19 were non-adjectival or grammatically indeterminate. Second, a random sample of 100 tokens was analyzed from the "no tag" category. This analysis found that 57/100 (or 57 percent) were attributive, 4 percent were predicative, and 39 percent were non-adjectival or grammatically indeterminate. Applying these proportions to the total number of "no tag" occurrences (i.e., 405 occurrences, see Table 4.1) we find:

$.57 \times 405 = 231$ attributive occurrences

$.04 \times 405 = 16$ predicative occurrences

We can now adjust the raw counts for *little* in conversation as follows:

	Attrib.	*Pred.*	*Source of information*
	2101	104	auto counts (from Table 4.1)
	+44	−63	adjusted counts of predicative
	+231	+16	adjustments for "no tag"
Total	2376	57	

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Table 4.2 *Percent use of* little *and* small *as predicative adjectives*

	Conversation	Academic prose
Based on automatic counts		
little	5%	6%
small	15%	11%
After hand-editing of files		
little	2%	<1%
small	23%	13%

Adjustments such as these were made for each of the adjectives in each of the registers.

With the adjustments made, we are now ready to figure the percentage of each of the adjectives used in each grammatical function. To see the impact of the hand adjustments, Table 4.2 displays the percentage occurrence in predicative position for each of the adjectives first based on the automatic counts and then after the adjustments.

As you can see from Table 4.2, the patterns are similar for both the automatic and edited results: *small* has a stronger association with predicative position than *little* does, both in conversation and academic prose. The more accurate adjusted counts show, however, that this pattern is even stronger than the automatic counts suggest. *Little* is extremely rare as a predicative adjective. Furthermore, the adjusted counts show that the greater association of *small* with predicative position is considerably stronger in conversation than in academic prose.

4.2.2 Interpretation of the grammatical association patterns for little *versus* small

The results at the end of the last section showed that *small* has a stronger association with predicative position in both registers than the adjective *little* does, and that this association is especially

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strong in conversation. It turns out that this grammatical association interacts in an interesting way with the collocational associations of these two adjectives.

In general, *little* is similar to *big* (see Chapter 2) in that it tends to co-occur with concrete, often animate, nouns (such as *thing(s)*, *boy(s)*, *girl(s)*). For example:

(11) Can you imagine a *little boy* being called Horace?

In contrast, *small* is similar to *large* in that it usually co-occurs with nouns indicating quantities (such as *quantity(ies)*, *amount(s)*, *number*, *proportion*). For example,

(12) a. It's only a *small proportion* of what it costs.
b. Even very *small quantities* can be troublesome.

Consideration of concordance listings shows that the frequent conversational use of *small* in predicative position does not follow these typical lexical associations, however. Instead, predicative *small* in conversation is often used to characterize physical size, similar to the use of *little* in attributive position. However, the predicative use of *small* and attributive use of *little* have different functions. With the use of *small* in predicative position, the main point of the utterance is to identify "smallness" as an important attribute of the person being described. In contrast, the use of *little* in attributive position provides an identifying characteristic of the noun being described, but the utterance itself has other purposes. Thus compare:

(13) *small* in predicative position:
a. She's *small* and really skinny.
b. He's really *small*, isn't he?

(14) *little* in attributive position:
a. She's known me since I've been a *little* girl.
b. Well, he's like any *little* kid I think.

Thus, although these two words are nearly synonymous when considered out of context, corpus-based analysis shows that they have important differences in both their grammatical and lexical association patterns. Further, these two sets of association patterns

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interact in important ways, resulting in the differing distribution of these words across registers.

4.3 The grammatical associations of nearly synonymous words: *begin* versus *start*

In this section we investigate the use of two nearly synonymous verbs: *begin* and *start*. In most situations these verbs can be used interchangeably (e.g., "After the race *started* . . ." versus "After the race *began* . . ."). In fact, these two verbs have exactly the same grammatical potential for combining with other clause elements, that is, the same "valency" possibilities. Traditionally, descriptive grammar books would stop with this observation, having noted the *potential* grammatical patterns for these two verbs. We are left wondering why the language has two verbs that mean nearly the same thing and are used in the same range of grammatical patterns. However, corpus-based research shows that this is not at all the end of the story: despite the similarities in meaning and grammatical potential, these two verbs are in fact typically used in very different structures. That is, each verb has its own strong preferred grammatical associations, which distinguishes it from the other verb. Thus, the discussion below shows that in terms of actual use, the two verbs are quite different.

Verbs can be grouped according to their "valencies," that is, their potential for combining with other clause elements. Transitive verbs, such as *carry* and *buy*, take a single direct object, while intransitive verbs, such as *come* and *sit*, do not take direct objects.[1] Some verbs – such as *begin* and *start* – can take both transitive and intransitive patterns:

(15) Transitive pattern – with a noun phrase as direct object (given in brackets):
Then they *started/began* [the quota system].

(16) Intransitive pattern – no direct object:
I had better issue a survival kit before we *start/begin*.

The direct object in transitive patterns can be a noun phrase, as in the above example. However, the direct object can also be a

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to-clause or *ing*-clause, and *begin* and *start* are further alike in that they can both take these variants:

(17) Transitive pattern – with an *ing*-clause as direct object (given in brackets):
They'd *started/begun* [leaving] before I arrived.

(18) Transitive pattern – with a *to*-clause as direct object (given in brackets):
One of the wheels had *started/begun* [to wobble].

In this section we investigate the association of *begin* and *start* with each of its potential valencies – intransitive, transitive with a noun phrase direct object, transitive with an *ing*-clause direct object, and transitive with a *to*-clause as a direct object. For this analysis, we use the fiction and academic prose registers in the Longman–Lancaster Corpus (c. 2 million words from each register). As with the analysis of *little* and *small*, we review several methodological steps before we move to the results and interpretation.

A computer program was written to code the valency patterns of the two target verbs in the fiction and academic prose texts. Some valency patterns are very difficult to identify accurately with an automatic program, however. Because of the expected inaccuracies, we planned to hand-edit the complete output files, rather than doing proportional corrections as with the analysis of *little* and *small*. To keep the editing task manageable for this sample investigation, a ceiling of 250 occurrences for each verb in each register was set. In addition, to ensure that the results were not influenced by idiosyncratic preferences of any individual text or writer, the program was written to code every third occurrence of the target verb. (If a register had a relatively low frequency of one of the verbs, then the total number of occurrences did not necessarily reach the 250 ceiling.)

The output files produced by this computer program are similar to a KWIC concordance file. For example, Figure 4.2 displays a sample of the unedited file for *begin* in fiction. In addition to providing the key word in context, the output shows the file name (e.g., 00002. FCT in the first concordance listing) and the valency pattern that was identified by the automatic analysis (<valency = ,

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00002 . FCT
<valency=TCLS
had begun to travel through the endless canefields , she had
= => begun
to feel afraid . For suddenly the skies had opened up

00018 . FCT
<valency=T
come close to shutting itself . "er, excuse me ,"
= => began
the youth . Garth stared at him without speaking."

00018 . FCT
<valency=I
hath her in thrall . "After a minute the trio
= => began
rather carefully to cross the room . The level of atmospheric

00021 . FCT
<valency=ING
stations , shops , roadhouses , all closed . A dog
= => began
barking and , having begun , went on . Other dogs

Figure 4.2 Unedited KWIC file for valency patterns of *begin* in fiction

with I for intransitive; T for transitive with a noun phrase; TCLS for transitive with a *to*-clause; and ING = transitive with an *ing*-clause).

These output files were hand-edited to correct any errors. For example, you can see that the third entry in Figure 4.2 (from text 00018.FCT) was automatically coded as an intransitive use; the program did not allow for two intervening adverbs, and therefore it failed to identify the *to*-clause as the direct object. The automatically assigned code was thus corrected to TCLS for this entry.

After all the entries had been edited, another computer program was written to tally the valencies used with each verb in each register. The results are presented in Table 4.3.

You can see from Table 4.3 that both verbs occur with all of the possible valency patterns. However, there are some striking differences in the relative use of the two verbs. For the sample study here, we briefly focus on two important patterns seen in Table 4.3:

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Table 4.3 *Valency patterns of synonymous verbs* begin *and* start *(percentages are within registers, rounded to the nearest whole number)*

		Transitive patterns			
	Intransitive	+ noun phrase	+ *to* -clause	+ *ing* -clause	Totals
begin					
Fiction	22% (54)	3% (8)	72% (178)	4% (10)	100% (250)
Academic	43% (82)	12% (22)	34% (66)	12% (22)	100% (192)
start					
Fiction	40% (101)	22% (55)	20% (50)	18% (44)	100% (250)
Academic	64% (91)	16% (22)	15% (21)	6% (8)	100% (142)

1. *start* is more commonly used as an intransitive verb than *begin* is;
2. *begin* is much more commonly used as a transitive verb with a *to*-clause as object than *start* is.

Table 4.3 shows that *start* is more commonly used as an intransitive verb in both registers, though the pattern is particularly strong in academic prose. 40 percent of the occurrences of *start* in fiction and 64 percent in academic prose are intransitive. In contrast, only 22 percent of the occurrences of *begin* in fiction and 43 percent in academic prose are intransitive.

Typically, the intransitive occurrences of *start* in academic prose describe the onset of some process, for example:

(19) a. . . . the soil formation process may *start* again in the fresh material.
b. Blood loss *started* about the eighth day of infection. . . .
c. Tillering *starts* about a week or earlier after broadling.

Often, as in the above examples, the subject of the sentence is a nominalized process, and an adverbial follows the verb. In fiction, more concrete and human entities are used as subjects, while adverbials still often follow the verb:

(20) a. As he *started* down the hill, he could see it . . .
b. . . . the train *had started* again . . .

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Table 4.4 *Proportional use of transitive* begin *and* start *with* to*-clauses in the Longman–Lancaster Corpus*

	+ *to*-clause	All other patterns	Totals
begin			
fiction	60% (1,871)	40% (1,267)	100% (3,138)
academic	36% (221)	64% (387)	100% (608)
start			
fiction	17% (294)	83% (1,462)	100% (1,756)
academic	14% (90)	86% (541)	100% (631)

A further interesting area for research would be the use of adverbials with the intransitive occurrences of these verbs.

Table 4.3 also shows that, in both registers, *begin* is more commonly used as a transitive verb with a *to*-clause than *start* is. In fiction, 72 percent of the occurrences of *begin* are completed with a *to*-clause, and in academic prose 34 percent; while only 20 percent (in fiction) and 15 percent (in academic prose) of the occurrences of *start* have a *to*-clause.

This pattern can be easily verified from analysis of a larger corpus. That is, the automated computer program described above identified the *to*-clause pattern with a high degree of accuracy. We are thus able to compare the counts from the edited sub-sample to that of a larger corpus, to see if the patterns are consistent between the two. Table 4.4 reports the patterns found in a 10-million-word sample from the Longman–Lancaster Corpus.

Table 4.4 shows the same patterns of use in the larger corpus as in the edited sample from the sub-corpus (Table 4.3): the verb *begin* is strongly associated with a *to*-clause complement; *start* is more strongly associated with other valency patterns. This association does not necessarily mean that the *to*-clause pattern occurs in the majority of cases for *begin*: in academic prose the *to*-clause pattern accounts for only 36 percent of the occurrences. However, it is still true that, relative to *start*, *begin* has a greater preference for the *to*-clause pattern. (Only 14 percent of the occurrences of *start* in academic prose are followed by a *to*-clause.)

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The use of *to*-clauses allows writers to include two actions in a sentence, first with the verb *begin* and then with the verb in the *to*-clause, as in these examples from fiction:

(21) a. Then I *began* to *laugh* a bit.
b. They pulled away from the siding and *began* to *move* through the back yards of Worcester . . .
c. A cloud *began* to *cover* the sun slowly . . .

In academic prose, the use of *begin* with a *to*-clause provides a particularly striking contrast with the intransitive use of *start* described above. Rather than showing the onset of nominalized processes, the occurrences of *begin*+*to*-clause more commonly have human or other concrete agents who begin activities:

(22) a. First, *one begins* to *question* the need for frequent rote regurgitation of words . . .
b. Elsewhere, *sociologists have begun* to *disclose* the social conditioning of the natural sciences.
c. *The original mass of gas* cooled and *began* to *contract*.

Much more could be said about the valency patterns of *begin* and *start*, but even this brief review illustrates further the way that apparently equivalent words turn out to be used in quite different ways in actual texts. Although these two verbs have nearly synonymous meanings and identical valency potentials, their actual patterns of use are strikingly different. Furthermore, our intuitions are not reliable guides to such patterns of use. Native speakers are often unable to even predict the existence of systematic association patterns, let alone predict which verbs are associated with which valency. In contrast, corpus-based investigations show that nearly synonymous words are rarely equivalent at all, when considered in terms of their association patterns of use.

4.4 The lexical associations of nearly equivalent grammatical constructions: *that*-clauses versus *to*-clauses

We noted in the introduction to this chapter that lexicogrammatical associations can be considered in either "direction."

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The preceding sections have shown how grammatical associations can be used to distinguish among nearly synonymous words. In the present section, we address the opposite research question, showing how nearly equivalent grammatical constructions can be distinguished in terms of their differing lexical associations.

As discussed in Chapter 3, nearly equivalent grammatical constructions are similar to nearly synonymous words, in that they can be distinguished by their patterns of use. One of the examples in that chapter focused on a comparison of *that*-clauses and *to*-clauses. The analyses reported there showed that these two types of complement clause have very different distributions across registers: *that*-clauses are particularly common in conversation but relatively rare in academic prose; *to*-clauses have about the same frequency in conversation and academic prose, but they are considerably less common than *that*-clauses in conversation. Chapter 3 also discussed how these two clause types have differing grammatical associations with extraposed and non-extraposed structures.

Another factor differentiating between *that*-clauses and *to*-clauses is their lexical associations. First of all, we know that there are many verbs that can control only one or the other type of complement clause. For example, the verbs *suggest*, *conclude*, *guess*, and *argue* can control a *that*-clause but not a *to*-clause; the verbs *begin*, *start*, and *try* can control a *to*-clause but not a *that*-clause. However, there are also a number of verbs, such as *decide*, *agree*, and *expect*, that can control both types of clauses. There is potential, then, for a wide range of associations, and the actual patterns of use are difficult to predict.

In this section, therefore, we continue the investigation of *that*-clauses and *to*-clauses from a lexico-grammatical perspective. The corpus used for the investigation remains the same: approximately 4 million words of academic prose from the Longman-Lancaster Corpus and approximately 5 million words of conversation from the British National Corpus. The computer programs described in Chapter 3 were also used to count the matrix verbs occurring with each type of complement clause.

Figure 4.3 displays the most common verbs taking each clause type. As you can see, it turns out that lexical patterns are one of the strongest factors that differentiate between the two types of

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Most common verbs controlling a *that*-clause

	Conversation	Academic prose
think	* * * * * * * * * * * * * * * * * * * *	*
say	* * * * * * * * * * * *	* *
know	* * * * * * * * *	*
see	* *	* *
show	*	* * *
find	*	* *
believe	*	*
feel	*	–
suggest	–	* *

Most common verbs controlling a *to*-clause

	Conversation	Academic prose
want	* * * * * * * * * *	–
try	* * *	*
like	* *	*
seem	*	*
tend	–	* *
appear	–	* *
begin	–	*
attempt	–	*
continue	–	*
fail	–	*

Figure 4.3 Most common verbs controlling complement clauses (each * represents 100 occurrences per million words; – represents less than 100 occurrences per million words)

clauses. That is, two completely different sets of verbs most commonly control *that*-clauses versus *to*-clauses.

Some of the verbs in Figure 4.3 (such as *want* and *try*) are grammatical controlling only one type of complement clause, and they have strong lexical associations with that structural type. Other verbs – such as *think*, *say*, and *know* – can control both types of complement clause; however, these verbs have strong association patterns with only one type of complement clause. Thus, although there is some overlap between the two types of complement clause in the controlling verbs that are grammatical, corpus-based analysis shows that there is in fact very little overlap in the commonly occurring lexical associations.

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Consideration of the lexical associations can help to explain the differing register patterns of the two clause types found in the analysis in the last chapter. In particular, the extremely high frequency of *that*-clauses in conversation is directly linked to the frequent co-occurrence of these clauses with three matrix verbs: *think*, *say*, and *know*. These are used to report two of the most important activities and states of humans: what they think/know, and what they said! For example:

(23) a. I **thought** *that she only wore jeans.*
b. She **thinks** *he's sweet.*

(24) a. It would be a long time before you **knew** *that you weren't getting anything.*
b. She **knew** *that I was teasing her.*

(25) a. I wouldn't **say** *I know most of the poetry of Eliot . . .*
b. She **said** *you wouldn't notice it.*
c. Archbishop Temple used to **say** *that the modern world reminded him of a shop window.*

Other common verbs used with *that*-clauses in conversation, such as *believe*, *feel*, *mean*, and *tell*, substantiate this interpretation. For example:

(26) a. She just really **felt** *that it was better for David to learn at school.*
b. I **believe** *it's something to do with speech therapy.*
c. Do you **mean** *that you were better equipped to answer some of the questions that Professor Hart put to you?*
d. I'**m told** *he's an absolutely grand chap.*

Given these most common lexical associations of *that*-clauses, and their typical use in reporting what was thought, felt, or said, it is not surprising that *that*-clauses are relatively rare in academic prose. That is, it is not common to report personal thoughts/feelings or speech in expository writing. Instead, when *that*-clauses do occur in academic prose, they are often used to report the inferences that can be drawn from research findings. For example:

(27) a. Laboratory studies **show** *that rocks vary in strength . . .*
b. The data **indicate** *that grain products are the major contributing source of sodium.*

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c. A recent study from Boston **suggests** *that these patients have a better immediate prognosis . . .*
d. When organisms are examined, it **is seen** *that they form natural groups with features, usually morphological, in common.*

In contrast, the most common verbs occurring with *to*-clauses come from a wider range of semantic domains. In conversation, two verbs are particularly common with *to*-clauses expressing desire (*want* and *like*), while a third is used to express effort (*try*). All three verbs typically occur with a human as subject, for example:

(28) a. I **want** *to get some minced beef.*
b. Zoe **wants** *to do it.*
c. I would **like** *to see it all actually.*
d. Glynn **is trying** *to copy Tommy's work.*

In contrast, academic prose has a larger number of moderately common verbs controlling *to*-clauses, although none of them are extremely common. These verbs fall into two major semantic domains: those expressing some degree of probability (or hedging), and those that are aspectual in that they mark a stage in a process. Notably, both sets of verbs typically occur with inanimate subjects. For example:

(29) Marking probability:
a. The "societal response" **seemed** *to determine the outcome.*
b. Classical physics **tends** *to express things in terms of second order differential equations.*
c. The Great Lake **appears** *to function as a sort of gigantic regulator.*

(30) Marking stage in a process:
a. Matter **began** *to drift toward the center.*
b. Traditional beliefs **continue** *to be important . . .*
c. Much of the seed will therefore **fail** *to germinate.*

In sum, although these two structures might be considered grammatically equivalent in that they both serve as complements to verbs and adjectives, empirical investigation shows that they are not at all used interchangeably. The analyses in this chapter, together with those in Chapter 3, show that *that*-clauses and *to*-clauses differ in their register associations, grammatical associations, and lexical associations. Taken together, these association

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patterns allow us to piece together the complex but distinct patterns of use for each clause type.

4.5 Conclusion

In this chapter we have illustrated that studies of the associations between words and grammatical structures can provide important information for understanding language use. The sample studies showed that lexico-grammatical associations are particularly useful for understanding differences between very similar words or nearly equivalent grammatical structures. Synonymous words can have very different patterns of use with respect to their grammatical associations – as we saw with the different associations between the adjectives *little* and *small* and predicative position, and the different associations between the verbs *begin* and *start* and their valency patterns. Likewise, similar grammatical structures can have very different patterns of use with respect to their lexical associations – as we saw with the different sets of verbs that occur most commonly with *that*-clauses and *to*-clauses.

All three of these sample studies also illustrate the usefulness of corpus-based analyses. For example, without the aid of computers it would have been a daunting task to analyze the thousands of occurrences of *little* and *small* or to accurately keep track of the frequency of every verb controlling a *to*-clause or *that*-clause. Even when some files need to be edited by hand, corpus-based techniques make it possible to do new types of investigations and to conduct research on a scope previously unfeasible.

Note

1. Ditransitive verbs, such as *bring* and *give*, take two objects.

Further reading

Many of the lexical and grammatical studies listed at the end of Chapters 2 and 3 also present lexico-grammatical findings.

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5 The study of discourse characteristics

5.1 Studying discourse characteristics

Discourse analyses focus on language characteristics that extend across clause boundaries. As a result, discourse characteristics are more difficult to identify and analyze than lower-level lexical or grammatical features. However, such analyses are important for both descriptive and applied linguistics. In fact, it turns out that the use of many lexical and grammatical features can only be fully understood through analysis of their functions in larger discourse contexts.

Most discourse studies identify salient discourse structures and exemplify those structures with illustrative text excerpts – such as identifying turn-taking structures in conversation or tracking the "themes" in a written text. However, it has proven difficult to apply these techniques to texts in a way that allows for generalizable results. Thus, although nearly all discourse studies are based on analysis of actual texts, they are not typically corpus-based investigations: most studies do not use quantitative methods to describe the extent to which different discourse structures are used, and relatively few of these studies aim to produce generalizable findings that hold across texts.

As a result of the lack of generalizable findings, we still know surprisingly little about the discourse similarities or differences across texts and registers. However, as the present chapter shows, corpus-based analyses can make a significant contribution in this area. Such techniques can be applied to a large body of texts to accurately describe the discourse characteristics of selected

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registers, as well as the extent to which any individual text conforms to the expected discourse patterns of its register.

There are several reasons why discourse studies have generally not been corpus-based in the past. First, many discourse features cannot be identified automatically. The analysis of such features is often labor-intensive, requiring detailed consideration of language features in their larger textual contexts. For example, one goal of discourse analysis is to classify the kinds of information in a text, usually focusing on noun phrases as the primary carriers of referential information. Such analyses attempt to determine which pieces of information are already "known" by the reader/listener, versus those noun phrases that present "new" information. It is impossible to make distinctions of this type automatically; the discourse analyst must consider the previous textual context, and in some cases analysts even consider the background knowledge that readers/listeners use to understand a text. As a result, it requires a large commitment of time and energy to analyze extended texts in this way.

Furthermore, commercially available corpus analysis tools are not very helpful for investigations of discourse-level features. Standard concordancing packages are designed to produce a listing of specified target words with their immediate sentential contexts. Because such tools are not designed for complex grammatical or semantic analysis, they are also not suitable for discourse analyses. For example, concordancing packages provide no means for identifying all the nouns in a text, let alone classifying those nouns as known versus new referents.

Difficulties such as these might be taken to suggest that the corpus-based approach is not useful for discourse studies, and that it is simply not feasible to attempt broader studies with generalizable results. However, there are two major ways in which a corpus-based approach can be used to investigate discourse features. First, it is possible to develop and use interactive computer programs (similar to a spellchecker) to analyze discourse characteristics. Such programs can identify certain discourse characteristics more reliably and faster than humans can, while at the same time providing a means for the researcher to make judgements about areas that cannot be analyzed automatically. Second, it is possible to use

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automatic analyses to track the use of surface grammatical features over the course of a text. These analyses actually map the development of discourse patterns through texts; they can be used to compare texts, to find the typical patterns for a register, or to see how a particular text compares to the general pattern for the register.

These two types of analyses are exemplified in the present chapter through the following research questions:

1. How are references marked in different ways in different kinds of texts?

   This question is addressed in Section 5.2 with an investigation of the use of nouns and pronouns in four registers of English. Some specific questions relating to this issue are: What factors influence the choice between nouns and pronouns in a text? Which noun phrases present "given" (or "known") information, and which present "new" information? How are given and new referents distributed across texts?

2. How does the sequence of verbs within a text develop with respect to the marking of tense and voice?

   Some specific questions relating to this issue are: To what extent is there a prototypical sequence of verbs – or a "discourse map" of verbs – for all the texts in a register? To what extent do such discourse maps correspond to the underlying rhetorical divisions marked within a text?

   This second research question is illustrated in Section 5.3 with an investigation of the tense and voice of verbs in scientific research articles.

5.2 Reference types in spoken and written registers

Noun phrases are the major grammatical device used to refer to people, objects, or other entities in texts. However, texts from different registers often differ dramatically in the use of these "referring expressions." For example, consider the following two samples from news reportage and conversation, with all noun phrases italicized:

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Text Sample 5.1: News reportage
Thortec International Inc. said *it* reached *agreements* with *an investor group* and *Wells Fargo Bank* under which *it* will receive *loans* and *an equity infusion* in return for *stock* that will reduce *the number of shares in public hands* by as much as 85 percent. *The engineering and consulting firm*, which has been plagued by *losses* for *five years*, said *the restructuring* is required to relieve *its debt burden* and *"acute shortage of cash."*

Text Sample 5.2: Conversation
A: Right, *I*'m ready. Have *you* locked *the back door*? [pause] *I* thought *we* were walking.
B: Well do *you* want to walk or do *you* want to go in *the car*?
A: Well *I* have to go to *the paper shop*.
B: Well *I*'ll drop *you* at *the paper shop* while *I* go round.
A: Oh *that's a good idea*.

One clear difference between these two text samples concerns the form of the noun phrases. The sample from news reportage relies primarily on full noun phrases (*Thortec International Inc.*, *agreements*, *an investor group*, etc.), while the conversation sample uses pronouns (*I*, *you*, *we*, *that*) more commonly.

In addition to this difference, it is probably also obvious that these samples tend to use different types of reference. In particular, the conversation sample has a high percentage of "exophoric" – or text-external – reference, with the pronouns *I* and *you* referring directly to the speaker and addressee, rather than some previous entity in the text (see Section 5.2.1 for further explanation). The news sample does not include this kind of reference. Furthermore, because of the greater reliance on exophoric reference, more of the referents in the conversation sample are already known by both of the participants even when they are first mentioned (e.g., *I*, *you*, *the back door*, *the paper shop*), while more of the referents in the news sample are initially unfamiliar (e.g., *agreements*, *an investor group*).

In this section we illustrate how corpus-based analyses can be used to investigate the characteristics of referring expressions and to determine how registers differ in their use of reference. We present an analysis of the use of nouns and pronouns across four registers of English – examining their use to present given

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and new information and different kinds of reference. We use two spoken registers from the London–Lund Corpus and two written registers from the LOB Corpus: conversation and public speeches from the London–Lund, and news reportage and academic prose from the LOB.

The analytical procedures involved in this investigation are introduced step by step throughout this section. In Section 5.2.1, we provide an explanation of the informational characteristics that are included for analysis. In Section 5.2.2 we then describe the interactive computer techniques used to analyze those characteristics for all the nouns and pronouns found in the texts. Finally, Section 5.2.3 discusses the results of the analysis.

5.2.1 Characteristics of referring expressions

There are many characteristics of referring expressions that could be examined in order to better understand their use across texts and registers. For this sample analysis, we concentrate on four parameters:

- status of information: given versus new
- for given information, type of reference: anaphoric, exophoric, or inferrable
- for anaphoric reference, form of the expression: pronoun, synonym, or repetition
- for anaphoric reference, the distance between the anaphoric expression and its antecedent

Each of the noun phrases in a text can be classified according to the type of information that it presents: given or new. For example, if you look back at Sample 5.1 from a newspaper article, you can see that many of the noun phrases present new information, identifying a person or thing that has not previously been referred to in the text. Noun phrases of this type include: *Thortec International Inc., an investor group, Wells Fargo Bank, loans, an equity infusion, stock*. Other referring expressions present given information, representing an entity that has already been identified. In the first sentence of Sample 5.1, the pronoun *it* is used twice to mark a "given" referent, referring to the company *Thortec International Inc.*

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Expressions presenting given information represent three kinds of reference relationships. Many of these expressions are "anaphoric." That is, they refer to a person or thing that has already been referred to in the text, i.e., the "antecedent." For example, the pronoun *it* in the first sentence of Sample 5.1 is anaphoric, referring to *Thortec International Inc.*, which is its antecedent.

However, other referents present given information because they refer to some person or thing in the external context. For example, in the conversation excerpt in Sample 5.2, the pronouns *I* and *you* refer directly to the speaker and addressee. *The back door*, *the car*, and *the paper shop* refer to physical objects present in the extended physical situation which are clearly understood by both participants. These are "exophoric" referents. Exophoric referents are given because their identity is known from the physical situation. In contrast, anaphoric referents are given because their identity is known from preceding textual references.

In addition to anaphoric and exophoric reference, there are some other expressions of given information which are more difficult to classify. For example, in Sample 5.1 the existence of *a restructuring*, referred to in the second sentence, is "inferrable" from the events described in the first sentence, but this noun does not have an anaphoric relation to any single preceding noun phrase nor does it refer to the external context. Similarly, the existence of a *debt burden* can be inferred from the fact that the company has been "plagued by losses," but again this does not represent an anaphoric relation. Thus, a category of "inferrable" is also necessary when classifying referents.[1]

Our third area of interest is to consider the different forms used to present anaphoric referents. Anaphoric referents are often expressed as a pronoun, as in the case of *it* used to refer to *Thortec International Inc.* in Sample 5.1. However, anaphoric referents can also be synonymous expressions, such as the use of *the engineering and consulting firm* in the second sentence of Sample 5.1 to refer to *Thortec International*. Finally, anaphoric referents can be a direct repetition of the original expression.

The fourth area that we examine in this illustrative analysis is the distance between the referring expression and its antecedent. For example, in the news reportage example above, the pronoun

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it occurs relatively close to the antecedent *Thortec International Inc.* The fuller synonymous expression *The engineering and consulting firm* has a greater distance from the original reference to this company.

Taken together, these four parameters can reveal many patterns in the use of reference in different registers. As you can see, however, there is a great deal of information to keep track of here, and analysis of even a couple thousand words of text could be very time-consuming. The next section explains how corpus-based analytical techniques can assist in studying the characteristics of referring expressions.

5.2.2 Interactive analysis techniques: coding the characteristics of referring expressions

In this section we describe the use of an interactive analysis program to code characteristics of referring expressions in texts. For the illustrative analysis here, a relatively small sample of the texts from the London–Lund and LOB corpora were coded: the first 200 words in forty texts (five texts from conversation, nine texts from public speeches, ten texts from news reportage, and sixteen texts from academic prose).[2]

Our investigation focused on the four informational characteristics reviewed in the last section, in addition to the form of the expression and the register of the text. Thus, the interactive program was designed to analyze and record up to six characteristics for each noun phrase:

1. register of the text
2. nominal form: pronoun versus full noun
3. information status: given versus new
4. if given, type of reference: anaphoric, exophoric, or inferrable
5. if anaphoric and a full noun, type of expression: synonym versus noun repetition (pronouns were already identified in step 2)
6. if anaphoric, the distance between the target referring expression and its antecedent

An interactive text analysis program is similar to a spellchecker in a word processor. In fact, to get a quick idea of the advantages

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of using an interactive text analysis program, remember what it is like to proofread a paper without a spellchecker. Not only are spellcheckers many times faster, they are also many times more accurate because it is easy to miss typos when proofreading by hand. The same benefits hold for the use of interactive text analysis tools.

The first step for the present analysis was to grammatically tag all texts (as described in Chapter 3 and Methodology Boxes 4 and 5). The interactive program then cycles through each tagged text, stopping when it reaches a noun or pronoun (as identified by the grammatical tagger), and prompting the user to select the correct codes for that noun phrase. As with the spellchecker, there are several advantages to using this interactive text analysis program: first, the program identifies noun phrases automatically (so the analyst does not have to read through the text trying to spot noun phrases); second, the program provides an initial analysis of the informational characteristics of the noun phrase – when the initial analysis is correct, the user simply accepts that code; finally, the interactive program provides a list of other possible correct analyses to choose from, so that the user needs only to select the number corresponding to the correct analysis (if the initial analysis is not correct).

Figure 5.1 gives an example of a typical screen from the interactive program, showing how the codes can be accepted or edited. The referring expression currently being coded is presented in context and identified with an arrow (in Figure 5.1, the pronoun *them* is being coded). Underneath the text excerpt, the automatically assigned code is listed (in this example, "ANAPHORIC"), and then the alternative codes are listed. To choose one of these alternatives, the user simply types the number of the choice, or – as would be the case for this example – the user can push "ENTER" to accept the automatically assigned code.

The interactive text analysis program relies on the computer to perform those parts of the analysis that can be done automatically, while retaining human decision-making for those difficult analyses that involve meaning distinctions. Specifically, to code the six characteristics listed above, the program operates as follows:

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```
* * * Code Check * * *   (processing file 00057 . TEC;   word #366)

   impressive that quantum mechanics can take that in its stride
   . The problems of interpretation cluster around two issues ;
   the nature of reality and the nature of measurement .
   Philosophers of science have latterly been busy explaining that
   science is about correlating phenomena or acquiring the power
   to manipulate
= = =>   them
   . They stress the theory – laden character of
   our pictures of the world and the extent to which
   scientists are said to be influenced in their thinking by
   the social factor of the spirit of the age .
      Such accounts cast doubt on whether an understanding of reality

Automatically assigned code is:   REF= ANAPHORIC

ALTERNATE CODES ARE:
1)   REF= ANAPHORIC          2)   REF= EXOPHORIC
3)   REF= INFERRABLE         4)
5)                           6)
7)                           8)

Type number 1–8 to select alternate code
Push <ENTER> to accept code;   * to terminate file;
c for more context   __
```

Figure 5.1 Sample screen from interactive program to code referring expressions

Characteristic 1. Register of the text

The register is recorded at the beginning of each text, and does not require any further analysis by the interactive program.

Characteristic 2. Nominal form: pronoun versus full noun

The program automatically records the grammatical category of the noun phrase (noun versus pronoun) using the information coded into the texts by the grammatical tagger.

Characteristic 3. Information status: given versus new

Pronouns are automatically coded as given information. For each noun, the program automatically checks whether there is an

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earlier occurrence of the same noun in the text. If there is, the repeated noun is automatically coded as given information. All other full nouns are pre-coded as new information. These nouns are then checked interactively to determine whether they actually represent given information.

Characteristic 4. If given information, type of reference: anaphoric, exophoric, or inferrable

The pronouns *I* and *you* are automatically coded as marking exophoric reference (i.e., referring directly to the speaker/writer or listener/reader). Third person pronouns are automatically labelled anaphoric but checked interactively to identify exophoric and inferrable occurrences.

Nouns with given informational status are automatically labelled anaphoric but checked interactively to identify exophoric and inferrable occurrences.

Characteristic 5: If anaphoric and a full noun, status as synonymous versus noun repetition

If nouns have been coded as anaphoric and an earlier occurrence of the same noun was found in the text, the referring expression is automatically identified as a noun repetition. Other anaphoric nouns are coded as synonymous.

Characteristic 6: Distance between the target referring expression and its antecedent

For this analysis, the antecedent of all anaphoric nouns and pronouns must be identified. For repeated nouns, the antecedent is automatically pre-coded as the earlier occurrence of the same noun; these antecedents are checked interactively to determine if there is a closer synonymous expression. For all other nouns and pronouns, the user of the interactive program must type in the antecedent.

The distance between the target referring expression and its antecedent can then be computed automatically. The program simply counts the number of intervening noun phrases occurring in between each referring expression and its antecedent.

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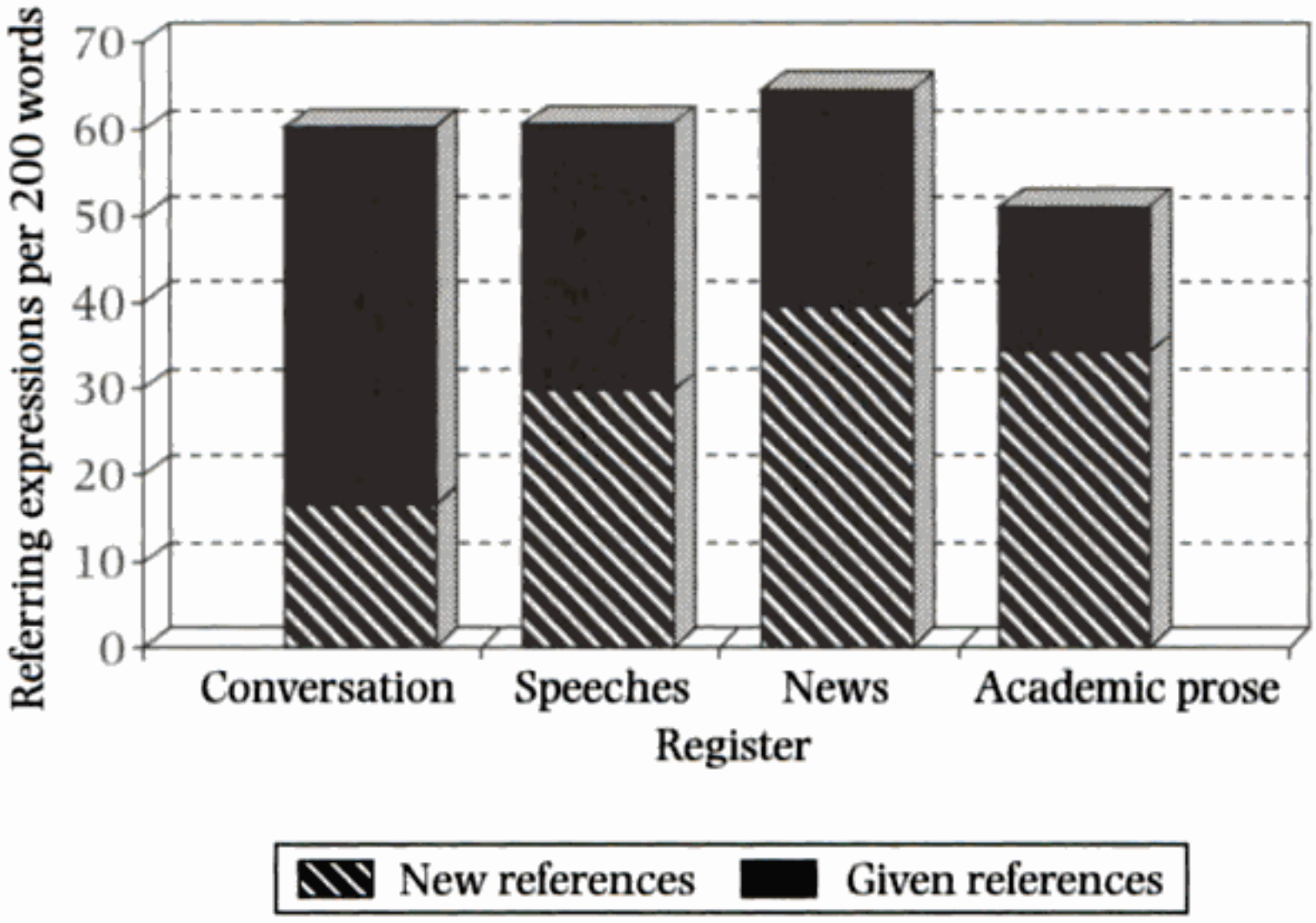

Figure 5.2 Frequency of given versus new referring expressions

As each noun phrase is analyzed, the codes are recorded in the text, with lines such as

<<<Ref = anaphoric

and

<<<Status = given

Another computer program is then used to analyze the coded texts and create a file listing the informational characteristics of each noun phrase. Finally, a statistical analysis package is used to compile frequency counts showing the interaction among these characteristics.

5.2.3 Patterns in the use of referring expressions in spoken and written registers

When the counts for the characteristics of all the noun phrases are compiled, the findings reveal a great deal about the use of different kinds of referring expressions in different registers. For example, Figure 5.2 plots the distribution of given and new referring

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expressions across the four registers. First, even considering only the overall frequencies of referring expressions, Figure 5.2 reveals some surprising differences. News reportage has the largest number of referring expressions of the four registers, while academic prose has the lowest. Interestingly, conversation and public speeches both have relatively frequent referring expressions, despite their general characterization as verbal rather than nominal.[3]

It is perhaps even more interesting that these four registers have striking differences in their reliance on given versus new references. At one extreme, over 70 percent of all referring expressions in conversation present given information. At the opposite extreme, over 65 percent of all referring expressions in academic prose present new information. News reportage shows the most frequent occurrence of new references, although they make up a slightly lower proportion than in academic prose.

The striking patterns of reference in conversation – both the surprisingly large number of total referring expressions and the extreme reliance on given references – can be better understood by considering the different types of given referring expressions. Figure 5.3 breaks down the occurrences of the given references to show the frequencies of exophoric pronouns, anaphoric pronouns, and anaphoric nouns.[4]

As you can see from Figure 5.3, exophoric pronouns account for over half of all given references in conversation. These pronouns are almost all used for reference to the speaker (*I*) or the hearer (*you*), although there are also exophoric references to third persons and objects present in the situational context (e.g., *she*, *he*, or *it*). Text Sample 5.3 illustrates the extremely common exophoric references typical of conversation. References to the speaker and hearer are italicized, while third-person exophoric references are marked in bold with square brackets (exophoric adverbial nouns are marked as well).

Text Sample 5.3: Conversation

A: What are *you* doing **[this afternoon]**?
B: *I*'m going **[home]**. *I*'ve got to teach about half past one. Can *you* pick **[your own trousers]** up?
A: no *I* don't think it'll be likely. *I*'ve got **[this meeting at three thirty]** and . . .

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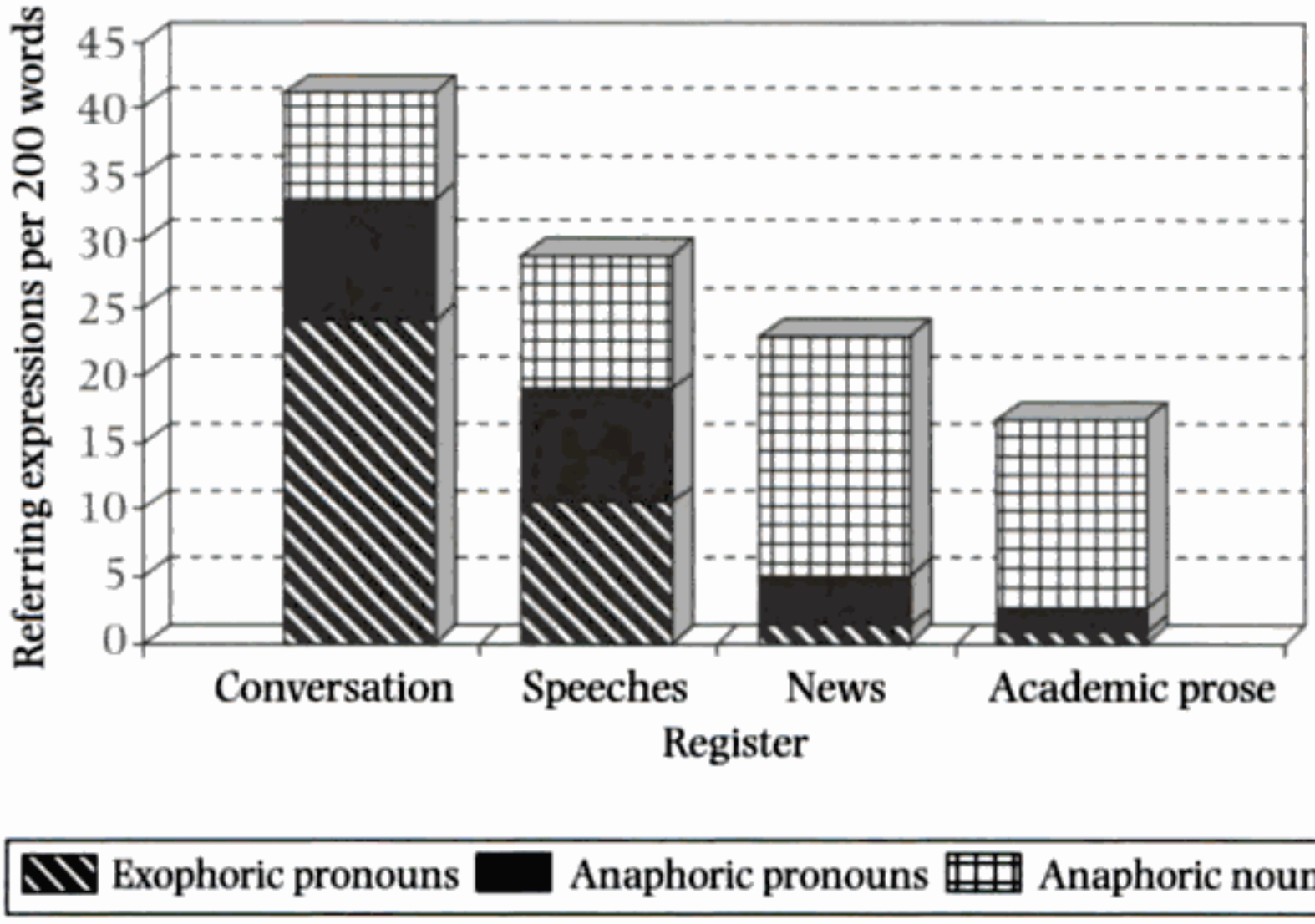

Figure 5.3 Frequency of exophoric and anaphoric referring expressions

B: well what are *we* doing **[this weekend]**? **[Tomorrow]** *I*'ve got **[dancing class]** in **[the morning]**
A: well *I*'ve nothing down anyway at all.
B: well **[they]**'re open **[tomorrow afternoon]** up till three o'clock. *you* remember **[last time]** *we* went, and **[they]** were closed.

In contrast, Figure 5.3 shows that this reliance on exophoric reference is not found in either written register; in fact, apart from an occasional reference to the author (*I*), exophoric references are almost completely absent from these written registers. Instead, the written registers have a greater reliance on anaphoric references. Further, Figure 5.3 shows that the preferred nominal forms used for anaphoric reference are different in the written registers from the spoken registers: conversation and speeches make about equal use of anaphoric pronouns and full nouns, while the written registers use full nouns most of the time.

Text Sample 5.4 illustrates the patterns typical of written exposition, including the reliance on full noun phrases and the high proportion of expressions marking new information. The only exophoric reference in this passage refers to the authors (*us*; also note the use of the possessive pronoun *our*). In the text sample

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below, full noun phrases marking new information are italicized; anaphoric references are given in bold face marked by brackets; and exophoric references are given in bold face **CAPS**:

> *Text Sample 5.4: Academic prose*
> *The NPL Russian–English automatic dictionary* is organized on *a stem-paradigm basis* wherein there is for *most nouns* and *adjectives a single entry* for all *their inflected forms* and for *most verbs* only *one or two entries*. **[This]** is in contrast to *the full-form type of dictionary organisation* wherein *each inflected form* of *every word* has *a separate entry*. The decision to organise **[our dictionary]** on **[this basis]** was made so as to be able to accommodate **[it]** on *the magnetic tape store* available to **US** on *the ACE digital computer* of *our laboratory* . . .

There are two types of anaphoric reference in the passage. The first type refers back to a specific object introduced earlier: for example, the *Russian–English automatic dictionary* is later referred to as *our dictionary* and *it*. The second type of anaphoric reference refers back to a concept introduced in a preceding chunk of discourse; for example, the pronoun *this* and the noun phrase *this basis* both refer to the organization of the Russian–English dictionary. In addition, noun phrases like *the decision* identify inferrable information – readers are expected to figure out that the researchers had made some kind of decision concerning the organization of the dictionary.

As noted above, the computer program also coded the distance between anaphoric referring expressions and their antecedents in a text, with distance defined as the number of intervening noun phrases. For example, in Sample 5.1 above, the company *Thortec International Inc.* is referred to four times – by the company name, twice by the pronoun *it*, and once by a synonymous phrase *the engineering and consulting firm*. The passage is repeated here for convenience with these referring expressions italicized:

> *Thortec International Inc.* said *it* reached [agreements] with [an investor group] and [Wells Fargo Bank] under which *it* will receive [loans] and [an equity infusion] in [return] for [stock] that will reduce [the number] of [shares] in [public hands] by as much as 85 percent. *The engineering and consulting firm* . . .

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Table 5.1 *Average distance measures for four registers*

	Average distance
Conversation	4.5
Public speeches	5.5
News reportage	11.0
Academic prose	9.0

The intervening referring expressions in this case are marked with square brackets. Thus, because there are no intervening expressions between *Thortec International Inc.* and the first *it*, this occurrence of the pronoun *it* has a distance of 0. The third reference to the company (again with a pronoun *it*) has a distance of 3, because there are three intervening referring expressions between the two occurrences of *it* (i.e., *agreements*, *an investor group*, and *Wells Fargo Bank*). Finally, the synonymous expression *the engineering and consulting firm* has a distance of 7 from the preceding reference to the company.

Based on such distance measurements for each anaphoric referring expression, it is possible to compute the average distance for each text and register. As Table 5.1 shows, the average distance scores are quite different across the four registers. In conversation and speeches, the average distance is considerably lower than in the written expository registers. This makes sense given the difference in the production and comprehension circumstances of written and spoken registers. Conversation and speeches must be comprehended on-line, and conversation must also be produced on-line. In those circumstances, co-references with short anaphoric distance are easier to understand. In contrast, writers and readers have as much time as they need to produce and comprehend written expository texts, and as a result, such texts have greater distances between co-referential expressions.

The frequent use of exophoric pronouns referring to the speaker or listener is also a major factor in the low average distance for conversation. For example, Sample 5.2 is repeated below with all exophoric references to Speaker A marked with italics and all

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Table 5.2 *Average distance measures for pronominal versus full noun anaphoric expressions*

	Average pronominal distance	Average full noun distance
Conversation	3.0	9.0
Public speeches	3.5	10.0
New reportage	3.0	13.5
Academic prose	2.5	10.0

exophoric references to Speaker B marked in bold **CAPS**. All other referring expressions are marked with square brackets.

A: Right, *I*'m ready. Have **YOU** locked [the back door]?
I thought *we* were walking.
B: Well do *you* want to walk or do *you* want to go in [the car]?
A: Well *I* have to go to [the paper shop].
B: Well **I**'ll drop *you* at [the paper shop] while **I** go round.

In this short excerpt, the many references to Speaker A have an average distance of less than 1, since many of these references occur in sequence with no intervening referring expressions. In contrast, references to Speaker B have a larger average distance of about 5 because of the many intervening referring expressions.

It turns out that the distance between two co-referential items is closely related to the form of the anaphoric referring expression: pronouns tend to occur much closer to their antecedent than repeated full nouns. Further, as Table 5.2 shows, this generalization holds across registers. That is, whether in conversation or expository writing, pronouns tend to occur relatively close to their antecedent, while full noun phrases tend to be used when the distance is greater. There is a natural explanation for this pattern: the greater the number of intervening referring expressions, the greater the chance for ambiguity and confusion over the intended reference of pronominal forms. Thus, full noun expressions are preferred for anaphoric reference over large distances, regardless of the register.

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The pattern seen in Table 5.2 (i.e., pronouns associated with shorter referential distances than nouns) helps to explain the overall register differences in average distance. That is, because conversation relies heavily on pronominal reference – in particular repeated references to the speaker and hearer – the overall average distance in that register is small. In contrast, because the written expository registers have many referring expressions presenting new information, a large number of different expressions will typically intervene between any two co-referential expressions. As a result, the average distance between co-referential expressions is large in news reportage and academic prose, and anaphoric full nouns are preferred to anaphoric pronouns.

Results such as these show the importance of including multiple factors in studies of the referential discourse characteristics of texts. Obviously, this analysis could be extended in many ways. A larger number of texts and longer text samples are required to insure generalizable results. In addition, a number of other distinctions could be investigated, such as considering the way in which the different kinds of referring expressions are distributed across grammatical contexts (e.g., in main clauses versus dependent clauses).

This sample analysis has also illustrated the usefulness of interactive computer analyses, in combination with automatic techniques, for analyzing a discourse system. The following section shows another application of corpus-based techniques to discourse studies: actually tracking the progression of discourse features within a text.

5.3 Linguistic correlates of rhetorical structure: discourse maps of verb tense and voice

A major concern in discourse analysis is the way that meaning unfolds over the course of a text. From a discourse perspective, the meaning of a text cannot simply be derived from the meanings of the individual sentences in the text. Rather, there are systematic ways in which the grammatical resources of sentences work together at a discourse level. Choices among nominal forms in relation to the marking of given and new information, discussed in the last section, is one such discourse system.

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The present section considers the marking of verb tense and voice, and the ways in which choices among the available tense/voice options can reflect larger rhetorical divisions within a text. That is, written texts are commonly divided into smaller sub-texts – chapters or sections – which are often marked overtly with titles. However, such divisions are not merely conventional ways of segmenting a text. Rather, they often reflect major shifts in communicative purpose within the course of a text.

It turns out that differences in communicative purpose also correspond to linguistic differences. However, linguistic features are more subtle indicators of purpose than titled section divisions, and as a result, it is more difficult for developing readers and writers to control such differences in use. Further, communicative purpose actually evolves continuously over the course of a text, rather than changing abruptly at marked section/chapter boundaries, making it even more important to understand the ways in which linguistic features reflect differences in purpose. In this section we illustrate this association by tracking the use of verb tense and voice over the course of a text.

Despite the central concern with structure larger than the sentence, it has been difficult in practice for discourse analysts to track the allocation of grammatical resources over the course of a text. One reason for this difficulty is simple bookkeeping: it is difficult to compile grammatical information from an entire text, and difficult to decide how to present such information once it is collected. Corpus-based analyses can help overcome these difficulties. Once the computer programs are developed for a given analysis, they can be applied to multiple texts of extended length with relatively little additional effort.

The following discussion illustrates the application of corpus-based methods to track the use of verb tense and voice across the major sections of research articles in experimental science. This is one of the few English registers that clearly distinguishes among internal purpose-shifts. That is, irrespective of discipline, experimental studies tend to follow a standard four-part organization: Introduction, Methods, Results, Discussion (I-M-R-D). Because each of these sections is overtly marked in the text and has distinct communicative functions, this register provides an ideal testing

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Table 5.3 *Mean scores (per 1,000 words) of selected linguistic features across the I-M-R-D sections of English medical research articles (N = 19)*

	Section			
Linguistic feature	Introduction	Methods	Results	Discussion
present tense	47.9	21.1	35.9	60.6
$F = 29.25$; $p < .001$; $r^2 = .549$ (54.9%)				
past tense	20.7	48.5	40.3	13.0
$F = 36.74$; $p < .001$; $r^2 = .605$ (60.5%)				
agentless passives	18.4	39.9	16.9	16.3
$F = 33.17$; $p < .001$; $r^2 = .580$ (58.0%)				

site for discourse-analytic techniques. In the present section, we illustrate such techniques through analysis of a few texts; however, these same techniques could be applied to much larger corpora.

A simple first step is to compare frequency counts of linguistic features across the text-internal sections, interpreting any linguistic differences in terms of the primary communicative purposes of each section. To do this, we treat each article section as a separate text and compute counts for each of those texts; then we are able to compare the average or "mean" frequency counts for each type of section. This is a different kind of research design from those used previously in this book: it treats each text rather than each occurrence of a linguistic feature as the unit of analysis. (For further information about determining the unit of analysis in corpus-based studies see Methodology Box 8.)

Table 5.3 presents the mean scores (normalized per 1,000 words of text) for three tense and voice features across the I-M-R-D sections of nineteen medical articles published in 1985. The texts are taken from the ARCHER Corpus (see Chapter 8 and the appendix for descriptions of the ARCHER Corpus); all articles are from either the *New England Journal of Medicine* or the *Scottish Medical Journal*. The texts were grammatically tagged and the counts compiled with a program written for that purpose.

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As Table 5.3 shows, present tense verbs occur most frequently in Discussion sections, and somewhat less frequently in Introductions. The use of present tense verbs in these sections reflects an emphasis on the current state of our knowledge and the present implications of research findings. In contrast, Methodology sections use considerably more past than present tense verbs, reflecting a focus on the reportage of past events and procedures. The Results sections in these articles have roughly equal frequencies of present and past tense verbs, although other studies based on science texts have found a much higher proportion of past tense verbs in Results sections (reporting the results of the study as past accomplishments rather than present findings). Finally, it should be noted that Methodology sections are also marked by their extremely frequent use of agentless passives, presenting events impersonally, with no acknowledged agent.

The following text samples, taken from an article published in the *Scottish Medical Journal*, illustrate these patterns in the use of verb tense and voice. Sample 5.5 comes from the Methodology section of the article and relies heavily on verbs in the past tense and passive voice to describe the procedures used in the experiment. Sample 5.6, in contrast, comes from the end of the Discussion section of the article. It relies almost exclusively on present tense verbs as it explains the implications for future studies.

> *Text Sample 5.5: Methodology section, medical research article* (past tense verbs are italicized, passive voice capitalized)
> Two hundred and fifty patients with rheumatoid arthritis who *required* change of NSAID *WERE* ENTERED into two consecutively run studies. In the first, 100 patients *WERE* randomly ALLOCATED to either feprazone (maximum dose 600 mg daily), or flurbiprofen (maximum dose 400 mg daily). In the second study, 150 patients *WERE* randomly ALLOCATED to ketoprofen (maximum dose 400 mg daily), benoxaprofen (maximum dose 600 mg daily), or fenbufen (maximum dose 900 mg daily). No additional information or advice *WAS* GIVEN other than that normally presented when non-steroidal anti-inflammatory drugs ARE CHANGED. . . .

> *Text Sample 5.6: Discussion section, medical research article* (present tense verbs are italicized)

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> It *is* certainly apparent that further examples of similar remedies in this category *are* not needed. There would seem little point in marketing yet another "me too" drug unless acceptability by this method *allows* more than 50 per cent of patients to tolerate therapy for six months. Only then would the major exercise necessary to seek rare side effects *seem* justified.

The statistics reported in Table 5.3 show that all three grammatical features (present tense, past tense, and passive voice) have significant differences across the I-M-R-D sections, and these patterns are relatively strong (with r^2 values over 50 percent; see Methodology Box 9 for more on the reporting of statistics). These findings support earlier research showing that there are systematic linguistic differences across the sections of experimental research articles.

Frequency counts like these provide useful average characterizations of each section, and by considering such patterns across all four sections, it is possible to obtain an overview of the discourse organization of an article as a whole. However, average frequency counts cannot be used to analyze the way that a text actually develops. For example, it would be useful to know whether article sections are in fact coherent internally, or whether there are systematic patterns of variation within sections. Further, we also need an analytical approach that can analyze the discourse structure of non-experimental texts, which do not have overtly marked sections.

One approach for tracking the sequence of verbs over the course of a text is to use a graphic display, drawing a "map" of the progression of verbs through a text. We illustrate that approach here with the analysis of two medical research articles in ecology. (The texts are taken from the Corpus of Writing in the Disciplines described in Chapter 6; both texts come from the journal *Ecology*.)

A computer program was written to cycle through grammatically tagged texts and record the tense and voice of each verb phrase. For this illustrative analysis, the program made only a two-way distinction for tense: past versus non-past. Non-past tense includes both present tense and verb phrases with modal verbs (in which case the main verb is not marked for tense). A two-way distinction

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was also made for voice: active versus passive. Non-finite clauses (i.e., infinitives and participial clauses) were excluded from the analysis.

The output of the program is set up in columns to show the movement between past and non-past tense, and between active and passive voice. When the program finds an occurrence of a verb phrase, it records the characteristics in the appropriate columns. The result is a visual display of the sequence of tense and voice as the text progresses.

Figure 5.4 displays the complete verb maps for the two articles in ecology. The sequence of verbs in each text moves from the top of the figure to the bottom, with each occurrence of a verb marked for tense and voice. Thus, in Text A the first verb is non-past and active (marked in the "NP" position for tense and the "A" position for voice); the second verb is non-past and passive (marked again in the "NP" position for tense, but the "PS" position for voice); and so on. The beginnings of the I-M-R-D sections are also marked on the figure.

As Figure 5.4 shows, the discourse progression of verbs in these two experimental articles is remarkably similar. Introduction sections in both articles are written mostly in the non-past tense (using present tense or modal verbs) and the active voice. In contrast, Methodology sections are written mostly in the past tense and passive voice. Results sections maintain the use of past tense but shift strongly to the use of active voice. Finally, Discussion sections maintain the predominant use of active voice but frequently switch back and forth among present tense, modal verbs, and past tense verbs.

These overall patterns correspond to our expectations from the frequency counts for each section. However, the maps can be additionally used to identify systematic departures from the expected patterns, leading to the identification of rhetorically salient shifts in the discourse. One such area of study relates to the transition zones between sections. That is, rather than shifting abruptly from one section to the next, these verb maps show that writers often begin a transition at the end of one section, or continue a transition into the beginning of the following section.

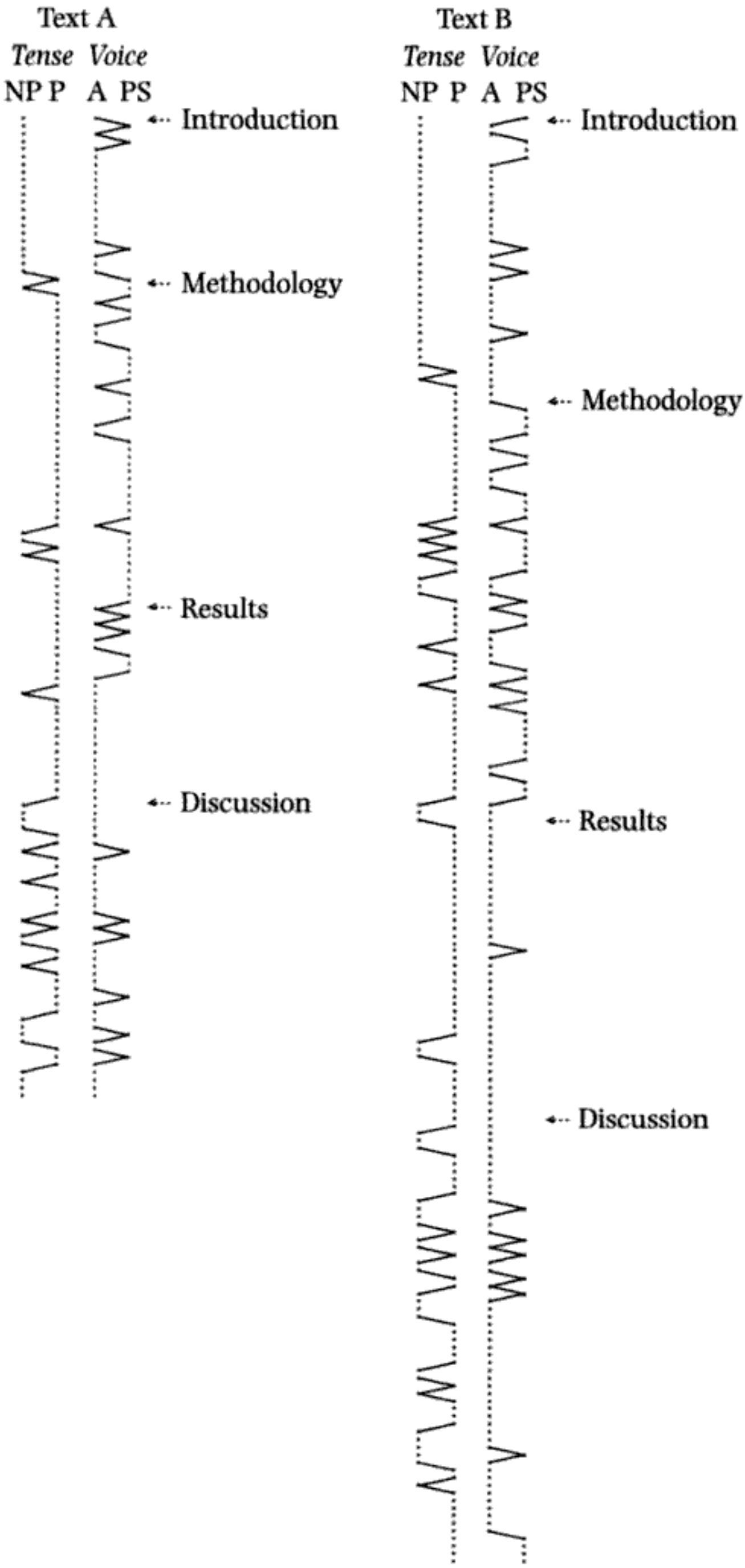

Figure 5.4 Verb maps of two research articles from ecology (NP = non-past; P = past; A = active; PS = passive)

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For example, Figure 5.4 shows that the end of the Introduction section in Text B shifts into the past tense with active voice, a marked pattern for this section. In addition, this marked pattern continues for the first sentence of the Methodology section, before shifting to the more expected use of past tense with passive voice.

Text Sample 5.7 gives the full text corresponding to this sequence of verbs in Text B. The first two sentences in this extract present typical introductory material using present tense verbs and modal verbs to provide background information. However, the last paragraph of the Introduction section in this article shifts to provide a brief synopsis of the methodological steps followed in the study, and this shift in purpose corresponds to a shift to the past tense. Unlike typical Methodology sections, this synopsis is written as a personal account, with first-person pronouns as subjects and verbs in the active voice. Interestingly, this marked pattern is maintained for the first sentence of the Methods section, after which the exposition shifts to the more expected pattern providing methodological details in the past tense with passive voice.

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Text Sample 5.7: Research article (non-past tense is marked by bold face; past tense is marked by italics; passive voice verbs are given in CAPS)

Introduction

. . . .

Because leaf area **IS LINEARLY RELATED** to sapwood cross-sectional area, if leaf area **is** static, sapwood volume **can only increase** if tree height **increases**. Because most height growth **occurs** early in stand development, sapwood volume and maintenance respiration **may not increase** greatly between intermediate and old-growth stands.

In this study, we *tested* the hypothesis that higher maintenance respiration for woody tissues **would reduce** net primary production in older stands. We *estimated* maintenance respiration (R) for woody tissues from sapwood volume, sapwood temperature, and a correction for diurnal temperature amplitude. We then *compared* estimates of R with measured changes in above ground wood production for a chronosequence of subalpine lodgepole pine (pinus contorta ssp latifolia) stands in Colorado.

> Methods
> To reduce site differences, we *selected* three adjacent stands growing within 300 metres. The youngest stand (40 years old in 1986) *WAS ESTABLISHED* when a plot *HAD BEEN CLEAR-FELLED* for a regeneration study . . .

Two other marked transition zones in these texts occur between the Methodology and Results sections. In Text A, the use of passive voice verbs (interspersed with active voice) continues over the beginning part of the Results section. Text B shows a different kind of marked transition at this boundary, shifting to the present tense and active voice at the end of the Methodology section. Similar to the example discussed above, marked shifts such as these identify particularly interesting parts of a text, deserving more detailed discourse analysis to understand the associated shifts in communicative purpose.

Research of this kind could be extended in many ways. For example, it would be useful to investigate the patterns of modal verbs in these texts, as well as the use of perfect and progressive aspect verbs. For these characteristics, too, corpus-based techniques can act like a pointer, locating those detailed portions of a text that are particularly interesting from a discourse-analytic perspective. They do this by providing a comprehensive analysis of the text. That is, to identify those parts of a text that are marked rhetorically (as reflected in unexpected linguistic/discourse characteristics), it is necessary to first study complete texts, tracking the details of linguistic form as they develop over the course of a text. Corpus-based techniques are ideal both for providing comprehensive analyses of this type and for highlighting text chunks that depart from the expected norms.

5.4 Conclusion

This chapter has illustrated the usefulness of corpus-based techniques for investigating issues in discourse analysis. Specifically, it showed how analyzing the characteristics of referring expressions revealed interesting differences across spoken and written registers, and how constructing verb maps provided important

insight into the rhetorical structure of research articles. For both analyses, larger-scale studies with more texts would be necessary to insure generalizable results. However, even the relatively small analyses in this chapter showed how such investigations can reveal important patterns in language use, while also pointing to areas within texts which would be interesting for more intensive, qualitative analysis (such as the rhetorical transition zones in research articles).

The examples in this chapter also showed that corpus-based techniques are useful for linguistic studies even when analyses cannot be done completely automatically or when information needs to be tracked throughout a text. Interactive computer programs and innovative output formats make it possible to investigate issues that have previously been considered intractable. In the future, therefore, as more studies exploit these techniques, we should be able to learn more about patterns of discourse that hold across texts and registers.

Notes

1. There is a fourth type of pronominal reference – cataphoric – in which the pronoun refers to a noun that occurs later in the text. For simplicity we exclude this type of reference from the sample analyses in this chapter.
2. A fuller presentation of related findings is given in Biber (1992). That study followed a somewhat different methodology, with some output files being coded by hand.
3. The overall distribution of referring expressions was confirmed by an analysis of all the texts in these four registers from the London–Lund and LOB corpora. The full analysis resulted in slightly higher counts for news reportage and academic prose, but the overall distributional pattern remained unchanged.
4. As noted earlier, "inferrable" is also an important category for given information. However, judgements about what is inferrable raise a number of methodological issues that go beyond the scope of this chapter, and we have chosen to exclude "inferrable" from the discussion of results. Interested readers can find a useful discussion in Prince 1981.

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Further reading

Because discourse studies often involve detailed interpretive analysis, there have been relatively few corpus-based investigations in this area. The following selected studies are corpus-based to differing extents: Biber (1992), Burges (1996), Chafe (1991), Fox (1987), Fraurud (1990), Givón (1983), Morris and Hirst (1991), Stenström (1987), Thavenius (1983), Youmans (1991).

Éléments sous droits d'auteur

Part II

Investigating the characteristics of varieties

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6 Register variation and English for Specific Purposes

6.1 Studying register variation

To this point in the book we have focused on the description of language characteristics, whether lexical, grammatical, or discourse features. However, researchers often have research questions relating to the description of "registers" rather than individual linguistic features. The term "register" is used here as a cover term for varieties defined by their situational characteristics. Some registers can be very specific, such as novels written by Jane Austen, or Methods sections in biology research articles. Other registers are more general, such as conversation or student essays.

Registers should be distinguished from "dialects." Registers are defined according to their situations of use (considering their purpose, topic, setting, interactiveness, mode, etc.). In contrast, dialects are defined by their association with different speaker groups (e.g., speakers living in a particular region or speakers belonging to a particular social group). In this chapter we focus on the description of registers; subsequent chapters in this section also include the study of dialect differences (e.g., essays written by L1-Navajo students versus L1-English students in Chapter 7).

Control of a range of registers is crucially important for any competent speaker of a language. It is probably accurate to say that no one controls only a single register; rather, during the course of any day, we all speak and write a wide range of registers. For example, the language that you use to write a term paper is different from the language you use when talking to your roommate;

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and those registers are both different from the language you use for a discussion with a teacher or to write a letter to your mother.

Given this constant need to switch among registers, the acquisition of register characteristics is fundamentally important at all developmental stages: whether for a child learning that the language used with playmates is different from the language used with adults; or for an elementary school student discovering how written narratives are different from conversation; or for a college student discovering how term papers in biology are different from written personal narratives; or for ESL students attempting to control the range of registers in their second language.

In order to understand the process of register acquisition and the way that teachers can facilitate that process, we must first describe the linguistic characteristics of different registers, identifying the most important ways in which registers are similar or different. Although researchers have long recognized the need for such descriptions, it has proven difficult to achieve them without a corpus-based approach. Difficulties arise without corpus-based techniques because comprehensive register studies have three important requirements: inclusion of a large number of texts, consideration of a wide range of linguistic characteristics, and comparison across registers.

First, inclusion of a large number of texts is important because register studies based on too few texts are likely to be inaccurate. The sample analysis in Section 6.2 illustrates this risk.

Secondly, register studies based on a few, selected linguistic features do not provide comprehensive register descriptions, and generalizations based on such studies are likely to be inaccurate. It is rare for a register to be identified by the occurrence of a distinctive linguistic feature found only in that register (like the "count" – balls and strikes – found only in broadcasts of baseball games). Rather, registers share many linguistic features – such as nouns, pronouns, verbs, adjectives, etc. – and they are distinguished by the relative use of these features. That is, systematic differences in the relative use of core linguistic features provide the primary distinguishing characteristics among registers.

Finally, register analyses require a comparative approach: we need a baseline for comparison to know whether the use of a

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linguistic feature in a register is rare or common. As discussed in previous chapters, a particular frequency is neither common nor rare in itself. For example, a frequency of 25 per 1,000 words would be extremely common for relative clauses, which typically have average frequencies between 1 and 10 per 1,000 words (depending on the register). In contrast, a frequency of 25 per 1,000 words would be almost impossibly rare for nouns, where the average frequencies are closer to 200 per 1,000 words.

Similarly, we can only interpret average frequencies for a register by comparison to other registers. A frequency count for relative clauses in a particular register is only meaningful in comparison to the frequency in other registers. Thus, register analyses are actually investigations of register variation – the distinctive ways in which linguistic features are relatively common or rare, when compared to the use of those features in other registers.

The three requirements for comprehensive register studies are difficult to achieve by hand: it is extremely time-consuming to analyze even a few texts by hand, considering even one or two linguistic characteristics. It is nearly impossible to identify the occurrences of multiple linguistic features on a single pass through a text. To get a feel for how difficult this task is, try the following exercise: pick a page of this book and read through it once, attempting to identify all occurrences of five linguistic characteristics: nominalizations, passive constructions, modal verbs, relative clauses, and prepositional phrases. Keep track of how long this exercise takes. Then, when you are finished, compare with a classmate to determine how accurate you were in your analyses. Finally, imagine how difficult it would be to keep track of dozens of different linguistic features at the same time, analyzing numerous texts from different registers!

6.1.1 Applying corpus-based techniques to register analysis

Corpus-based techniques make it much easier to carry out comprehensive register studies. Modern computers have made it possible to store a large number of texts, to analyze a large number of linguistic features in those texts, and to make comparisons of the results across registers. In this chapter we illustrate how

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corpus-based approaches can be used to address questions about register variation, beginning by comparing the distribution of selected grammatical features across registers and then turning to more comprehensive register characterizations. We also illustrate the analysis of registers at different levels of specificity. Four research questions are addressed:

1. How do spoken and written registers differ in their use of dependent clauses?

This research question illustrates corpus-based techniques used to study register differences with respect to a single type of construction – dependent clauses. In the past, many studies assumed that all kinds of dependent clauses had similar purposes and distributions. The illustrative study in Section 6.2 shows, however, that the use of dependent clauses varies across registers. In discussing the results of this sample analysis, we also show how inclusion of too few texts or too few linguistic features can lead to inaccurate register characterizations.

2. How is speech different from writing in English? Specifically, what patterns in the use of linguistic features are important in distinguishing among the major spoken and written registers? How are specific spoken and written registers similar or different in terms of those patterns?

For many years, differences between speech and writing have been a topic of interest to linguists. Answering this research question requires a much more comprehensive analysis of linguistic features than the first research question. We therefore introduce an approach to studying register variation called multi-dimensional analysis. Working within the framework of corpus-based study, multi-dimensional analysis allows investigation of the major patterns of variation in linguistic features across spoken and written registers. This research question is addressed in Section 6.3.

3. How do texts from different academic disciplines vary with respect to patterns of linguistic variation?

This research question addresses issues of concern in English for Specific Purposes (ESP). Specifically, the example analysis shows

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how research articles from two diverse disciplines (biology and history) vary in their patterns of language use, and how these linguistic patterns are related to the primary purposes and methods of each discipline. This question shows how multi-dimensional analysis, introduced for the previous research question, can be used to investigate more specific registers. It is addressed in Section 6.4.

4. How do the internal sections of texts within a single academic register vary linguistically?

This question illustrates analyses of very specific registers – in this case, the sections within biology research articles. Following from Research Question 3, this question applies the multi-dimensional framework to each of the sections within the biology research articles, focusing in more detail on the variation across Introduction, Methods, Results, and Discussion sections.

For all the above research questions and, indeed, all analyses of register variation, it must be emphasized that quantitative techniques are not sufficient in themselves. Rather, qualitative interpretations are necessary to examine the functional bases underlying patterns of linguistic features. Thus, each of the following sections illustrates both the use of quantitative analyses and the application of qualitative functional interpretations to study the patterns of register variation in English.

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6.2 Dependent clause use across spoken and written registers

This section examines spoken and written registers' use of dependent clauses. In the past, many studies have simply assumed that all kinds of dependent clause are functionally similar, representing structural elaboration and complexity. Based on this assumption, analysts have sometimes considered the distribution of a single type of dependent clause and then generalized their findings to all kinds of structural subordination.

For example, the following brief sample from a geology textbook shows a very dense use of relative clauses:

> *Text Sample 6.1: Geology textbook*
> By 1904 Jukes-Browne had completed his survey of the cretaceous rocks (1900–4), which includes description of sections west of the area dealt with in the sheet memoir, some of which do not appear to have received any attention since.

This single sentence includes three relative clause constructions: two *wh*-relative clauses (*which includes descriptions . . . , which do not appear . . .*) and a reduced relative clause (*the area dealt with*). All of these constructions are used to identify or provide elaborating information about a head noun. Based on the density of such features, many studies have concluded that written registers are generally more structurally elaborated than spoken registers.

However, there are many different kinds of dependent clauses in English. In this section, we will consider the distribution of three kinds of dependent clauses: relative clauses, adverbial clauses, and complement clauses.

As we have seen above, relative clauses are used to identify or elaborate a noun. In contrast, adverbial clauses are used to provide some kind of a frame for the proposition in the main clause, such as reasons, conditions, or temporal settings. For example, the following sentence has two adverbial clauses:

(1) Do ring Cathy if you feel like it, because I'm sure she won't mind.

The first adverbial clause, *if you feel like it*, provides a condition for the main clause *Do ring Cathy*, while the second adverbial clause, *because I'm sure she won't mind*, provides a reason.

Finally, complement clauses fill noun phrase slots. In the sentence *I think I can*, the complement clause *I can* completes the meaning of the verb, telling what it is that I think. (Complement clauses have been further discussed in Chapters 3 and 4.)

The corpus for this example analysis was composed of texts from two written registers from the LOB Corpus (80 academic prose texts and 14 official documents) and two spoken registers from the London–Lund Corpus (44 conversations, and 14 prepared speeches), for a total of 478,000 words. Occurrences of each of the three types of dependent clause were counted in the texts, which had been grammatically tagged. To keep this illustration simple,

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Table 6.1 *Average frequencies of three dependent clause types (per 1,000 words) in four registers*

Register	Number of texts	Relative clauses	Causative adverbial subordinate clauses	*that*-comp. clauses
Academic prose	80	6.8	.3	3.2
Official documents	14	8.6	.1	1.6
Conversations	44	2.9	3.5	4.1
Prepared speeches	14	7.9	1.6	7.6

only causative adverbial clauses – those showing reasons or causes – were counted; a full analysis would include many more kinds of clauses, of course.

The different types of dependent clause are distributed in very different ways across the registers, as shown in Table 6.1. Relative clauses are frequent in academic prose, official documents, and prepared speeches, but relatively rare in conversation. In contrast, causative adverbial subordination occurs most frequently in conversation and is quite rare in academic prose and official documents. Finally, *that*-complement clauses occur most frequently in prepared speeches, and with moderate frequencies in conversations and academic prose, but they are rare in official documents.

The different functions of the different types of subordinate clause correspond to their distribution across the registers. Academic prose, official documents, and prepared speeches are often focused on conveying information about particular referents in the text. Relative clauses are commonly used to perform this function, as shown in Text Sample 6.1.

Conversation, on the other hand, is more concerned with the interaction among participants, and this includes a concern with reasons and causes for actions, often conveyed with causative adverbials, such as the following:

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Text Sample 6.2: Conversation

a. I wouldn't want it before the end of June anyhow Reynard *because I'm going to Madrid on the tenth . . .*
b. I rushed into the kitchen *because I smelt something was burning . . .*

Finally, *that*-complement clauses often mark the stance of the speaker or writer (e.g., with the verbs *think, wish, hope, doubt*), and such statements of stance are more common in sermons, lectures, court oratories and other types of prepared speeches than in the other registers investigated here:

Text Sample 6.3: Prepared speeches

a. There are many people who think *that to be a Christian is to lead a soft option in life . . .*
b. We would hope *that our students would have a full understanding of the cultural differences . . .*

Even this short analysis shows that lumping together different clause types, or making generalizations based on one type, will lead to inaccurate conclusions concerning the relations among registers. For example, a study considering only relative clauses would conclude that subordination features were most common in expository (especially written) registers, while a study based only on causative adverbial clauses would conclude that subordination features were most common in the spoken registers (especially conversation).

In addition, this analysis can also illustrate how consideration of too few text samples can lead to dramatically inaccurate conclusions. For example, Text J30 from the academic prose category in the LOB Corpus differs radically from the overall mean scores for academic prose given in Table 6.1 (based on all 80 texts). Specifically, Text J30 has 50 relative clauses in a 2,000+ word sample, or about 25 relative clauses per 1,000 words – a frequency almost four times greater than the overall average frequency for the register. The following excerpt illustrates the extremely dense use of relative clauses in this text:

Text Sample 6.4: LOB Corpus; Academic prose, J30

Most Vale people also have kin ties with people *who* live in these areas and in other parts of south Wales with *whom* they

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> maintain effective social relations. A larger number of Vale people *who* do not work in the urban areas nevertheless visit them fairly regularly to see friends and relatives *who* live there or *who* are in hospital there . . .

A study of this text alone, or even with two or three additional texts, would find relative clauses to be much more common in academic prose than they actually are. In addition, a small study including this text would also result in grossly inaccurate conclusions concerning the frequency of *that*-clauses in academic prose – because Text J30 does not use a single *that*-clause in this 2,000-word sample! This is in contrast to the register average frequency of 3.2 *that*-clauses per 1,000 words, or over 6 *that*-clauses in a typical 2,000-word sample of academic prose.

Overall, then, this example analysis illustrates the importance of including a large number of texts in register studies and of looking at multiple linguistic features. For a research question such as this – about registers' differing use of dependent clause structures – decisions about the structures to include are quite straightforward. However, when we seek to characterize registers more broadly, deciding which linguistic features to include and how to study them becomes more challenging. These issues are discussed in the next section.

6.3 Variation among spoken and written registers

6.3.1 Co-occurrence patterns in linguistic features

In carrying out a broad comparison of spoken and written registers, it is difficult to know which linguistic features to compare. For example, a comparison of relative clauses (as above) would show academic prose to be quite different from conversation. In contrast, academic prose and conversation are relatively similar in their use of past tense verbs: they are relatively rare in both conversation (about 35 per 1,000 words) and academic prose (about 22 per 1,000 words), in comparison to fiction texts which typically use over 80 past tense verbs per 1,000 words.

For this reason, it is not possible to reliably distinguish among registers by considering the relative distribution of individual

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linguistic features – there are simply too many different linguistic characteristics to consider, and no way of knowing ahead of time which ones will be important in any given register comparison. Rather, we need to understand the ways in which linguistic features work together – or "co-occur" – in texts. From a linguistic perspective, registers are distinguished by the extent to which they utilize such different sets of co-occurring features.

Linguists have long recognized that groups of co-occurring linguistic features are instrumental in distinguishing among registers. In many cases, the need to consider co-occurring features is apparent even from single sentences, as shown by the following examples from conversation and academic prose:

Text Sample 6.5: Conversation
What you'd have to do, you know, you tell him what you need to know, he'd be able to tell you how to do it.

Text Sample 6.6: Academic prose
As has been repeatedly shown, cultural evolution is not a unilinear process, and it is possible that under certain conditions a simpler social formation may emerge out of a more complex one.

The most obvious characteristic of this conversational example is the fragmented sentence structure. However, there are a number of additional linguistic features co-occurring in this short extract: contractions, second-person pronouns (*you*), discourse inserts (*you know*), semi-modals (*have to, need to, be able to*), and *wh*-complement clauses.

In contrast, the academic prose extract has a different set of co-occurring features: frequent nouns and attributive adjectives, nominalizations and other specialized vocabulary items (e.g., *cultural evolution, unilinear process, certain conditions, simpler social formation*), passive constructions (*has been shown*), and extraposed constructions (*it is possible that . . .*).

Although linguists have generally recognized the importance of co-occurrence patterns, the empirical identification of these patterns has proven to be quite difficult. Only a few researchers have attempted to identify sets of co-occurring linguistic features. Until

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recently, these attempts have relied primarily on a qualitative inspection of the linguistic features occurring in selected texts from a few registers. While those analyses have been important in describing the functional associations of selected linguistic features, they have not succeeded in accurately identifying the sets of linguistic features that actually co-occur in texts.

In fact, this latter research goal can be achieved only through a comprehensive analysis of a representative corpus. One approach to this research question is called multi-dimensional analysis. In this approach, co-occurrence patterns are identified quantitatively, based on the actual distributions of linguistic features in a large corpus of texts. The approach was developed in the 1980s for the study of spoken and written registers in English (see Biber 1988), while more recently it has been extended to study register variation in a number of non-western languages.

6.3.2 Multi-dimensional analysis of English

Multi-dimensional (MD) analyses of register variation are based on corpora representing the full range of major co-occurrence patterns in a language. Such a corpus includes multiple texts from a wide range of spoken and written registers. (See Methodology Box 1 for a discussion of representativeness in corpus design.) The corpus analyzed for the early MD studies of English is composed of 481 texts (960,000 words): 340 texts from the Lancaster–Oslo/Bergen, covering the major written registers (e.g., academic prose, editorials, mystery fiction); and 148 spoken texts from the London–Lund corpora (e.g., face-to-face, public conversations, prepared speeches, etc.).

The first step in a MD analysis is to identify the set of linguistic features to study. The goal in this step is to include a wide range of the linguistic features that have functional associations. The features included in the MD analysis of English fall into sixteen major grammatical categories: (A) tense and aspect markers, (B) place and time adverbials, (C) pronouns and pro-verbs, (D) questions, (E) nominal forms, (F) passives, (G) stative forms, (H) subordination features, (I) prepositional phrases, adjectives, and adverbs, (J) lexical specificity, (K) lexical classes, (L) modals, (M) specialized

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verb classes, (N) reduced forms and dispreferred structures, (O) coordination, and (P) negation.

Second, computer programs are developed to identify and count the occurrence of each linguistic feature in texts. A grammatical tagger was developed to identify many of these linguistic features, while interactive programs are needed to accurately identify more complex features. (See Methodology Boxes 4 and 5 for a discussion of grammatical tagging.) All computational analyses must be checked by hand to ensure that the feature counts are accurate.

After the linguistic features are counted and normalized in all texts (see Methodology Box 6), the analyst is faced with an overwhelming amount of data – in the case of the English study, this included 67 different frequency counts for each of the 481 texts in the corpus. There is obviously no way to simply look at these counts and figure out which features co-occur in texts. However, a statistical procedure known as "factor analysis" can be used for this purpose. This is a correlational technique designed to identify sets of variables that are distributed in similar ways. In the present case, factor analysis shows which of the linguistic features tend to co-occur in texts. In other words, the factor analysis identifies the co-occurrence patterns among linguistic features – the sets of linguistic features that typically occur together in texts.

Each set of co-occurring features is called a "dimension" of variation. These are groups of linguistic features that co-occur with a high frequency in texts. One group of co-occurring features might be first-person pronouns, second-person pronouns, and *wh*-questions; a second group of co-occurring features might be nouns, prepositional phrases, and attributive adjectives. These groupings reflect the fact that these features are distributed in similar ways across texts. For example, when a text has a high number of nouns, it is likely to also have high frequencies of prepositional phrases and attributive adjectives; and conversely, if a text has notably few nouns, it is likely to also have few prepositional phrases and attributive adjectives.

After the linguistic features defining a dimension are identified through factor analysis, the dimension is interpreted functionally, in terms of the situational, social, and cognitive functions most widely shared by the linguistic features. This interpretation is based

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on the assumption that co-occurrence reflects shared function. That is, linguistic features co-occur in texts because they function in similar ways. A simple example is the way in which first- and second-person pronouns, direct questions, and imperatives tend to co-occur in texts because they all relate to interactiveness. Direct questions and imperatives require a specific addressee to respond, while first- and second-person pronouns refer directly to the speaker and that addressee. Similarly, contractions, false starts, and generalized content words (e.g., *thing*) are all related to real-time production constraints. Thus, these features tend to co-occur in conversation, where participants interact directly with one another and speakers must cope with the constraints of on-line production.

Five major dimensions of variation were identified in the MD analysis of spoken and written English registers. Table 6.2 summarizes the co-occurring features associated with each of these dimensions. As described above, these sets of co-occurring features are identified quantitatively by a factor analysis.

Most of the dimensions consist of two groupings of features, which represent sets of features that occur in a complementary pattern. That is, when the features in one group occur together frequently in a text, the features in the other group are markedly less frequent in that text, and vice versa.

For example, consider Dimension 1 in Table 6.2. The two groups are labeled "positive" and "negative" to indicate their complementary relationship. The features in the positive group on Dimension 1 (above the dashed line on Table 6.2) include private verbs (e.g., *think*, *feel*), *that*-deletions (in complement clauses), contractions, present tense verbs, second-person pronouns, etc. The features in the negative group are nouns, word length, prepositional phrases, type/token ratio, and attributive adjectives.

The groupings defining Dimension 1 can be broken down into three major distributional patterns: 1. the positive group of features tend to co-occur in texts; that is, when a text has frequent private verbs, it is also likely to have frequent *that*-deletions, contractions, etc.; 2. the negative group of features also tend to co-occur in texts; 3. the positive and negative groups of features have a complementary distribution. Thus, when a text has many

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Table 6.2 *Summary of the co-occurrence patterns underlying the five major dimensions of English (based on Biber 1988*[1]*)*

Dimension 1: "Involved versus informational production"	
private verbs	.96
that-deletion	.91
contractions	.90
present tense verbs	.86
second-person pronouns	.86
do as pro-verb	.82
analytic negation	.78
demonstrative pronouns	.76
general emphatics	.74
first-person pronouns	.74
pronoun *it*	.71
be as main verb	.71
causative subordination	.66
discourse particles	.66
indefinite pronouns	.62
general hedges	.58
amplifiers	.56
sentence relatives	.55
wh-questions	.52
possibility modals	.50
non-phrasal coordination	.48
wh-clauses	.47
final prepositions	.43
(adverbs	.42)
nouns	−.80
word length	−.58
prepositions	−.54
type/token ratio	−.54
attributive adjs.	−.47
(place adverbials	−.42)
(agentless passives	−.39)
(past participial postnominal cls.	−.38)
Dimension 2: "Narrative versus non-narrative discourse"	
past tense verbs	.90
third-person pronouns	.73
perfect aspect verbs	.48
public verbs	.43
synthetic negation	.40
present participial clauses	.39
(present tense verbs	−.47)
(attributive adjs.	−.41)
Dimension 3: "Elaborated versus situation-dependent reference"	
wh-relative clauses on object positions	.63
pied-piping constructions	.61
wh-relative clauses on subject positions	.45
phrasal coordination	.36
nominalizations	.36
time adverbials	−.60
place adverbials	−.49
adverbs	−.46
Dimension 4: "Overt expression of argumentation"	
infinitives	.76
prediction modals	.54
suasive verbs	.49
conditional subordination	.47
necessity modals	.46
split auxiliaries	.44
(possibility modals	.37)
[no negative features]	
Dimension 5: "Impersonal versus non-impersonal style"	
conjuncts	.48
agentless passives	.43
past participial adv. clauses	.42
by-passives	.41
past participial postnominal clauses	.40
other adverbial subordinators	.39
[no negative features]	

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occurrences of the positive features (private verbs, contractions, etc.), it will usually have notably few occurrences of the negative features (nouns, prepositional phrases, etc.). Conversely, if a text has many occurrences of the negative features, it will usually have notably few private verbs, contractions, etc.

The functional interpretation of each dimension is based on: 1. analysis of the communicative function(s) most widely shared by the set of co-occurring features defining the dimension, plus 2. analysis of the similarities and differences among registers with respect to the dimension.

The first step in the interpretation of a dimension is to assess the functions shared by the co-occurring features. On Dimension 1, the interpretation of the negative group of features is relatively straightforward. Nouns, word length, prepositional phrases, type/token ratio, and attributive adjectives all indicate an informational focus and a careful integration of information in a text. These features are associated with written texts that have an informational focus. Such texts also typically provide ample opportunity for careful integration of information and precise word choice. The following text sample illustrates these co-occurring linguistic characteristics in an academic journal article:

> *Text Sample 6.7: Academic prose*
> Apart from these very general group related aspects, there are also individual aspects that need to be considered. Empirical data show that similar processes can be guided quite differently by users with different views on the purpose of the communication.

This text sample is typical of written expository prose in its dense integration of information: frequent nouns and long words, with most nouns being modified by attributive adjectives or prepositional phrases (e.g., *general group related aspects*, *individual aspects*, *empirical data*, *similar processes*, *users with different views on the purpose of the communication*).

The positive group of features on Dimension 1 is more complex, although all of these features have been associated in one way or another with an involved, non-informational focus, related to a primarily interactive or affective purpose and on-line production circumstances. For example, first- and second-person pronouns,

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wh-questions, emphatics, amplifiers, and sentence relatives can all be interpreted as reflecting interpersonal interaction and the involved expression of personal feelings and concerns.

Other positive features on Dimension 1 mark a reduced surface form, a generalized or uncertain presentation of information, and a generally "fragmented" production of text; these include *that*-deletions, contractions, pro-verb DO, the pronominal forms, and final (stranded) prepositions. In these cases, a reduction in surface form also results in a more generalized, less explicit content. Text Sample 6.8, from a formal conversation (an interview), illustrates the use of the linguistic characteristics that co-occur in the positive group on Dimension 1:

Text Sample 6.8: Formal conversation

B: come in . come in – – ah good morning
A: good morning
B: you're Mrs Finney
A: yes I am
B: how are you – my names Hart and this is Mr Mortlake
C: how are you
A: how do you do .
B: won't you sit down
A: thank you – –
B: mm well you are proposing . taking on . quite something Mrs Finney aren't you
A: yes I am
B: mm
A: I should like to anyhow
B: you know what you'd be going into
A: yes I do

This text sample is typical of face-to-face interactions in showing frequent questions, first- and second-person pronouns, and other direct references to the immediate participants. The sample also illustrates the effects of on-line production, as reflected in frequent contractions and the use of general referring expressions (e.g., *taking on quite something*). The negative features grouped on Dimension 1 are notably absent in this text sample. There are few lexical nouns (as opposed to pronouns) in this interaction, and the ones that do occur are often part of formulaic expressions (e.g.,

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The relations among registers shown in Figure 6.1 confirm the interpretation of Dimension 1 as distinguishing among texts according to involved versus informational production. Conversational texts are largely interactive and involved, since participants typically are not inclined to highly informational purposes, nor do they have time for careful, highly informational production. Conversations thus have frequent occurrences of present tense verbs, private verbs, first- and second-person pronouns, contractions, and the other positive features on Dimension 1, coupled with relatively infrequent occurrences of nouns, prepositional phrases, long words, and the other negative features on Dimension 1. Registers such as public conversations (interviews and panel discussions) are intermediate because they have a relatively informational purpose but participants are constrained by on-line production and actively interacting with one another. Finally registers such as press editorials, academic prose, and official documents are non-interactive but extremely informational in purpose, produced under highly controlled and edited circumstances.

The other dimensions summarized in Table 6.2 are defined by different groupings of linguistic features. As a result, each dimension identifies different similarities and differences among registers, and each reflects different functional associations. The interpretive labels listed in Table 6.2 summarize the functional interpretations.

Dimension 2 is interpreted as distinguishing between narrative and non-narrative types of discourse. Narrative discourse (especially fiction) is marked by frequent reference to past time (past tense and perfect aspect verbs), together with third-person referents and reported speech (public verbs like *say*). Non-narrative discourse, which can be expository, descriptive, or conversational, is marked by frequent reference to non-past time.

Dimension 3, "Elaborated versus situation-dependent reference," is interpreted as characterizing highly explicit, context-independent reference versus situation-dependent reference. *wh*-relative clauses (including pied-piping constructions) are used to specify the identity of referents within a text in an explicit and elaborated manner. Time and place adverbials, on the other hand, are usually used for text-external references to the physical context of the discourse.

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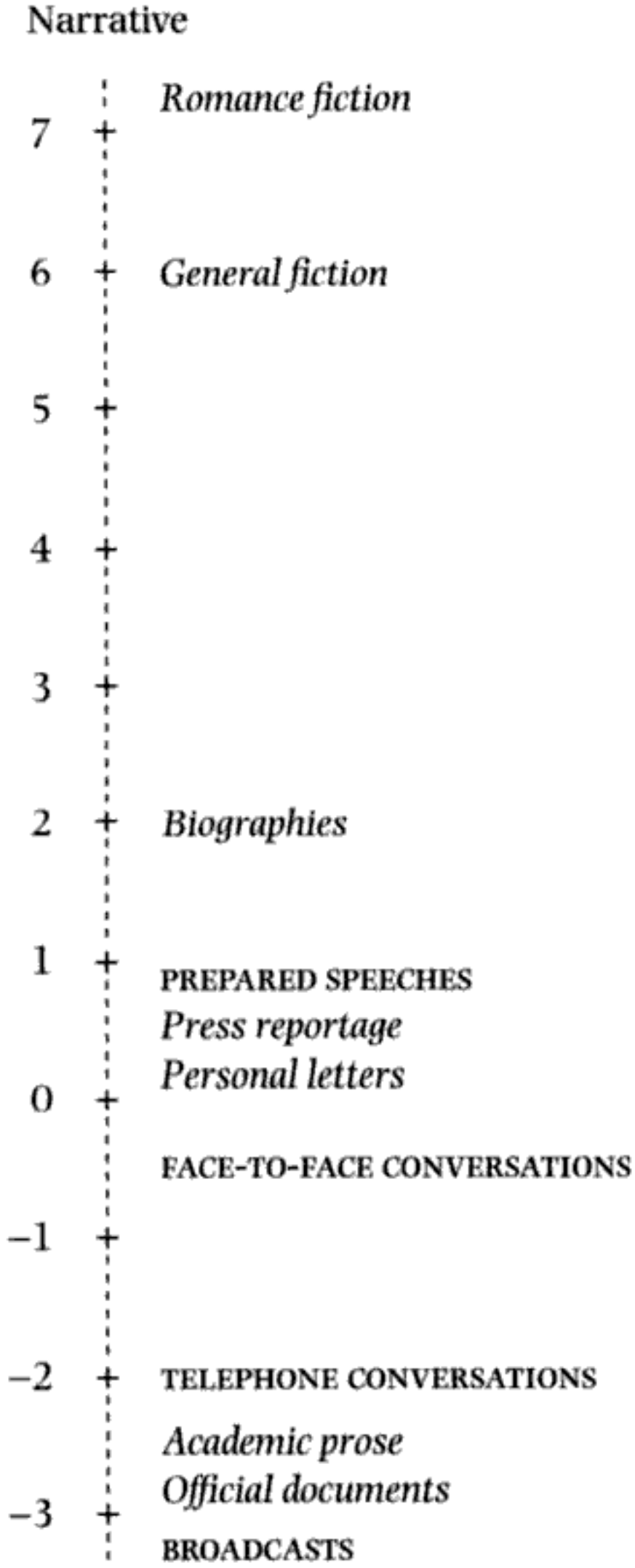

Figure 6.2 Mean scores of English Dimension 2 for eleven registers: "Narrative versus non-narrative discourse"
($F = 32.3$, $p < .0001$, $r^2 = 60.8$ percent; spoken registers in small capitals, written registers italicized)

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The interpretive label "Overt expression of argumentation" is used for Dimension 4. This dimension marks the degree to which persuasion is marked overtly, whether marking the speaker's point of view, or the speaker's attempt to persuade the addressee.

Finally, the co-occurrence of conjuncts, passive constructions, and past participial clauses on Dimension 5 marks informational discourse that is impersonal, technical, and formal in style versus other types of discourse. The interpretive label "Impersonal versus non-impersonal style" can thus be proposed for Dimension 5.

6.3.3 Comparing spoken and written registers

Each of the dimensions described above identifies a distinct set of similarities and differences among registers. They therefore provide multiple perspectives on the relationships among spoken and written registers.

For example, Figure 6.1 showed that the two poles on Dimension 1 distinguish between spoken and written registers. Specifically, the registers with extreme scores for involved production are spoken registers, while the registers with extreme scores for informational production are expository written registers. This distribution corresponds to the dimension's associations with interactiveness and production circumstances. The need for precise, dense packaging of information, as well as the time for planning and editing such prose, is characteristic of written, expository registers, while fast communication with direct interaction among participants is characteristic of spoken registers such as conversation.

Dimension 2, on the other hand, reveals a different relationship between spoken and written registers. Figure 6.2 plots the Dimension 2 scores for eleven registers. In contrast to the register characteristics seen along Dimension 1, Figure 6.2 shows that fiction is most marked with respect to Dimension 2 characteristics. Thus, consider the following excerpt from romance fiction:

> *Text Sample 6.9: Romance fiction*
> But Mike Deegan was boiling mad now. When the inning was over he cursed the Anniston catcher all the way into the dugout . . . The Anniston manager came right up to the dugout in front of Mike. His face was flushed. "Deegan," the manager

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> said, his voice pitched low, quivering, "That was a rotten thing to do." "For God's sake," Mike said, waving the manager away, "Stop it, will you? Tell your guys not to block the plate!"
> "You didn't have to ram him."
> "That's what you say."

This text sample shows the frequent use of past tense and perfect aspect verbs, together with the absence of present tense verbs. It also illustrates the use of speech act verbs, especially for direct dialogue. As Figure 6.2 shows, fictional texts from several sub-registers are marked by the frequent use of these narrative features.

In addition, Figure 6.2 shows that Dimension 2 is fundamentally different from Dimension 1 in that the distribution of registers along this dimension has no relation to speech and writing. That is, written registers occupy both extremes along Dimension 2, with fiction being extremely narrative, while registers such as academic prose and official documents are extremely non-narrative. Several spoken registers are also markedly non-narrative in nature, including broadcasts (which report events actually in progress) and telephone conversations (which deal primarily with immediate concerns). Non-narrative discourse can thus have many purposes with different structural organizations; these registers are similar in their Dimension 2 characteristics in that they lack a narrative organization.

In this introductory treatment, there is space for only a brief introduction to these dimensions of variation. A fuller discussion would show how each dimension is unique in its defining linguistic characteristics, its functional associations, and the similarities and differences among registers that it identifies. One of the main points of the discussion here has been to show how register variation must be considered in a multi-dimensional space – that register comparisons with respect to individual linguistic features, or along a single dimension of variation, are inadequate for a comprehensive understanding of the relationships among registers.

In addition, this discussion has shown how a variety of analytical techniques, applied to a representative text corpus, are required to identify the underlying dimensions of variation in a language. The following sections show how this framework can also be applied to register analyses at different levels of specificity, addressing

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issues which are of concern to teachers and researchers in English for Specific Purposes.

6.4 The need for register characterizations in ESP

In the field of ESP – English for Specific Purposes – researchers and practitioners seek to understand the linguistic characteristics of specialized registers in English. One major goal of such research is to design the best possible materials and activities to help students comprehend and produce these registers appropriately. For example, much attention has been paid to scientific research articles, studying the language forms – such as verb tense and aspect – that are commonly used in them. Traditionally, such studies have been text-based, analyzing the language used in authentic texts. Few, however, have used a corpus-based approach as defined here, using a large, principled collection of texts and combining quantitative and computational techniques with qualitative interpretations.

In addition, most studies in ESP have focused on one or a small number of language features. While these studies give us details about the use of certain forms, they have the drawback discussed earlier in this chapter: studies of a few, selected linguistic features do not provide comprehensive register descriptions, and at times they may even be misleading in their characterization of the relationships among registers. In contrast, the multi-dimensional approach described in the last section provides a more comprehensive description of registers. It therefore can facilitate a major goal of work in ESP.

In this section we illustrate the application of the multi-dimensional analysis of English to ESP studies, examining the relationships between biology and history research articles along two dimensions of variation. Using the MD model of variation described in the last section, we are able to consider the language in these registers from two perspectives. First, the dimensions are used to investigate the linguistic differences between the academic disciplines. These differences are interpreted in relation to a range of situational characteristics, such as subject matter, evidence, and

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methods of presentation. Second, to give a wider perspective on how these specialized registers compare with other registers of English, the biology and history research articles are compared with two more general registers of English: fiction and conversation.[2]

6.4.1 Investigating differences between biology and history research articles

To this point, the analyses in this book have made use of existing corpora to analyze general register categories, such as conversation and academic prose. For studies in ESP, however, a broad sample of texts from academic prose is too general. The analysis in this section focuses on more specific registers, controlling both the academic disciplines and the type of publications that are included. When very specific register comparisons such as this are being made, the first step must often be to compile an appropriate corpus.

The sample analyses in this section use parts of the Corpus of Writing in the Disciplines, a corpus compiled specifically for investigating professional and student writing in different academic disciplines (see Conrad 1996a for details). Biology and history were chosen as the first two disciplines to investigate because they differ from each other in many ways, including subject matter, evidence, and typical methodologies. They thus encompass many of the differences that exist across disciplines. However, within disciplines there are also differences across sub-disciplines (e.g., differences in the kinds of organisms and experiments used in microbiology and ecology), so two subdisciplines were chosen as the focus of study – ecology and American history.

The corpus includes approximately 600,000 words from ecology and American history, distributed over research articles, university-level textbooks, and undergraduate and graduate student writing. Only the research articles were used in the analyses in this section (Table 6.3). For each ecology article, samples were taken from the four major sections: Introduction, Methods, Results, Discussion. The history articles had no consistently marked sections though many had introductions set off from the rest of the article; therefore, samples were taken from the beginning and middle of each article.

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Table 6.3 *Composition of the sub-corpus of biology and history research articles*

Category	No. of texts	Approx. no. of words
Ecology research articles (from *Ecology*, *Journal of Ecology*, and *Journal of Animal Ecology*)	20	64,000
American history research articles (from *The Journal of American History* and *The Western Historical Quarterly*)	20	32,000

The texts in the corpus were scanned onto disk, grammatically tagged (see Methodology Boxes 4 and 5), and edited for errors. Then, the features used in the multi-dimensional analysis of English were counted, normed, and standardized for each of these texts (see Section 6.3 and Methodology Boxes 6 and 10). In this case, there was no need to run a separate factor analysis to identify the dimensions of variation. That is, the dimensions described in Section 6.3 provide a general model of adult speech and writing in English. Since the academic disciplines considered in this section are a specialized sub-domain within that general model, it is possible to analyze the texts from those disciplines in terms of these same dimensions; and it makes sense to compare these specialized registers to the more general registers described above (such as fiction and conversation). Thus, the mean scores for each type of disciplinary text were plotted on each dimension of variation. For the illustration here, only two dimensions are described (see Conrad 1996a for a discussion of all five dimensions).

Figure 6.3 displays the mean scores for the ecology and American history texts along Dimension 2, "Narrative versus non-narrative concerns," compared to overall mean scores for conversation and general fiction.

The Dimension 2 scores reveal two particularly interesting points about the relationship between these registers. First, there is a large difference here between American history articles and

Figure 6.3 Mean scores of ecology and history research articles on Dimension 2, "Narrative versus non-narrative concerns"

ecology articles: the American history texts are more narrative in focus than the ecology texts. That is, the American history articles contain a higher frequency of the positive features on this dimension – past tense verbs, third-person pronouns, perfect aspect verbs, public verbs, etc. (see Table 6.2) – features which are associated with narrative concerns. A second important point, however, is that the history register has only an intermediate score on this

dimension. Although these history articles are more narrative in focus than the ecology texts, their mean scores are far less narrative than registers like general fiction. This finding is particularly noteworthy because previous work has often emphasized the importance of narration in history writing.

Overall, then, this dimension highlights both a difference between the two academic disciplines and the intermediate score of history relative to other more general registers. The interpretive step of multi-dimensional analysis – looking at the communicative function associated with the features – provides insight into the functional reasons for these differences.

First, the difference in the use of narrative language features between the two disciplines seems largely related to the disciplines' subject matter. By definition history is concerned with the past, and all these history texts focus on human events. It is thus logical that history would make use of past tense and third-person pronouns to refer to the past and people's actions. The following sample from a history research article illustrates a typical passage with a high frequency of past tense verbs and third-person pronouns (in italics):

> *Text Sample 6.10: History research article*
> Entertainer Josephine Baker *posed* a special problem for the government. During *her* international concert tours in the 1950s, *she* harshly *criticized* American racism. The United States government could not restrict *her* travel by withdrawing *her* passport because *she carried* the passport of *her* adopted nation, France.

This passage is much like the fiction passage in Text Sample 6.9, concerned with telling the past actions of people.

In contrast to history, the subject matter of ecology is natural-world phenomena. The ecology texts tend more to emphasize general processes and findings about plants, animals, and their environments. Few human subjects appear in these texts, even when the impact of humans is discussed. The ecology texts therefore tend to have fewer third-person pronouns and to use more present tense verbs as they make generic statements. The following example illustrates the use of present tense (in italics) in describing the interactions of animals and their environment:

Text sample 6.11: Ecology research article
The effects of herbivores on boreal forests *are* potentially large and long lasting. How herbivores *affect* nutrient cycles in these forests *is* particularly important because nutrient availability *is* generally low, and changes in nutrient availability *are* major factors driving succession. Furthermore, populations of boreal herbivores *fluctuate* drastically between years and decades . . .

In this sample, interactions are described exclusively in the present tense. Though parts of ecology research articles do use past tense, overall the frequency of narrative features is low compared with history articles.

So far, the findings on this dimension have shown how the subject matter of a discipline can be related to its use of linguistic features. In addition, as noted above, this dimension shows that history articles do not have a particularly high frequency of narrative features when compared with a register like fiction. Rather, the intermediate score of the history articles suggests that they include both narrative and non-narrative concerns.

To better understand how narrative and non-narrative concerns are intermixed in history texts, consider the following two passages from a research article about bison and the Plains Indians. The article opens with a passage that has a high frequency of narrative features (in italics):

Text Sample 6.12: History research article
In bright spring light on the Great Plains of two centuries ago, governor Juan Bautista de Anza *failed* in the last of the three crucial tasks that *his* superiors *had set him* as part of *their* effort to reform New Mexico's Comanche policy. Over half a decade, Anza *had followed* one success with another. *He had* brilliantly *defeated* the formidable Comanche nomnekaht (war leader) Cuerno Verde in 1779, and as a consequence in 1786, *he had* personally *fashioned* the long-sought peace . . .

In essence, this article begins with a story, a recounting of Anza's failure with the Comanche, and typical features of narrative storytelling are used. As the article continues, however, the author identifies a major question he wants to address: whether the Plains ecosystem was in equilibrium. As the following excerpt shows, the

discussion of this problem and how to estimate bison population takes place in the present tense, without third-person pronoun subjects.

> *Text Sample 6.13: History research article continued*
> Answering that question [whether the Plains tribes had established an ecological equilibrium] involves an effort to come to grips with the factors affecting bison populations, the factors affecting Indian populations, and the cultural aspects of Plains Indians' utilization of bison. Each of the three aspects of the question presents puzzles difficult to resolve.
>
> In modern, protected bison herds on the Plains, bison are a prolific species whose numbers increase by an average of 18 percent . . .

In this passage, the author is not presenting historical information, but rather explaining the current problem of the study. None of the features of narrative concerns are present. Similar passages, discussing the problems being addressed or the current importance of historical events, are found in all of the history articles in this corpus. Thus, the intermediate score of the history articles on Dimension 2 reflects the different purposes of passages within these texts. The texts interweave narration of past events with explanation of present problems and significance.

Another perspective on the differences across disciplines can be gained from looking at Dimension 5, "Impersonal versus non-impersonal style." The positive features on this dimension, as listed in Table 6.2, include several passive constructions (*by* passives, agentless passives, past participial reduced relatives), as well as conjuncts and adverbial subordinators (such as *since*, *while*, and *whereas*). The mean scores for the ecology and American history articles, as well as for conversation and fiction, are shown on Figure 6.4. The research articles from both disciplines are highly impersonal in style, with means ranging from 4.0 to 6.7. However, the registers are again differentiated by discipline: the ecology articles are more impersonal in style than history.

The differences in the mean scores again reflect differences in the non-linguistic characteristics of the disciplines. As with Dimension 2, the subject matter differences are important; however, differences in how the disciplines report procedures and

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previous section, history is focused on human events, and many passages concentrate on people or social groups. Therefore, many of the history texts lend themselves to using-active voice verbs, especially with humans as subjects. Such is the case, for example, in Text Samples 6.10 and 6.12 above, about Josephine Baker and de Anza.

As their mean score shows, however, the history articles do use features of impersonal style. Past participial reduced relatives are often used to elaborate a referent, and passive voice is used rather than active voice when abstractions are discussed or when human agents are not the focus of concern. The following passage, for instance, comes from an article that discusses advertisements about runaway servants. The content of the advertisements (rather than the writers) is focused on, and as a result many sentences are in passive voice (passive constructions are in italics):

> *Text Sample 6.14: History research article*
> It is impossible to overstate the role of dress in these advertisements. While no one item of apparel *was noted* as persistently as certain social and physical characteristics, the intensity with which clothing *was denoted* is astonishing. More than three-quarters (75.9 percent) of the cases *sampled* made some mention of garments . . .

We noted above that ecology had less emphasis on the actions of particular human agents. In addition, ecology is distinct from history in giving more explicit accounts of procedures and evidence. The methods and evidence of studies are overtly reported – and this seems related to the ecology texts' more frequent use of features of impersonal style. At first glance, it may be surprising that this is so. You might expect an explicit account of a procedure to be written in active voice, describing the researcher's actions. However, making procedures and evidence overt does not mean that the agents of sentences must be stated. Rather, the agents are often left understood. In fact, passives are particularly salient in methodology sections of research articles.

Consider the following sample, typical of a procedural description in an ecology research article. The authors do at times use first-person pronouns and mention themselves as agents. However, passive voice is the norm (passive constructions are in italics):

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> *Text Sample 6.15: Ecology research article*
> Seeds *were taken* from canopies of hosts *not normally used* and *transferred* into canopies of normal hosts within a 500 m^2 area of the study site. In these experiments, seeds of the other host *were offered* to beetles in their own host microenvironment. Seeds *were transferred* with and without potential competitors present (adults, eggs, and larvae). We also left seeds in donor and recipient plants as controls.

In the last sentence of this passage, the authors use a first-person pronoun subject and active-voice verbs. However, all the preceding sentences (and the sentences following the excerpt here) are in passive voice. It is simply understood that the researchers were the ones to take seeds from canopies of hosts, to transfer them, to offer them to beetles, etc. Although one first-person pronoun is used, the overall impression given by this passage is typical of procedural descriptions in ecology texts: the actors are incidental to the main point, which is to focus on the procedures themselves.

Even this quick review of Dimension 5 shows the usefulness of looking at another dimension of variation for understanding differences across registers. For Dimension 2, subject matter differences were found to relate to history's greater use of narrative features. Examining this dimension has shown that subject matter also relates to history's greater use of active voice. In addition, the ecology articles' greater use of impersonal features is associated with explicit reporting of methods and evidence. The following section shows that a fuller understanding of the ecology texts is possible from considering registers at an even more specific level.

6.4.2 Relationships among the sections of research articles

An important characteristic of many scientific research articles is their explicit organization, with Introduction, Methods, Results, and Discussion sections clearly marked. Though a research article overall has a purpose, each of the sections also has its own sub-purpose. Understanding the variation among the sections and the ways that language use is related to the sections' sub-purposes is

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competition between two insects, and the second is from an article examining differential survival in male and female wild horses (passive-voice verbs are in italics):

> *Text Sample 6.16: Ecology article discussion sections*
> a. Competition for space between these two study insects *was readily documented* for two reasons . . .
> b. The tendency for sex ratios of adults to *be skewed* towards females, therefore, can *be attributed* to differential survival.

In addition, when Discussion sections make statements about other researchers' findings, they often use passive voice, as in this sample from an article investigating variation in flower production:

> *Text Sample 6.17: Ecology article discussion section*
> A comparable pattern of intraplant flower production *has been observed* in the monoccious species Acer gradidientatum . . . Variation within plants in the number of flowers produced per inflorescence *was also reported* by Webb and Bawa . . .

Furthermore, in comparing their findings to previous work, authors often use conjuncts to mark their logical progression explicitly (another of the features of impersonal style). Consider the following passage in which the author compares his results on mice breeding to the results of other studies (conjuncts are italicized):

> *Text Sample 6.18: Ecology article discussion section*
> *However*, their field data from nine females indicate an average of 1.8 litters per season (McCabe and Blanchard 1950: p. 118). Number of litters per season does not appear to vary considerably among Peromyscus, but *rather* the days between litters correlates positively with length of breeding season (Millar 1989). The primary breeding season in this study was around 8 months long (Fig. 1a and b) and average interbirth interval within a season was 60.3 days. *In contrast*, interbirth intervals for other Peromyscus species range from 25 to 30 days . . . Data from P. Californicus *thus* agree with the general trend among Peromyscus for an increased interval between successive litters with increased length of breeding season. There was, *however*, considerable variation in interbirth intervals . . .

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The author repeatedly marks the logical connections he is making, both for contrasts (*however*, *in contrast*, *rather*) and results (*thus*). The connection between his own work and previous work is thus signaled explicitly.

Even this brief illustration focusing on only one dimension shows how we can learn more about registers by considering them at increasingly specific levels. For example, few people would expect ecology research article Discussion sections to use features of impersonal style as commonly as Methods sections. Using multi-dimensional analysis, we see how passive constructions and conjuncts are used together to meet the purposes of Discussion sections.

6.5 Conclusion

Understanding how language varies when we use it for different purposes in different situations is the central research goal of register analysis. This chapter has shown that multi-dimensional analysis, with its corpus-based and computational techniques, is an effective tool for such studies for several reasons. First, it results in more complete characterizations of registers than approaches that focus on individual linguistic features. Even considering just two or three dimensions, as in the examples here, it is possible to see that complex relationships hold among registers, and that a comprehensive understanding requires multiple perspectives. Considering all five dimensions would provide an even more thorough characterization of register variation.

In addition, using multi-dimensional analysis provides a baseline for comparison with other registers. For example, we saw how the comparison of history and ecology research articles required multiple perspectives. When compared to each other, history is considerably more narrative than ecology; but when compared to general fiction, both academic registers are relatively non-narrative. Without the wider perspective on registers in English, it would be tempting to characterize history articles as very narrative – but such a characterization would be misleading. Furthermore,

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7 Language acquisition and development

7.1. Introduction

Research in language acquisition and development has focused on three major areas: 1. the first-language acquisition of very young children; 2. later language development, such as the acquisition of literacy skills, by students at various stages; and 3. second-language acquisition, by children and adults. In all three areas, studies have traditionally been of two major types: experimental or observational. Experimental studies involve the use of a specially designed instrument to try to test directly what a speaker knows at a given stage. In contrast, observational studies are based on the natural language production of learners. As such, observational studies have usually used a kind of corpus-based approach – collecting a series of texts from different stages of acquisition or from different learners, and then analyzing the linguistic patterns in those texts.

For example, observational studies of early first-language acquisition have often been based on recordings of a child made at regular intervals over a period of several months. The transcripts of the child's speech thus comprise the texts for the study. Studies of later language development have sometimes been based on collections of texts from a larger number of students, such as essays written by students in a particular class.

For the most part, however, these observational studies have not employed modern corpus-based techniques. That is, few studies have collected a systematic corpus of texts, entered those texts into

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a computer, and used computational tools to aid in the analysis of language use patterns. As a result, the scope of most earlier studies has been limited in certain ways. First, many studies have adopted a case study approach, focusing on one or two individual learners and providing no basis for generalizations across groups of learners. In addition, studies have typically focused on a small number of linguistic features, leaving the majority of features and co-occurrence patterns among features unexamined. Finally, most studies have also been restricted in the registers that they include, usually concentrating on a single register, such as conversation or narrative writing.

The present chapter illustrates how these limitations in research in first- and second-language development can be overcome with the application of corpus-based techniques. In particular, we emphasize three advantages of the corpus-based approach. First, by including a relatively large collection of texts, corpus-based analyses provide the basis for generalizations concerning groups of speakers and writers at different stages. Second, by investigating the association patterns among sets of linguistic features, corpus-based analyses enable more comprehensive descriptions of language use at different developmental stages. Third, by including texts from multiple registers, corpus-based analyses facilitate broader perspectives on language development.

We illustrate the application of corpus-based techniques with analyses of elementary student[1] speech and writing, covering five research questions. We begin with four questions that focus on first-language issues. The questions move from more traditional language development research – studying individual linguistic features – to a more comprehensive characterization of elementary student language, including multiple linguistic features and multiple registers. This more comprehensive analysis is then applied in two ways – to investigate developmental trends across different levels of students and to compare the language of elementary students with the language of adults. Finally, the fifth research question changes the focus of the chapter to second-language acquisition issues, investigating the errors of non-native English-speaking students.

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Section 7.4.3. The dimensions of variation found for the fifth-grade registers are compared to the adult model of spoken and written register variation that was described in Chapter 6.

5. How do the error patterns of non-native English speakers compare to those of native speakers?

Finally, in Section 7.5 we change the focus of the chapter to second-language development issues, illustrating an area of research of direct interest to ESL instructors and students. We investigate the distribution of selected errors by comparing essays written by native English and native Navajo speakers, in third and sixth grade.

Obviously, there are numerous other areas of research that could be considered in this chapter. Furthermore, given space constraints, even these research questions can be addressed only briefly. Nevertheless, the illustrative analyses give diverse perspectives on the ways that corpus-based techniques can be applied to acquisition and development issues, enabling more generalizable and comprehensive investigations than previously possible.

7.2 Corpora of student language

Most analyses reported in this book are based on publicly available corpora. However, finding corpora suitable for investigating the language of children and students can be difficult. One important source of data for child and student language is the CHILDES database, a collection of corpora for the study of language acquisition. The corpora in CHILDES are particularly useful in two main areas: the language of young children, in the early stages of acquiring their first language; and language collected in controlled settings.

However, there are relatively few publicly available corpora of spoken and written texts produced by language learners in natural settings, and even fewer corpora produced by older children and students. To help fill this gap, Grabe and Reppen developed the Arizona Corpus of Elementary Student Writing, at Northern Arizona University. This corpus includes over 5,000 essays

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Table 7.1 *Corpus of elementary student speech and writing*

	Number of texts	Approx. number of words
Texts published for children		
Basal readers	23	5,000
Science textbooks	25	5,000
Social studies textbooks	24	5,000
Children's fiction	53	16,000
Texts written by children		
Student writing	179	14,000
Texts spoken by children		
Interaction	21	9,000
Monologue	14	8,000
Totals	339	62,000

Children's fiction: from Cornell Corpus, Hays 1988 (CHILDES database). Student writing: from Arizona Corpus of Elementary Student Writing. Spoken texts: from Carterette and Jones 1974, Hicks 1990 (CHILDES database)

collected from forty classes of students living in fifteen towns across the state of Arizona. Students are from three language groups: native English, native Navajo, and native Spanish. Each class of students wrote on nine different topics over the course of each year. These topics were organized by task – for example, expressing a persuasive opinion (e.g., by arguing how much TV children should be allowed to watch); narrating events (e.g., by retelling a story from a series of pictures); and presenting information (e.g., by explaining how to play a game). Students wrote the essays during class time, with no editing by teachers or peers.

In addition, Reppen constructed a separate Corpus of Elementary Student Speech and Writing, comprised of language produced by and for fifth-grade students. This corpus, summarized in Table 7.1, includes texts commonly read by fifth graders and texts

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Table 7.2 *Essay length and average T-unit length (in number of words) across grade levels*

	Third grade		Sixth grade	
	mean	std. dev.	mean	std. dev.
Total words	76.8	48.6	113.5	32.2
T-unit length	9.6	3.9	10.8	3.0

of T-units in a sample of third- and sixth-grade essays from the Arizona Corpus of Elementary Student Writing: 241 third-grade essays and 238 sixth-grade student essays. The topics for the essays were the same across the two grades.

Because student punctuation is not a reliable guide to grammatical structure, it was necessary to mark the T-unit boundaries by hand as the essays were entered into the computer. A computer program was then developed for these counts. The total number of words in each essay was divided by the total number of T-units, to compute the average number of words per T-unit for each essay. The total word count and average number of words per T-unit were then recorded for each essay, and the mean word count and T-unit length for each grade level computed at the end of the analysis.

Table 7.2 presents descriptive statistics for essay length and T-unit length, comparing the essays from the third and sixth grades. This table shows that students in the sixth grade wrote essays that were considerably longer (113.5 words versus 76.8 words). In contrast, average T-unit length barely increases from the third to sixth grade – just a little over one word per unit, from 9.6 to 10.8.[2] Thus, while these students are developing to produce much longer essays, they are not producing more complex structures at the clause/sentence level – at least to the extent that T-unit length measures clause/sentence complexity. It is also interesting to note that at the higher grade level the standard deviation decreases for both measures; that is, students' performance is more tightly clustered around the mean score in sixth grade, while there is more variation among students in the third grade.

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Table 7.3 *Summary of the linguistic features defining each dimension in the model of student register variation (features in parentheses are not used to compute dimension scores)*

Factor 1	
nouns	.719
word length	.701
nominalizations	.558
passives	.427
attributive adj.	.423
prepositions	.415
(type/token ratio	.367)
(once-occurring words	.330)
initial *ands*	−.503
other initial words	−.483
time adverbials	−.453
third-person pronouns	−.413
(adverbials	−.327)
(*it* pronouns	−.323)
(first-person pronouns	−.320)
(participials	−.305)
Factor 2	
once-occurring words	.704
type/token ratio	.671
past tense	.670
public verbs	.666
(perfect aspect	.335)
(verb complements	.320)
present tense verbs	−.540
infinitives	−.376
Factor 3	
causal subordination	.501
that-deletion	.467
private verbs	.447
(present tense	.443)
contractions	.438
do as pro verb	.402
(first-person pronouns	.331)
(necessity modals	.323)
(general emphatics	.322)
(time adverbials	−.405)
Factor 4	
prediction modals	.971
be as main verb	.814
[no negative features]	
Factor 5	
second-person pronouns	.883
conditional subordination	.623
possibility modals	.353
(first-person pronouns	−.305)

The factor analysis for elementary student language identified five important dimensions, summarized in Table 7.3. There is space here only to describe a few of the most interesting findings resulting from this analysis; readers are referred to Reppen (1994) for a fuller discussion.

These dimensions are interpreted in the same way as the dimensions of adult register variation, considering the communicative functions shared by each group of co-occurring features, together with the distribution of registers along the dimension.

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Interpretive labels are assigned to each dimension, summarizing the functional interpretation. All five labels are given below, though our discussion focuses on only the first two dimensions:

Dimension 1: Edited informational discourse versus on-line informational discourse
Dimension 2: Lexically elaborated narrative versus non-narrative
Dimension 3: Involved personal discourse versus non-personal uninvolved discourse
Dimension 4: Projected scenario
Dimension 5: Other-directed idea justification/explanation

Dimension 1 is interpreted as representing "Edited informational discourse versus on-line informational discourse." The positive features on Dimension 1 (see Table 7.3) reflect a dense informational focus. Nouns and nominalizations are the primary referring expressions in English, while attributive adjectives and prepositional phrases provide more information about nominal references and integrate information, creating a denser text. Longer words tend to be more specific in content than shorter words, and passives tend to be common in texts with an informational rather than personal focus. In addition, these features reflect production circumstances that allow for carefully produced text. Thus, the positive end of Dimension 1 is labelled "Edited informational discourse," reflecting the careful integration of ideas with an informational focus.

In contrast, the negative features of Dimension 1 are indicative of on-line production. For example, ideas are often strung together through the use of sentence-initial *and* or *then*. However, as Figure 7.1 shows, the registers with frequent use of the negative features are not necessarily involved or interactive. In fact, spoken monologue – which is focused on conveying information rather than interacting with others – has the most extreme negative score along this dimension. Thus, the interpretive label "On-line informational discourse" can be proposed for the negative end of Dimension 1.

Figure 7.1 shows the distribution of student registers along Dimension 1. Science and social studies textbooks have the largest positive scores, being highly informational and carefully edited. In

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contrast, student essays have a more "on-line" characterization (though they are not as extreme as the spoken registers). Text samples 7.1 and 7.2 illustrate the differences between textbook language and student essays with respect to this dimension. In these texts, selected positive and negative features have been highlighted: among the positive features, passive constructions appear in bold face, and prepositional phrases appear in square brackets; among the negative features, sentence-initial *and* and third-person pronouns are italicized.

Text Sample 7.1: Social studies textbook
Slavery [in the Americas] grew out [of a need] [for cheap labor]. The large farms or plantations that **had been founded** [in the Americas] needed many workers. [At first] the Spanish tried to force American Indians [into slavery]. When this did not work *they* brought slaves [from Africa]. [By 1600], there were 40,000 Black slaves [in Spanish America]. [In 1619] the first Africans **were brought** [to English settlements] [in America].

Text Sample 7.2: Student essay
One day some kids went [to a circus]. *And* a clown was catching plates. *And he* did not drop one. *And* then the kids put up the plates where *they* belonged. *And* then the kid thought *he* could do it. *but he* was not as good as the clown who did it. These were *his* mom's plates. *And he* tried. *He* got it going. *And* then *he* dropped *them*. *And they* all broke. *His* mom came in. *She* was very mad.

The student essay has an extremely high number of initial *and*s and third-person pronouns (which are often unclear in their intended reference). Even though this is a written text, it has many characteristics of on-line production. In contrast, the social studies text has none of these characteristics of on-line production, but it does have a dense use of prepositional phrases and passive constructions. Ideas are carefully integrated into this text, reflecting the informational focus combined with careful production circumstances.

The on-line characteristics of student essays can be related in part to the fact that they are all in-class writings, produced in a single sitting. (No take-home or multi-draft essays were included, to avoid the confounding influence of teachers modifying student

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analysis of individual features are clear. We can obtain a more thorough characterization of elementary student language, seeing the patterns of use for sets of linguistic features and comparing the language used in student essays with other registers. In the next section we show how this multi-dimensional analysis can be applied to study language development for student writing.

7.4.2 Writing development considered from a multi-dimensional perspective

In the latter part of Chapter 6, we showed how the adult multi-dimensional model of register variation could be used to describe specialized registers that were not included in the original analysis. In the same way, the student model of register variation (described in the last section) can be used to describe and compare other student registers. Perhaps the most interesting application of this type is to compare student writing from different developmental stages. In the present section, we illustrate this type of analysis by comparing essays written by third- and sixth-grade students.

A repeated theme throughout this book has been the centrality of register: that different registers are typically distinctive in systematic ways. This same perspective can be applied to the study of student writing, by considering essays written in response to different tasks. That is, whereas many earlier studies considered only a single task (e.g., narratives), the corpus used for the study of student writing is structured around several different writing tasks (e.g., narration, descriptions, explanations, persuasive opinions). In the present section, we focus on three specific assignments representing different tasks: "T.V. opinion essay," "Favorite game explanation," and "Picture series narrative." These assignments represent a range of the different types of writing that students encounter in school. The first assignment involved taking a position and providing reasons for the position; the second assignment involved explaining a game to someone; and the third assignment involved telling a story based on a series of pictures.

As described in the previous section, both Dimensions 1 and 2 in the student model define basic oral/literate oppositions: they

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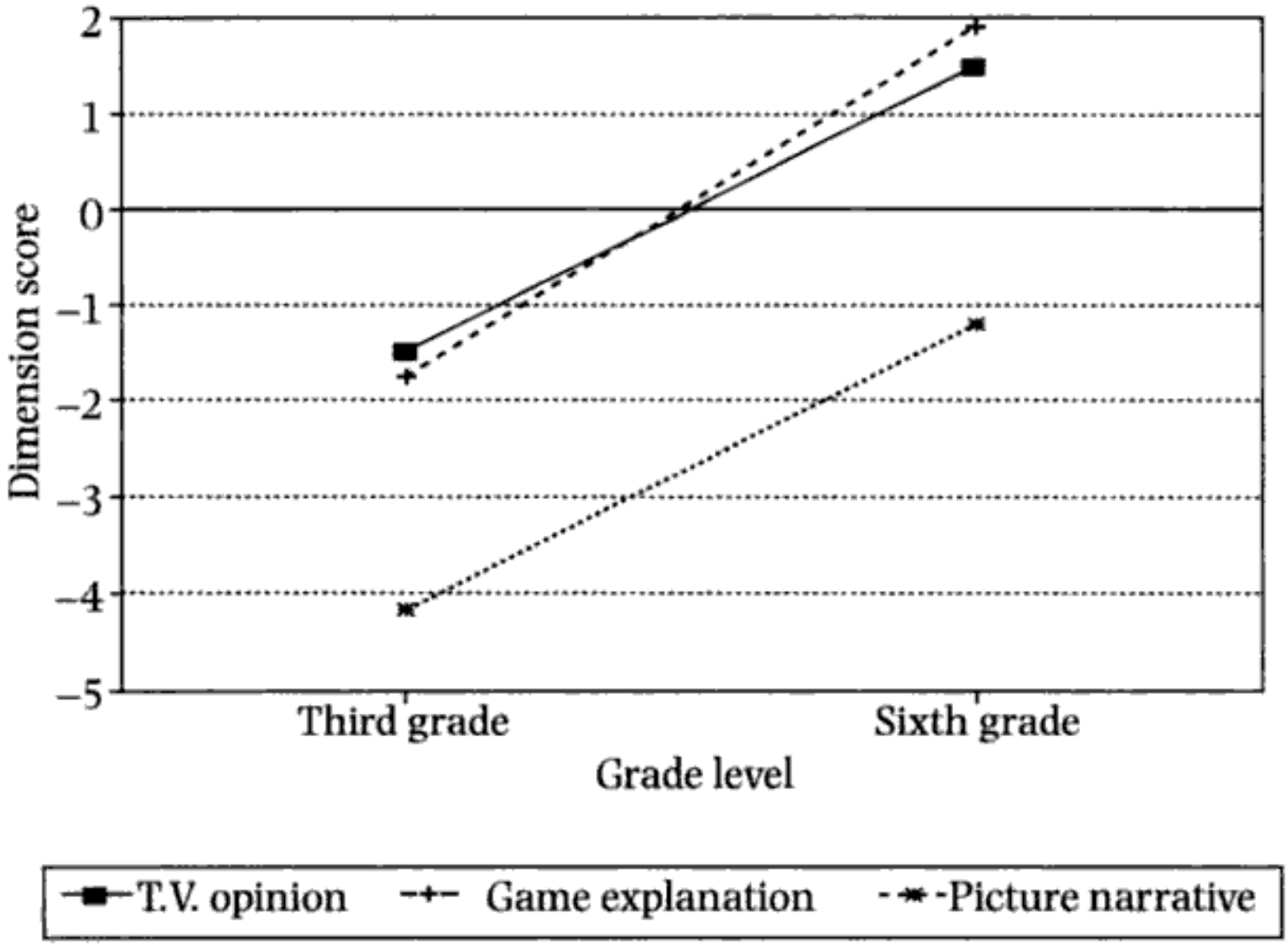

Figure 7.3 Dimension 1 scores for three tasks, comparing third- and sixth-grade essays

distinguish between published written registers at one pole and spoken registers at the other pole. Overall, student writing tends towards the oral pole of both dimensions. For these reasons, these two dimensions should also prove to be useful for comparing student writing across grade levels, investigating whether students show a developmental trend from oral to literate styles.

Figures 7.3 and 7.4 plot the change in Dimension 1 and Dimension 2 scores between third- and sixth-grade essays. The figures show that the developmental trend is indeed towards the use of a more "literate" style. This trend is most consistent along Dimension 1 (Figure 7.3). On all three writing tasks, students show a strong shift to more informational and carefully edited styles from the third to the sixth grades (i.e., to higher positive dimension scores).

The changes along Dimension 2 are slightly more complicated (Figure 7.4). Students show a marked developmental change in writing the T.V. opinion and game explanation essays, decreasing their use of non-narrative characteristics and increasing in lexically elaborated narrative characteristics (i.e., moving towards more

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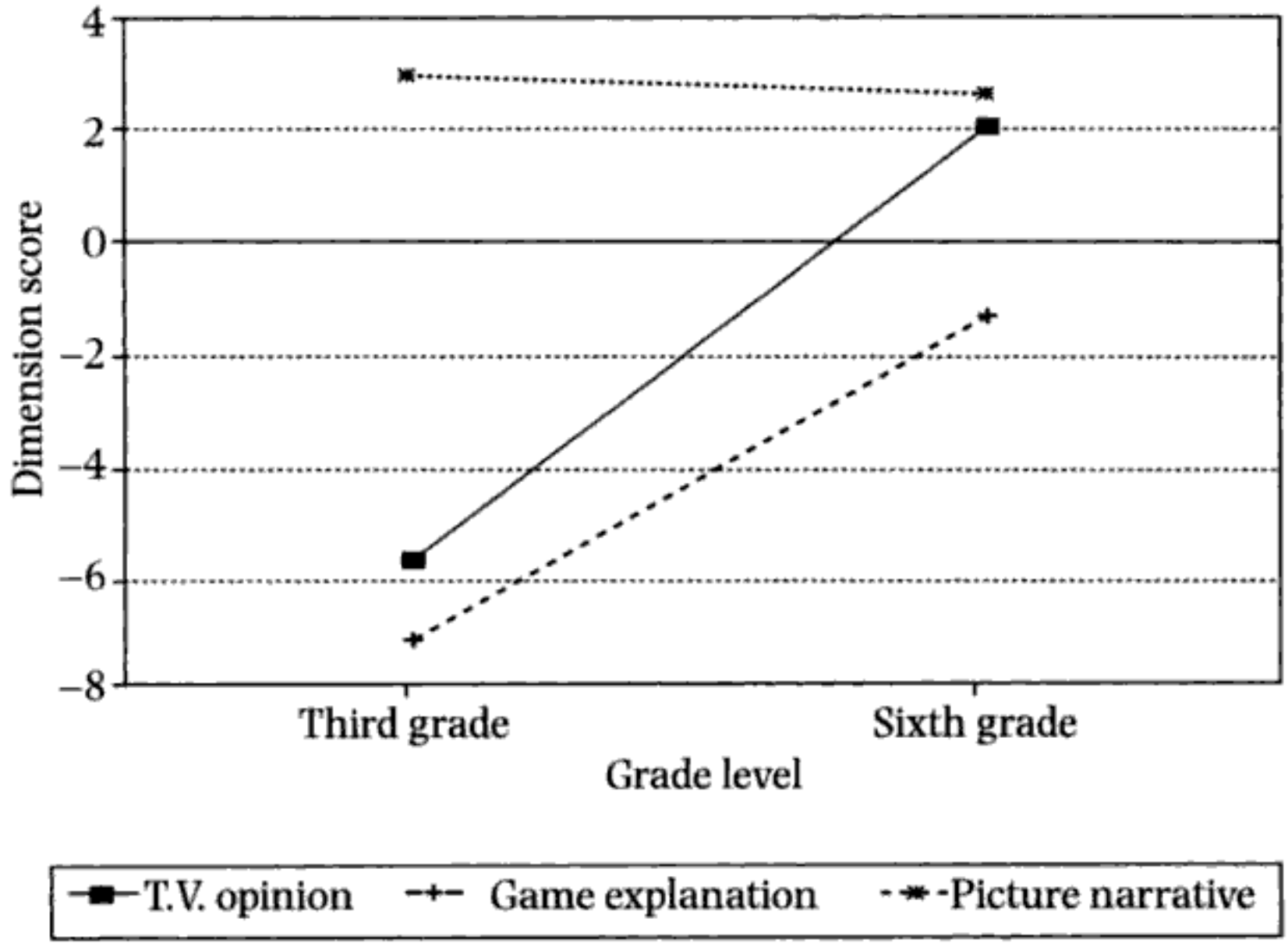

Figure 7.4 Dimension 2 scores for three tasks, comparing third- and sixth-grade essays

positive scores on the dimension). Most of this change can be attributed to the development of greater lexical elaboration by the sixth grade. However, students in the third grade are already using extensive narrative features for the picture series task, and students in the sixth grade continue to employ similar characteristics for that task.

These developmental changes along Dimensions 1 and 2 can be illustrated from the following two game explanation essays.

Text Sample 7.4: Third-grade favorite game explanation
My favorite game is Namo. You write at least 20 names on a board. And then you write 5 names on paper. Then someone puts a checkmark on a name on the board. And if you have that name on your paper mark it with a checkmark. Then keep on doing it until someone gets all the names are marked off. And after someone has done it then they say Namo.

Text Sample 7.5: Sixth-grade favorite game explanation
I am going to explain how to play a game called Capture the Flag. You can have as many players as you want to. You put

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> a flag at either side of the court. There is an imaginary line dividing the court. The people on the 2 teams are trying to get each other out. They do this by tagging each other. You can only tag someone if they are on your side of the court. Once you tag people, they are in your jail. The only way that they can get out is to have one of their teammates come and tag them.
>
> The object of the game is to get the opposite team's flag back to your side safely.

The third-grade essay is minimally developed, with most sentences beginning with a simple sentence-initial coordinator (*and* or *then*). Some references in the essay are vague (e.g., the use of *doing it* and *has done it* in the last two sentences), and there is considerable lexical repetition. Characteristics such as these result in the relatively "oral" scores for third-grade game explanations along both Dimensions 1 and 2.

In contrast, the sixth-grade essay has none of the "on-line" features from Dimension 1, but it has a considerable number of the "edited, informational" features, such as attributive adjectives (e.g., *imaginary line, opposite team*) and prepositional phrases (e.g., *at either side of the court, on the 2 teams, in your jail, to your side*). This essay also has a relatively elaborated vocabulary, with a number of longer words (e.g., *imaginary, teammates, opposite*) and considerable lexical variety.

Figures 7.3 and 7.4 reveal an additional important fact about writing development: that students as early as the third grade have acquired some control over the patterns of register variation, being able to distinguish among the linguistic demands of different writing tasks. Thus, by the third grade the "Favorite game explanation" and the "T.V. opinion essay" are linguistically quite different from the "Picture series narrative," with respect to both Dimensions 1 and 2. The two essays below are written by the same third-grade student; they illustrate awareness of the differences between the explanation and narration tasks and the ability to control the linguistic resources required for these tasks. For example, the "Picture series narrative" begins with the classic marker of a story – "Once upon a time" – and it is framed in the past tense, while the "Game explanation" is framed entirely in the present tense. (Past tense forms in the following two text samples are italicized.)

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Text Sample 7.6: Third-grade favorite game explanation
My favorite sport is soccer. And this is how it's played. First there are the positions. Soccer has a goalie, defenders, halfbacks, and forwards. In soccer kids can't touch the ball with hands. Only the goalie can touch the ball. And if the ball goes past a line a player has to throw it in. And the goalie tries to stop the ball. And you also goal kick.

Text Sample 7.7: Third-grade picture series narrative
Once upon a time four boys *were* watching a clown juggle plates. Danny one of the four boys *went* home fascinated. Another boy named Josh *came* home with Danny. When the boys *got* home Danny *was* going to try to juggle plates too. Danny *thought* he could do it because he could juggle balls. He *started* off good. But he accidently *dropped* the plates. Then Jamie, Danny's mom *walked* in the kitchen. And Jamie *was* furious. Danny *was* grounded for 1 week.

Additional aspects of writing development could be discussed relative to the patterns along Dimensions 3–5. However, even consideration of the first two dimensions has identified several important developmental changes that occur from third to sixth grade. For example, along both dimensions there is a steady progression away from more oral-like language and towards more literate forms. These dimensions also show that third-grade students are already beginning to control some of the linguistic resources needed to write for different purposes.

7.4.3 Comparison of elementary student and adult dimensions of variation

The multi-dimensional model of elementary student language provides a view into the discourse universe that elementary students experience – the sets of linguistic features that co-occur, the functional and linguistic parameters that are important in distinguishing among texts, the registers that have marked characteristics relative to other registers. The multi-dimensional model of adult English (described in Chapter 6) provides a similar view into the universe of adult discourse. It is thus possible to compare the elementary student and adult domains of discourse in terms of

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these models, providing a broader perspective on the relationships between adult and student discourse than has been possible before.

In this section we briefly illustrate how different multidimensional models can be compared. Specifically, we show how interesting insights can be gained both from comparisons of particular dimensions as well as from a comparison of the models overall.

Table 7.4 presents an overview of the first two dimensions in the adult and elementary student models, summarizing the functions, linguistic features, and registers that characterize each dimension. As you can see, in a general sense these two dimensions are similar in the two models.

In both models, Dimension 1 distinguishes between oral and written registers with respect to informational focus and planning considerations.[3] Functionally, Dimension 1 is associated with the careful production of highly integrated information, on the one hand, and on-line production on the other. Linguistically, the negative pole of this dimension in both models is defined by features such as nouns, attributive adjectives, prepositions, and long words. And in terms of register distribution, both dimensions differentiate between spoken texts versus informational written exposition.

Similarly, Dimension 2 is similar in many respects across the two models. In both models, this dimension distinguishes between narrative and non-narrative discourse, in terms of many of the same linguistic features: past tense and speech act verbs versus present tense. Furthermore, in both models, fiction is particularly marked as a narrative register.

However, even these similar dimensions also show that there are important differences between the two domains of discourse. In particular, each of these dimensions incorporates functions in one domain that are not present in the other. For example, Dimension 1 in the adult model distinguishes between interactive and non-interactive discourse, both in terms of the defining linguistic features and the register distribution. In contrast, Dimension 1 in the student model has no relation to interactiveness at all. It lacks those linguistic features with interactive functions, and spoken interactions are not especially marked along the dimension.

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Table 7.4 *Comparison of two adult and elementary student dimensions of variation*
(The polarity of Dimension 1 in the student model has been reversed – see Note 3.)

	Dimension 1	
	Adult	Elementary student
Functions		
Positive pole:	Interactive (Inter) personal focus Involved Personal stance On-line production	On-line production Non-interactive Informational Careful production Informational focus Integrated information
Negative pole:	Monologue Careful production Informational focus Integrated information	
Linguistic features		
Positive pole:	1st and 2nd-person pronouns, questions, reductions, stance verbs, hedges, emphatics, adverbial subordination	initial *ands*, initial non-*ands*, time adverbials, 3rd-person pronouns
Negative pole:	nouns, adjectives, prepositions, long words	nouns, long words, nominalizations, passive, attributive adjectives, prepositions
Characteristic registers		
Positive pole:	conversations personal letters public conversations	monologue student essays
Negative pole:	informational exposition (e.g., official documents, academic prose)	social studies and science textbooks

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Table 7.4 *Cont'd*

	Dimension 2	
	Adult	Elementary student
Functions		
Positive pole:	Narrative	Narrative Lexical elaboration
Negative pole:	Non-narrative	Non-narrative
Linguistic features		
Positive pole:	past tense, perfect aspect, 3rd-person pronouns, speech act (public) verbs	past tense, once-occurring words, high type-token ratio, speech act (public) verbs
Negative pole:	present tense, attributive adjectives	present tense, infinitives
Characteristic registers		
Positive pole:	fiction	fiction basal readers
Negative pole:	exposition broadcasts conversation	monologues opinion and explanation essays

Dimension 2 illustrates the opposite case from Dimension 1; that is, the student model includes a function absent from the adult model. Specifically, Dimension 2 in the student model reflects lexical elaboration in addition to narrative focus, while Dimension 2 in the adult model reflects only a narrative focus. Thus, features of lexical specificity (once-occurring words and a high type-token ratio) appear on this dimension in the student model, but such features are not on this dimension in the adult model.

Comparison of these two dimensions across the models raises several interesting issues. The Dimension 1 differences, for example, raise questions about elementary students' control of

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Text Sample 7.9: Sixth-grade English L1 student essay
I watch a little tv a day and some of my friends watch a lot of t.v. Now to think of it I watch a – lot of tv. I watch it at night about 3 hours of it. I stay up real late. and I watch it after school.

These two essays are roughly equivalent in overall fluency or other quality measures. Thus, the greater error-rate by native Navajo speakers does not necessarily reflect a lower literacy competence overall. Nevertheless, the presence of errors such as these can be distracting and often leads to a lower overall evaluation by teachers.

In addition to simple frequency counts, a variety of other measures could complete a study such as this one. A useful next step, for example, would be to determine the distribution of errors across students. This, too, is easy to do with computational analyses, since the computer can record the number of errors in each text. Questions that could be addressed include: Are the errors evenly distributed across students or do a few students account for most of the errors? Does the distribution differ between the third and sixth grades? Examination of the errors in context (e.g., with KWIC files) would also provide valuable information for questions such as: In what grammatical or phonological contexts do these students make noun and verb morphology errors?

Though we end this sample analysis here, it is clear that the analysis could be expanded in many ways to increase our understanding of error patterns in second-language student writing. Furthermore, studies such as this one can be done on a larger scale, investigating more error types, different language groups, error distribution across different registers, the effect of different kinds of instruction, and a host of other issues.[5]

7.6 Conclusion

This chapter has illustrated the use of corpus-based analyses to investigate linguistic issues relating to language acquisition and development. We initially concentrated on first-language development, moving from a traditional approach that investigates individual linguistic features to a more comprehensive multi-dimensional model of elementary student language. The model

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Further reading

Relatively few corpus-based studies have been carried out to investigate children's first-language acquisition, and even fewer corpus-based studies have focused on elementary student language or second-language acquisition. MacWhinney (1991) lists the sub-corpora included in the CHILDES database and summarizes studies based on those corpora. This manual contains many resources for corpus-based studies. In addition, a few studies have used a multi-dimensional analysis to investigate variation among elementary student spoken and written registers (Biber 1991; Reppen 1994, 1995).

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8 Historical and stylistic investigations

8.1 Stylistic and historical studies

Some of the earliest applications of corpus-based analytical techniques were for the study of literary style. By the late 1960s, a number of literary scholars were applying computational techniques to on-line text collections, to investigate the styles of authors, genres, and historical periods. This early history is somewhat less surprising when we consider the fact that literary scholars have always been interested in the language of texts. It was thus a natural development to move those texts onto computer, allowing the use of concordancing software and other computational tools for analysis.

More recently, corpus-based analytical techniques have become popular for studies in historical linguistics. This, too, has been a natural development, given that historical linguists have always relied on text collections from earlier periods to trace historical change. However, a major problem for corpus-based historical investigations has been the absence of representative historical corpora. While literary scholars have often focused on the works of an individual author, historical linguists require corpora that represent a range of texts from multiple genres, across historical periods. Compiling such historical corpora has presented many challenges (see Methodology Box 2).

Diachronic text corpora enable a multitude of investigations. In fact, any of the kinds of research questions that have been discussed in the previous chapters of this book can also be studied from a historical perspective using diachronic text corpora.

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Specifically, a diachronic corpus can be used to study lexicographic, grammatical, and discourse features, as in Chapters 2–5, as well as registers and dialects, as in Chapters 6–7. Stylistic perspectives can also be included by focusing on individual authors.

In this chapter we illustrate several of these different kinds of analysis. For the first research question, we consider a combination of lexical and grammatical issues:

1. How have the patterns of use for modal verbs versus semi-modal verbs shifted over the last three centuries (in the domain of obligation/necessity modality)? In addition, do the patterns of use for individual verbs differ from these overall patterns?

This question is addressed in Section 8.2 by considering first the overall classes of obligation/necessity modal verbs and semi-modal verbs, and then breaking down the analysis for individual verbs – *must* and *should* as modal verbs, and *have to*, *got to*, *need to*, *ought to*, and *be supposed to* as semi-modals.

Then, we consider two research questions investigating varieties, one relating to register variation, and one relating to a type of dialect variation (gender):

2. How have written and speech-based registers changed in their linguistic characteristics over the last three centuries?

Specifically, in Section 8.3 we consider the development of medical research articles relative to drama (representing a speech-based register).

3. How has the language of women and men changed over the last three centuries?

Specifically, we analyze changes in the linguistic characteristics of personal letters written by men and women, also considering whether the letter was addressed to a woman or man. This question is addressed in Section 8.4.

Finally, we briefly consider a research question relating to individual style:

4. What are the linguistic characteristics of particular texts written by an author, when considered relative to a range of other texts from the same period?

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In Section 8.5, we compare passages written by Johnson with the range of fiction written by his contemporaries in the eighteenth century.

8.2 Diachronic shifts in the use of grammatical and lexical features

Diachronic corpora have proven to be particularly influential for lexical and grammatical studies in historical linguistics. For example, although work on compiling the Helsinki Corpus began only in 1984, by 1993 Kytö could already list about 100 published articles and monographs that were based on analysis of the Helsinki Corpus – and that number has increased dramatically over the last few years. For studies of this type, a diachronic corpus provides a number of invaluable resources, including: a large collection of texts from across historical periods; texts from several different registers; and the ability to search for selected grammatical and lexical constructions using computational tools.

The present section provides a brief example of this type of study, considering the parallel development of modal verbs and semi-modal verbs marking obligation/necessity. Two modal verbs are used in this semantic domain: *must*, marking a strong sense of personal obligation or logical necessity, and *should*, marking a weaker sense. For example:

(1) Marking obligation:
   a. Now we *must* look to the next stage of development. (News)
   b. You *shouldn't* go to bed too early. (Conversation)

(2) Marking logical necessity:
   a. As Sherlock Holmes said, "When you have eliminated the impossible, whatever remains, however improbable, *must* be the truth." (Academic prose)
   b. He is able to specify a rule for what that amplitude *should* be. (Academic prose)

Semi-modals can be used to mark similar meaning distinctions: the semi-modals *have to*, *got to*, and *need to* mark a strong sense of personal obligation or logical necessity, and the semi-modals *ought to* and *be supposed to* mark a weaker sense. For example,

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Table 8.1 *Corpus used in the analysis of modal and semi-modal verbs over four centuries*

Category	Number of texts	Approx. number of words
Historical texts		
News (ARCHER)	100	100,000
Fiction (ARCHER)	110	100,000
Drama (ARCHER)	95	100,000
Contemporary texts		
News (BNC)	280	5,000,000
Fiction (Longman–Lancaster)	145	5,000,000
Conversation (BNC)	165	5,000,000

	Modal total	Semi-modal total
News		
Century		
17th	**	–
18th	****	*
19th	*****	–
20th	*******	*
1990	****	****
Fiction		
Century		
17th	********	–
18th	*********	*
19th	*******	*
20th	******	*****
1990	******	*****
Drama		
Century		
17th	**********	*
18th	**********	–
19th	**********	**
20th	**********	******
1990 conv.	******	**********

Figure 8.1 Relative frequencies of modal and semi-modal grammatical classes marking obligation/necessity across four centuries, in three registers (each * = more than three per 10,000; – = occurs less than three per 10,000)

Szerzői jogi védelem alatt álló anyag

News

Century	must	should	have to	got to	need to	ought to	supposed to
17th	*	.	.		–	–	–
18th	*	**	.	.	–	.	.
19th	**	***	.	–	–	.	–
20th	***	***	*	–	–	–	–
1990	.	***	**	–	.	–	–

Fiction

Century	must	should	have to	got to	need to	ought to	supposed to
17th	**	*****	–		.	.	–
18th	****	*****	.		–	.	–
19th	***	****	*	–	–	.	.
20th	****	**	**	.	.	*	–
1990	***	**	***	.	.	.	.

Drama

Century	must	should	have to	got to	need to	ought to	supposed to
17th	**********	***	.		–	*	–
18th	*******	*****	.		–	.	–
19th	**********	***	*	.	–	.	.
20th	********	***	***	**	.	.	.
1990 conv.	**	***	******	**	.	.	.

Figure 8.2 Relative frequencies of individual modals and semi-modals marking obligation/necessity across four centuries, in three registers (each * = more than three per 10,000; . = occurs less than three per 10,000; – = attested in the *OED* but not occurring in ARCHER)

centuries, but that relationship has been changing, with semi-modals becoming increasingly common. The frequencies for the modal verbs have remained relatively high across the periods, although there has been a slight decrease in the present century (especially in drama/conversation). In contrast, semi-modals have shown a gradual increase across the centuries, with a more notable increase more recently. By the contemporary period (1990), the overall frequencies for semi-modals are as high as that for modals in news and only slightly less in fiction. In conversation, semi-modals have actually become more common than modal verbs.

However, when we take a more lexicographic perspective on these verbs, as in Figure 8.2, we find that they do not all follow the same patterns of use. Further, the forms that are attested earliest are not the most common forms at present. Specifically, the

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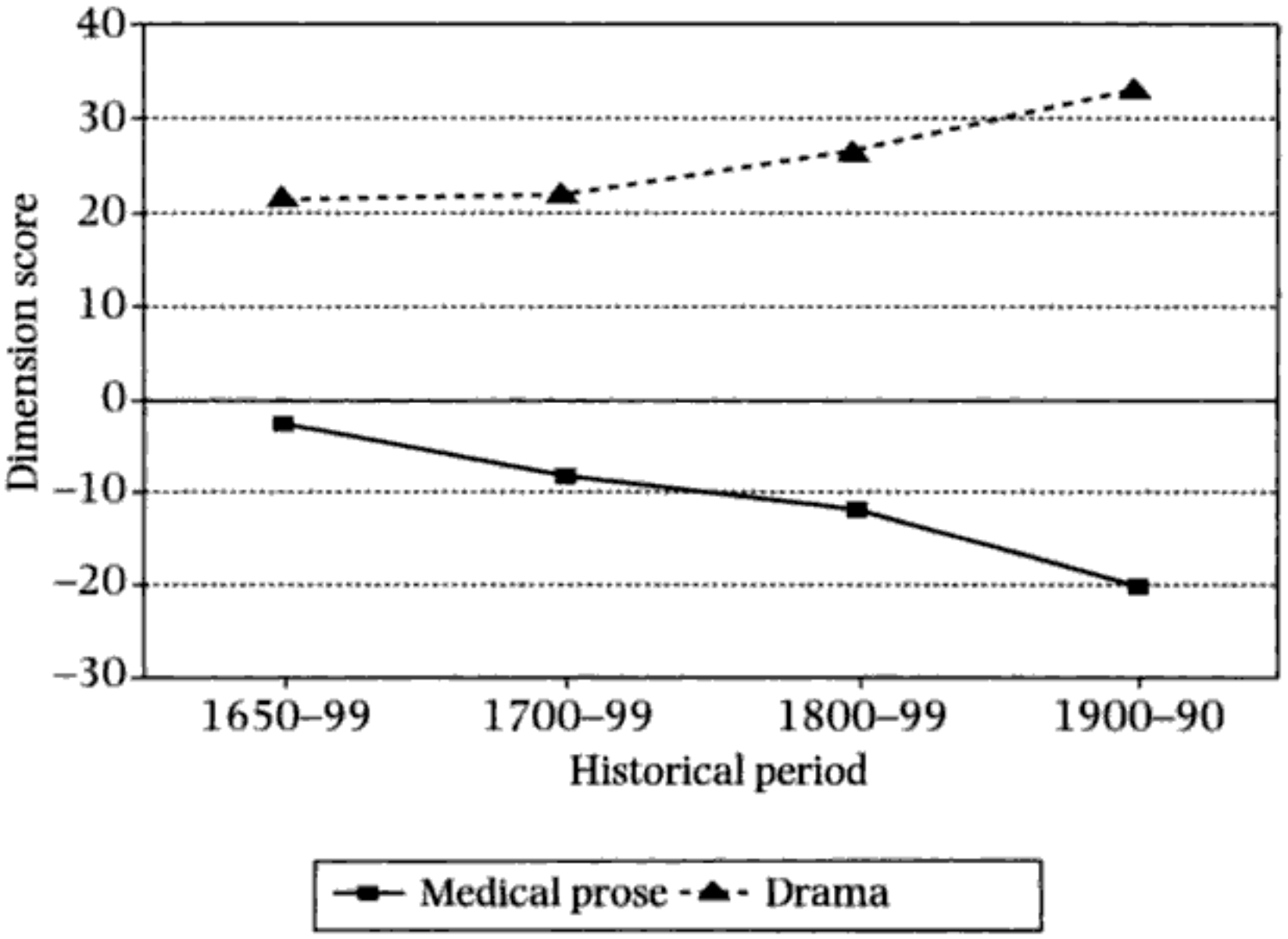

Figure 8.3 Historical change along English Dimension 1, "Involved versus informational style"

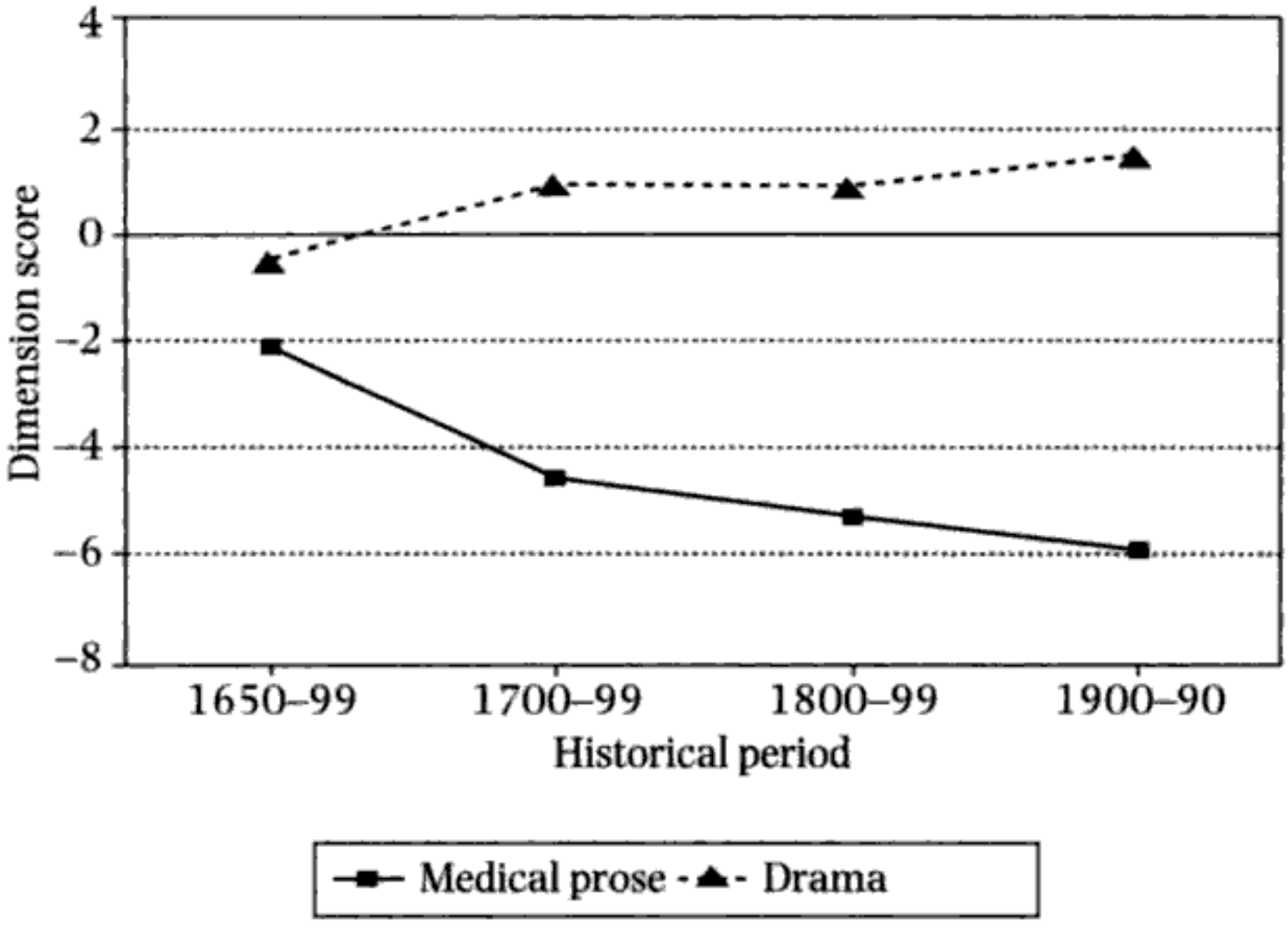

Figure 8.4 Historical change along English Dimension 5, "Non-impersonal versus impersonal style"[1]

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Medical prose has evolved in the exact opposite direction from drama: towards a very dense packaging of information and the extensive use of passive constructions. However, seventeenth-century medical prose was much more colloquial in style. In fact, many seventeenth-century medical texts were written as letters submitted to the editor and published with little alteration.

For example, Text Sample 8.7 is from a letter written in 1685 by Brown, published under the title of "A Remarkable Account of a Liver, appearing Glandulous to the Eye" in the *Philosophical Transactions of the Royal Society of London*. This sample illustrates many of the typical features of seventeenth-century medical prose. It has an intermediate position along Dimension 1, combining the use of involved, interactive features (such as first- and second-person pronouns, and verbs of cognition like *thought*) with a carefully selected vocabulary and a relatively dense use of nouns and other informational features. With respect to Dimension 5 features, seventeenth-century medical prose was already moderately impersonal in style, as illustrated by the use of passive constructions in Sample 8.7 (*was accurately taken*; *was observed*).

Text Sample 8.7: Seventeenth-century medical prose

Sir,

I send you here the figure of the Liver of an Hydropical Person, a Patient of our Hospitall, (as it was accurately taken by Mr Faithorn,) which I thought so curious as to be worthy the notice of the Royal Society, and therefore have presumed to communicate it to you.

The Person was about 25 years of age, a Soldier in one of his Majesty's Regiments here in Town; who contracted his distemper by drinking much water, when he could not stirr from his duty, and catching cold at nights in being upon the guard: He was under the care of our Physicians for some time, by whose directions his swellings did by times abate; but afterwards it was observed, that the method which had been beneficiall to others, had not here the like success, his swellings returning upon him as before . . .

By the twentieth century, medical prose developed extreme characterizations along both Dimensions 1 and 5: a very dense packaging of information and an extensive use of passive constructions.

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For example, Text Sample 8.8 is from an article published in the *New England Journal of Medicine* in 1985. There are essentially no references to the writer or reader in this article but a very dense use of nouns, prepositional phrases, and an extremely varied, highly technical vocabulary. At the same time, passive constructions are commonly used in this text (*are split*; *are designated* – twice; *is rapidly cleaved*).

> *Text Sample 8.8: Twentieth-century medical prose*
> During the sequential activation of the complement components of the alternative pathway by a variety of surfaces, such as new cuprophane membranes or yeast-cell walls, peptides are split from the third (C3) and fifth (C5) components of complement. These peptides are designated C3a and C5a, respectively, and the latter has the capacity to induce leukopenia and pulmonary leukostasis in vivo and to aggregate granulocytes in vitro. The carboxy terminal amino acid arginine, of both peptides C3a and C5a, is rapidly cleaved by the carboxypeptidase N of the blood, and the resultant stable peptides are designated C3a desArg and C5a desArg, respectively.

Register comparisons such as these would simply not be possible without the use of a representative diachronic corpus, coupled with the use of computational tools for analysis. A complete analysis of historical register variation would be based on a full range of written and speech-based registers, considering all dimensions of variation. However, even the restricted analyses presented in this section show the dramatic changes that can occur within and across registers over time.

8.4 Dialect change: historical changes in the language of women and men

In addition to register studies, a diachronic corpus can also be used to investigate the changing characteristics of dialects over time. Dialect studies have typically focused on geographical differences, usually analyzing phonological rather than grammatical characteristics. However, the use of corpus-based techniques enables consideration of a full range of grammatical characteristics (in addition to phonological features), and it opens the possibility

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Table 8.3 *Texts included in the analysis of personal letters by females and males across four centuries*

	Number of texts for each century			
Category	17th C	18th C	19th C	20th C
F to F	2	8	8	7
F to M	5	8	6	7
M to F	3	24	24	30
M to M	14	52	4	36
Approx. no. of words/century	12,000	38,000	40,000	33,000

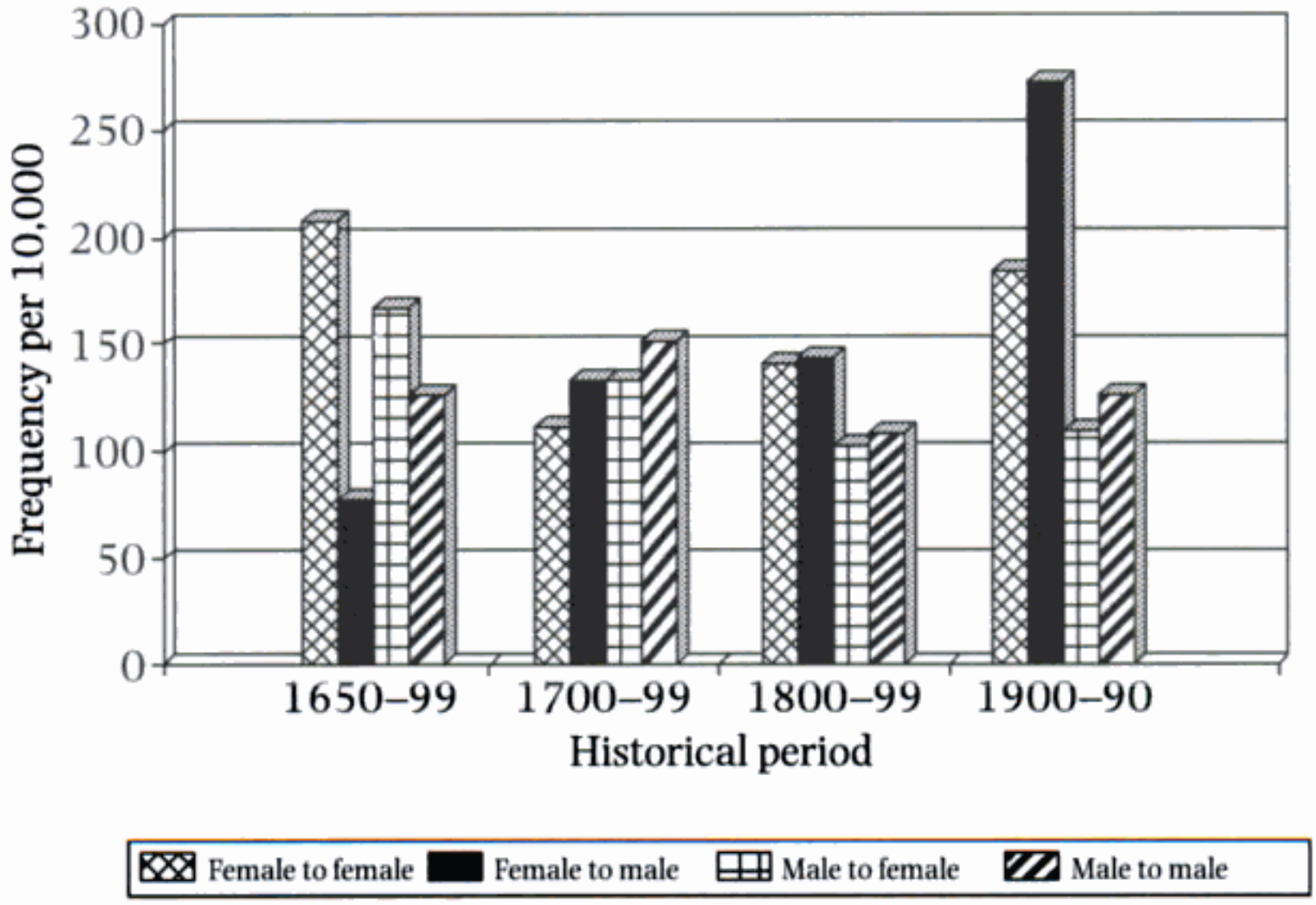

Figure 8.5 Historical change in the use of general emphatics, by gender

Figure 8.5 displays the frequency of emphatics in the four categories of letters across the four historical periods. Certain trends are apparent from the figure. First, in the nineteenth and twentieth centuries, letters written by females do contain a higher frequency of emphatics. This is true both for letters addressed to other females and to males. The difference is far greatest for the

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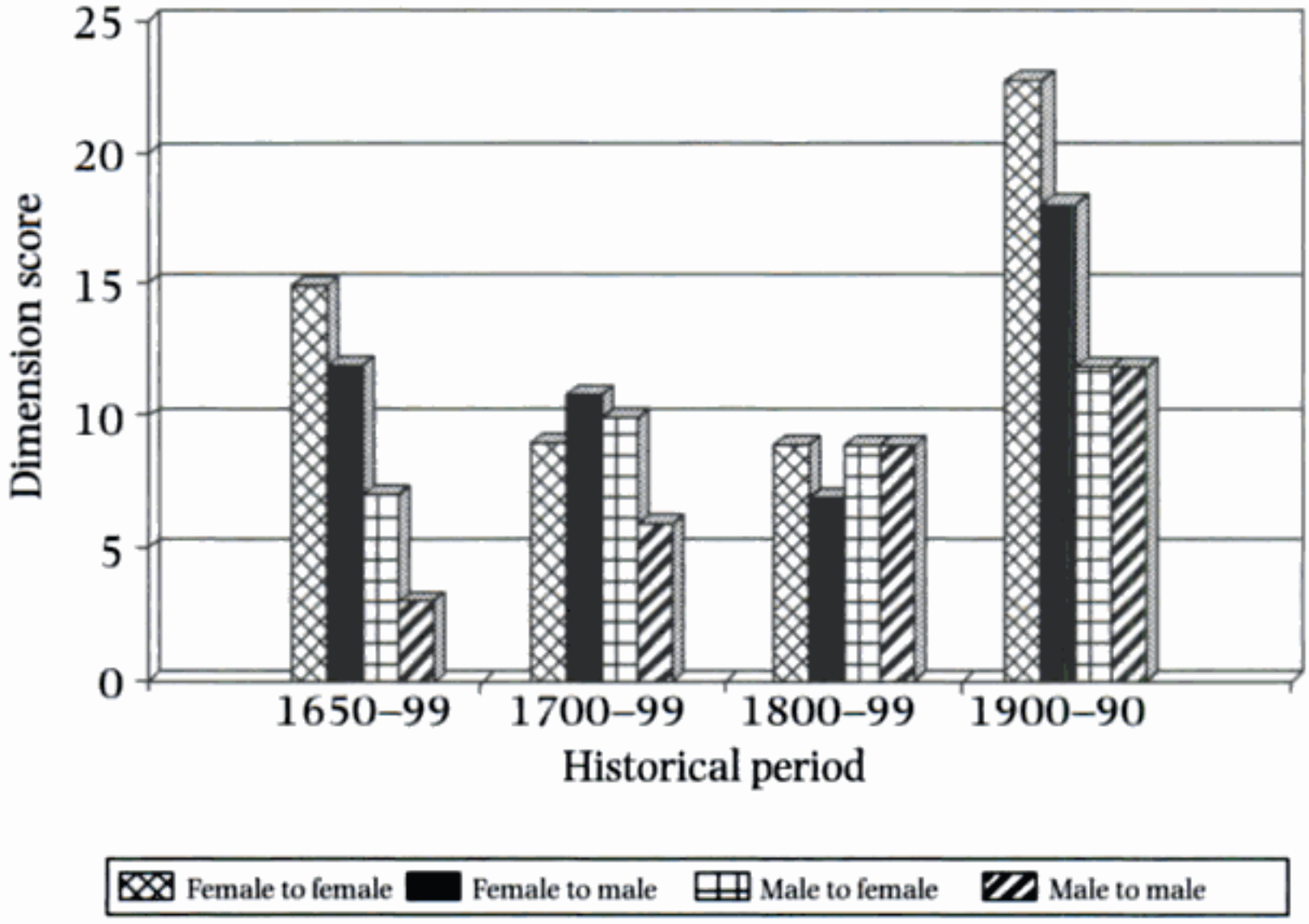

Figure 8.6 Historical change in letters, by gender, along Dimension 1, "Involved versus informational style"

periods have positive Dimension 1 scores, reflecting the fact that they are all relatively "involved." Thus, these letters all have relatively frequent occurrences of private verbs, *that*-omissions, contractions, present tense verbs, first- and second-person pronouns, etc. (i.e., the positive features on Dimension 1), together with relatively infrequent occurrences of nouns, long words, prepositions, etc. (i.e., the negative features on Dimension 1). Historically, this pattern becomes more pronounced in the twentieth century, when letters from all four gender categories become more involved than in earlier centuries.

Beyond those overall patterns, Figure 8.6 also shows systematic differences among the four gender categories. In particular, letters from the seventeenth century and those from the twentieth century show two important patterns of variation: letters written by women are more involved than letters written by men; and, holding author gender constant, letters written to women are more involved than letters written to men. In the seventeenth century, these factors result in consistent differences across all four gender groups, with female-to-female letters being most involved,

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and male-to-male letters least involved (and most informational). Female-to-male letters are somewhat less involved than female-to-female letters, while male-to-female letters are somewhat more involved than male-to-male letters. These same patterns are found across the gender groups in the twentieth century, except that men show the same degree of involvement whether writing to women or men. (In the eighteenth and nineteenth centuries, gender differences among these letters were less consistent and generally less striking in their magnitude.)

Text Samples 8.10 and 8.11 illustrate the seventeenth-century differences among the gender categories. Sample 8.10, illustrating female-to-female patterns, is taken from a letter from Margaret Cavendish to her sister in 1666. It is marked by extensive use of first- and second-person pronouns, pronoun *it*, *wh*-clauses, and a generally high reliance on verbs and clauses rather than elaborated noun phrases. In contrast, Sample 8.11 illustrates male-to-male patterns in the seventeenth century, taken from a letter from Sir Thomas Browne to Mr. Elias Ashmole in 1658. This letter is marked by a much more extensive use of the informational features on Dimension 1: generally longer words, a more varied vocabulary, and frequent nouns modified by attributive adjectives or prepositional phrases (e.g., *humble thancks*; *courteous letter*; *hopefull recoverie*; *worthy endeavors*; *Hermeticall philosophy*; *the good newes of the hopefull recoverie of Mr Dugdale*; *tracts of that subject*).

Text Sample 8.10: Seventeenth-century female-to-female letter

Dear Sister Pye,

Distance of Place, nor Length of Time, cannot Lessen my Natural, or rather Supernatural Affection to you, for certainly my Love for you is more than a Sisters Love, nay, such a Love, as when I Lived with you, it could not choose but be somwhat Troublesome, by reason my Love was Accompanied with such Fears, as it would neither let you Rest, Pray, nor Eat in Quiet, for though it was a Watchful Love, yet it was a Fearful Love, for I remember I have oftentimes Waked you out of your Sleep, when you did Sleep Quietly, with Soft Breathing, fearing you had been Dead, and oftener have I laid my Face over your Mouth, to feel if you Breath'd . . .

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Text Sample 8.11: Seventeenth-century male-to-male letter
Most worthy Sr,

I returne you humble thancks for your courteous letter and the good newes of the hopefull recoverie of Mr Dugdale . . . It can very hardly fall into my apprehension how I can afford any addition unto your worthy endeavors, notwithstanding I have enclosed a list of such tracts of that subject wch I have by mee, most whereof I receaved from Dr Arthur Dee, my familiar freind, sonne unto old Dr Dee the mathematician; hee lived many yeares and dyed in Norwich, from whom I have heard many accounts agreable unto those wch you have sett downe in your annotations, concerning his father and Kelley; hee was a persevering student in Hermeticall philosophy and had noe small encouragement, having seen projection made . . .

Similar differences among the gender categories are found in the twentieth century, although letters written by both men and women are generally more involved than in the seventeenth century. Text Sample 8.12, written by Jean Rhys to her daughter in 1952, illustrates the extremely involved characteristics of female-to-female letters in this century: very frequent first- and second-person pronouns, contractions, private verbs (e.g., *think*, *guess*, *hope*), *that*-omissions, present tense verbs, pronoun *it*, etc.

Text Sample 8.12: Twentieth-century female-to-female letter
My dear Maryvonne,

I was awfully glad to have your letter written on Ruthie's birthday. But it was not dated and I couldn't distinguish the postmark so I'm still doubtful about exact day. September anyhow. I have now become a great numerologist – at least I do think it odd the way certain numbers turn up over and over again in people's lives. Two is your number I know, six is mine. All nonsense I daresay.

She sounds a great darling and I am sure you are awfully good with her. Perhaps you can guess how much I'm longing to see you both so I'll say no more about that except that I hope you'll have a grand time and it must be fun planning for it . . .

About Ruthie. Don't think I'm getting all grandmotherly if I remind you that Calcium is awfully good for teeth – It certainly worked with you. It isn't nasty only a bit dry . . .

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In contrast, Sample 8.13 – from a letter by Harry Truman written to his daughter in 1955 – is surprisingly un-involved: although the letter is directly interactive (marked by frequent references to *I* and *you*), it reads almost like a historical report rather than a personal letter, with a relatively specialized vocabulary and frequent nouns, attributive adjectives, and prepositional phrases (e.g., *civil government*; *violent protest*; *Republic of the Philippines*; *protests of special interests*; *the history of home rule and self-government*).

> Text Sample 8.13: Twentieth-century male-to-female letter
> Dear Margie:
>
> I sent you a letter yesterday but it was written in a big hurry. There was a Mennonite preacher waiting to give me a book he'd written and to ask me about Guam where he was going to teach history. The book is good but I couldn't reciprocate because I've never been to Guam. I did try to give it civil government, and did – over the violent protest of the Navy.
>
> I also gave Puerto Rico home rule and insisted on setting up the Republic of the Philippines over protests of special interests. I appointed a native to be governor of the Virgin Islands too. Ike sent an Iowa Negro to the Virgins and he failed dismally, and now he's sent a California one. He just doesn't know the history of home rule and self-government.

As in the previous sections of this chapter, a complete investigation of gender differences would require comparisons along additional dimensions of variation, as well as comparisons of female and male language in additional registers. Similar kinds of comparisons could also be carried out for other social differences (relating to class, education, ethnicity, etc.). The analyses presented here are thus not comprehensive; however, they clearly illustrate the effectiveness of a corpus-based approach for the investigation of such dialect differences.

8.5 Analyses of author style

Any discussion of historical and stylistic investigations would not be complete without considering the use of corpus-based techniques to investigate author style. Some of the earliest applications in computational linguistics involved the analysis of literary texts. Several of these studies focused on authorship attribution:

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Johnson's texts are being compared to those of authors of similar stature.

For this analysis of author style, we have singled out passages from Johnson's *The History of Rasselas: Prince of Abyssinia*. Two passages are included – one narrative and one descriptive. We analyze the stylistic characteristics of these texts with respect to three of the dimensions of register variation (see Chapter 6 and Section 8.3): "Involved versus informational production" (Dimension 1), "Situation-dependent versus elaborated reference" (Dimension 3), and "Non-impersonal versus impersonal style" (Dimension 5).

Figures 8.7–8.9 plot the dimension scores of both Johnson text samples relative to the comparable texts in the eighteenth century. The spread of scores within fiction for the other periods – seventeenth century, nineteenth century, and modern – provide a further standard of comparison. In the figures, the mean score for fiction in each period is marked by "*," and the range of scores is shown by the vertical bar.

All three of these dimensions identify Johnson's descriptive passage from *Rasselas* (J-D) as being the most stylistically distinctive of not only the eighteenth-century texts, but of all the historical periods. In fact, this text has characteristics that are more typical of informational expository prose than fiction. With respect to Dimension 1 (Figure 8.7), Johnson's text is extremely informational; along Dimension 3 (Figure 8.8), this text is extremely elaborated in reference; and along Dimension 5 (Figure 8.9), this text is extremely impersonal in style. Text Sample 8.14, from the very beginning of this novel, illustrates these extreme stylistic characterizations along all three dimensions. Relative clause constructions, associated with the elaborated reference dimension (D3), are given in small capitals; while passive constructions, associated with the impersonal style dimension (D5), are italicized.

> *Text Sample 8.14. Samuel Johnson*: The History of Rasselas, Prince of Abyssinia (eighteenth century)
>
> Ye WHO listen with credulity to the whispers of fancy, and pursue with eagerness the phantoms of hope, WHO expect that age will perform the promises of youth, and that the deficiencies of the present day *will be supplied* by the morrow, attend to the history of Rasselas, Prince of Abyssinia. Rasselas was the fourth

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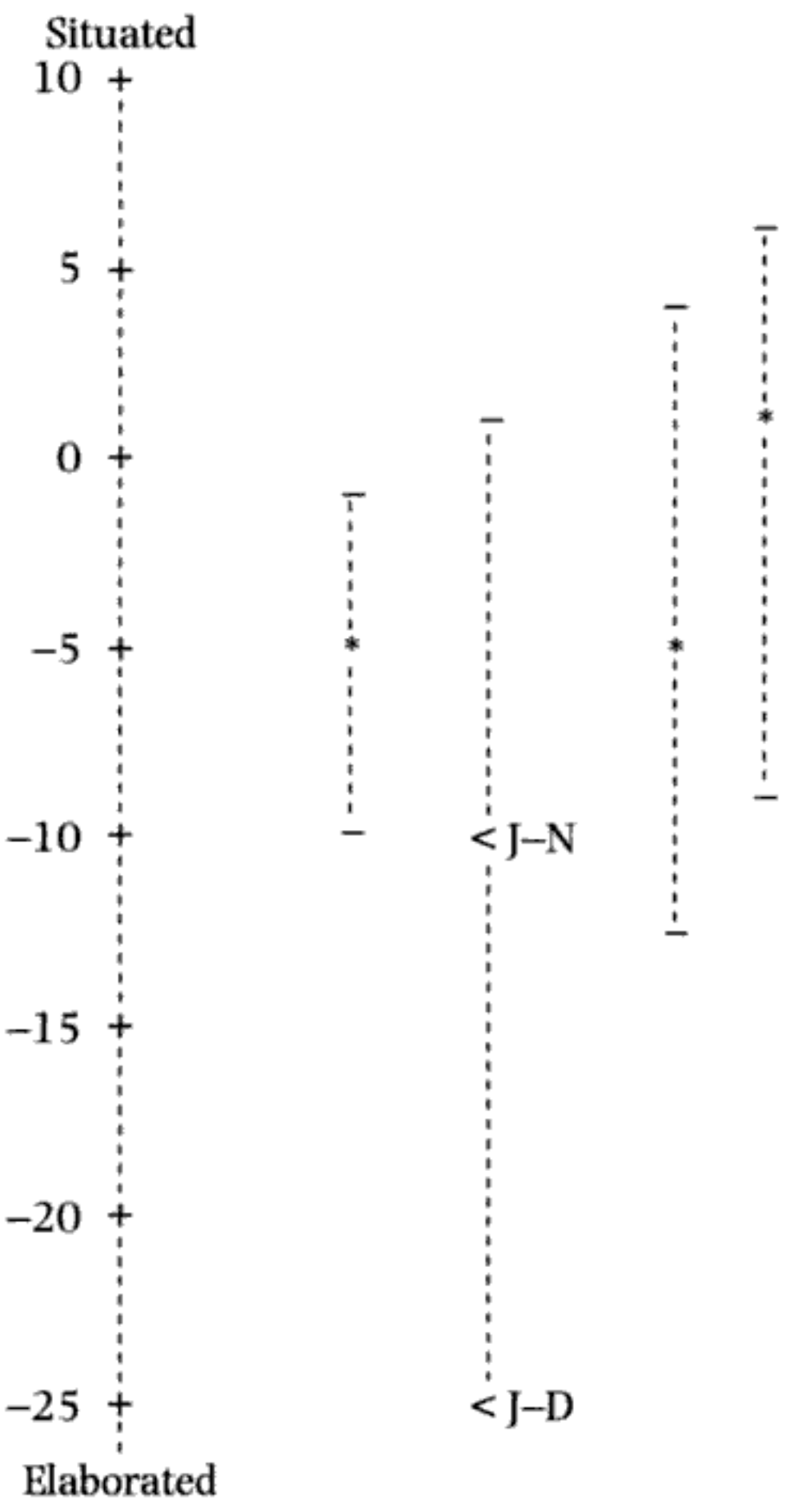

Figure 8.8 Characterization of fiction texts along Dimension 3, "Situation-dependent versus elaborated reference"[1]
(* marks the mean score in each century; J-N = Johnson narrative text; J-D = Johnson descriptive text)

– investigations of grammatical and lexicographic features, examining the use of modals and semi-modals across registers over four centuries;
– an investigation of register development, considering changes in the language use patterns of spoken and written registers over three centuries;

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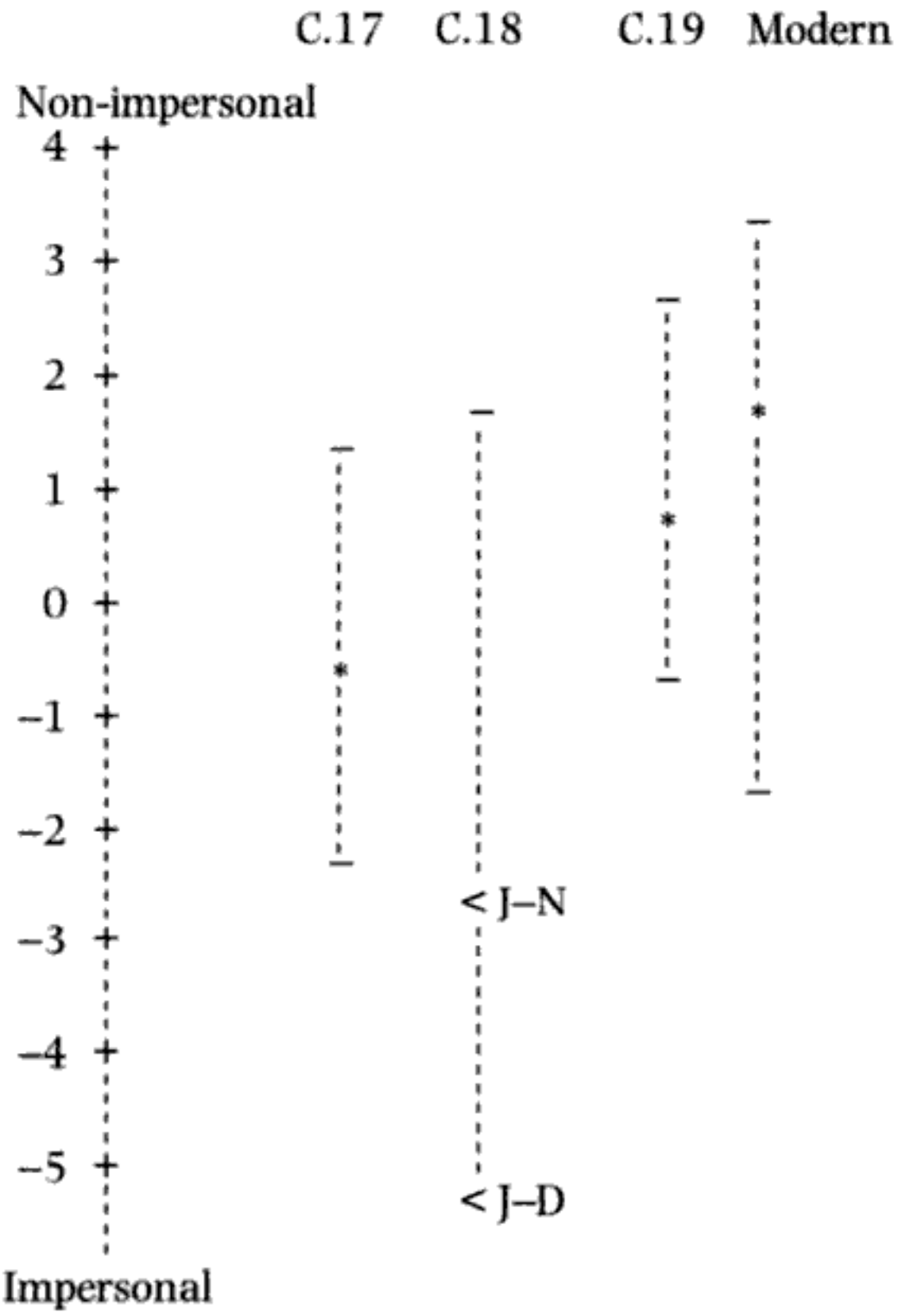

Figure 8.9 Characterization of fiction texts along Dimension 5, "Non-impersonal versus impersonal style"[1]
(* marks the mean score of each century; J-N = Johnson narrative text; J-D = Johnson descriptive text)

– examination of dialect differences, focusing on the language used by women and men in personal letters over four centuries;
– and investigation of individual author style, relative to other fiction texts across historical periods.

Doing these historical and stylistic analyses by hand would not only have been time-consuming; it would also have ruled out the investigation of complex co-occurrence patterns among linguistic features. Thus, though all the examples in this chapter have been brief, each one has shown how on-line corpora and computational techniques can facilitate our understanding of language use across time and styles.

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Note

1. In this chapter, the polarity of Dimensions 3 and 5 have been reversed in order to facilitate register comparisons across dimensions.

Further reading

Numerous historical linguistic studies have been carried out based on the Helsinki Corpus, including Kytö (1991), Nevalainen (1991), Rissanen (1986), and the volume edited by Rissanen, Kytö, and Palander-Collin (1993). Kytö (1993) includes a bibliography of studies based on the Helsinki Corpus. Kytö, Rissanen, and Wright (1994) offer a collection of papers describing other historical corpora, while Nevalainen and Raumolin-Brunberg (1996) describe studies based on the Corpus of Early English Correspondence. Multi-dimensional studies of historical register variation include Atkinson (1992, 1996), Biber and Finegan (1989, 1997), Biber, Finegan, and Atkinson (1994), and Biber (1995, especially Chapter 8).

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Éléments sous droits d'auteur

Part III

Summing up and looking ahead

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Éléments sous droits d'auteur

9 Conclusion

9.1 Contributions of the corpus-based approach

Throughout this book we have emphasized the usefulness of the corpus-based approach for studying how speakers and writers use the linguistic resources available to them in their language. This approach takes advantage of: computers' capacity for fast, accurate, and complex analyses; the extensive information about language use found in large collections of natural texts from multiple registers; and the rich descriptions that result from integrating quantitative findings and functional interpretations. For these reasons, the corpus-based approach has made it possible to conduct new kinds of investigations into language use and to expand the scope of earlier investigations.

These advantages apply to the study of individual linguistic features as well as the characterization of language varieties. Thus, for example, the application of corpus-based techniques in lexicography make it possible to study the collocations of words in a comprehensive way, identifying differences in the preferred senses of related words. Similarly, many researchers in the past have been interested in differences between speech and writing, but until the development of corpus-based methods it was impossible to discover the patterns of co-occurring features and dimensions of variation that characterize the two modes.

As an approach to linguistics, corpus-based analysis provides a new perspective on language use. The findings of the example analyses presented in this book, as well as the findings of other corpus-based studies, show that there are strong, systematic patterns in

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the way that language is used. In fact, one of the most consistent findings in corpus-based studies concerns the importance of register variation. In this book, you have seen how linguistic features from all levels – including lexical collocations, word frequencies, nominalizations, dependent clauses, and a full range of co-occurring features – have patterned differences across registers. Therefore characterizations of "general English" are usually not characterizations of any variety at all, but rather a middle-ground that describes no actual text or register.

Findings about the strong association patterns in language, and about the importance of register, show how the corpus-based approach identifies important characteristics that need to be included in a full description of language use. The corpus-based approach can thus bring to the fore aspects of language use that have not received attention in traditional studies.

9.2 Additional research with the corpus-based approach

Our primary goal in the present book has been to illustrate the wide range of research questions within linguistics that can be addressed with the corpus-based approach. Thus, we included investigations of individual words, grammatical constructions, discourse patterns, varieties of texts, language acquisition, and historical and stylistic issues. However, the breadth of work that is possible with corpus-based studies is far greater than we have been able to illustrate in the example analyses here.

First of all, for the areas of research included in the book, many additional investigations could be done asking related research questions and applying similar analytical techniques. For example, in grammar, you might be interested in the distribution of other word classes or structures across registers, or you might be interested in other structural variants, such as the factors influencing the choice between agentless passives and by-passives. For discourse analyses, you might be interested in tracking the use of other kinds of linguistic features, such as markers of stance. In the area of register analysis, you might be interested in registers used in the business world – investigating memos, reports, etc. – or

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the investigation of dimensions of variation in other languages, comparing registers and use of features across different languages (see, e.g., Biber 1995).

Finally, the corpus-based approach can be used to study specific situations that are important in society, in order to learn more about the language used in those contexts. Workplace or classroom discourse, for example, are fruitful arenas for a range of studies, examining language use across speakers, cultural groups, genders, and positions of differential power. As with other areas, analysis of a large body of authentic language can show the actual language patterns being used – rather than having to rely on intuition or anecdotes, which may or may not be accurate reflections of the dynamics between teachers and students or managers and employees.

9.3 The corpus-based approach and language education

One of the areas that we have not addressed in detail concerns the application of the corpus-based approach to language education. However, many of the findings of corpus-based studies, as well as the techniques in corpus-based research, have valuable applications in educational situations. In this section, we consider three ways that the corpus-based approach can inform education: the dissemination of findings from corpus-based studies, the development of educational materials, and the design of classroom activities.

In some chapters of this book, we have noted studies that are particularly applicable to ESL or ESP teachers. For example, in Chapter 6 we discussed the usefulness of our multi-dimensional analysis of disciplinary research articles for ESP teachers. However, findings of corpus-based studies can often be useful in other types of programs, too.

For example, many universities are now trying to incorporate a writing-across-the-curriculum component into their curricula. In such programs, it is often difficult to determine what kinds of writing and reading to include, how to sensitize students to

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differences across disciplines, or even how to make disciplinary experts overtly aware of how they use language in their disciplines. The designers of such programs are often left to rely on their intuitions about variation among texts. Findings from register studies, such as the multi-dimensional comparison of biology and history research articles, are directly applicable to these problems. The results of such corpus-based studies give concrete evidence of the similarities and differences across disciplines, and they can also show the relationship among different kinds of texts within a discipline (e.g., book reviews, review articles, textbooks, etc.).

In certain chapters, the usefulness of the corpus-based approach for materials development has already been discussed. We saw in Chapter 3, for example, that there are strong patterns in the conditions of use for subject position *that*-complement clauses. A review of several ESL textbooks showed, however, that students often get no explanation of these conditions of use. Thus, the results of corpus-based studies are an important resource for textbook writers – especially when the textbooks are aimed at non-native speakers who are unlikely to have the same intuitive sense of how to choose between grammatical variants.

Dictionaries and thesauruses for language learners (and native speakers, too) are also being improved with the inclusion of collocation information and grammatical preferences extracted from corpus analysis. This information provides students with a much greater understanding of how a word is used than a definition alone or a list of synonyms.

Finally, the techniques and results of corpus-based studies can be used to have students do their own language analyses (both for second-language students and for students training to be language teachers). As a concrete example, consider a classroom activity designed from the investigation of *that*-complement clauses, as described in Chapters 3 and 4. Figure 9.1 displays part of a handout to help students understand the use of *that*-clauses across two registers. The activity asks students to identify *that*-complement clauses and the verbs that control them in concordance listings from conversation and academic prose. (The concordance listings also illustrate the frequent omission of *that* in conversation, though that is not the focus of this activity.) Students then categorize the

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Instructions to students:

Below are concordance lines from two registers of English – conversation and academic writing. In this activity, you will investigate patterns in the use of *that*-clauses as verb complements in these two registers.

1. First, make a list of all the verbs that are followed by *that*-clauses in the conversation concordance listings below. Then, do the same for academic writing.
2. Categorize the verbs in each list by their functions. What do the verbs do – report what people said, present evidence, etc.?
3. Write down the patterns that you see in the use of *that*-clauses in conversation and academic writing. What are the similarities? What are the differences?

Concordance lines from conversation:

to cut down on your eggs. I think	0	that's the problem.
into the hospital, I know	that	he's afraid obviously cause
going no I don't mean	that	he's gonna fight you or
I thought	0	that was harder though cos you had to
He said	that	you'll say that I'm his sister
I can see	that	this one's still not on his feet.
yeah, she thinks	0	she's so cool though she doesn't
Oh that reminds me, did you say	0	you rang David this morning?
a bit of a reputation. I suppose	0	everyone has but
She knows	0	we love the pink champagne.
Well I suppose	0	a game of tennis doesn't last as long

Concordance lines from academic writing:

Laboratory studies show	that	rocks vary in strength and
Lyon and Wood (1977) have suggested	that	the constraints of time
They found	that	the acid formation is the
present philosophers have assumed	that	ethics can be based upon a
Therefore the physician must ensure	that	these social services
The dates indicate	that	grain products are the
Swan (1938) showed	that	the period of egg development
A recent study from Boston suggests	that	these patients have a
they undertook to ensure	that	each decision, one by
It could be said	that	the French Revolution of 1789

Figure 9.1 Sample language analysis activity

handling each of these areas, though different kinds of investigation require different amounts of expertise.

Many of the types of investigations conducted in the earlier chapters of the book can be done with concordancing software and corpora that are available for no charge or a nominal fee. If

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Further reading

There are numerous edited collections that contain corpus-based studies covering a wide range of topics; these include Aarts and Meijs (1986); Aarts and Meyer (1995); Aijmer and Altenberg (1991); Armstrong (1994); Fries, Tottie, and Schneider (1994); Johansson and Stenström (1991); Svartvik (1992); Thomas and Short (1996); and Flowerdew and Tong (1994).

In addition, two other textbooks on corpus linguistics have recently appeared, covering many of the topics referred to in this chapter from somewhat different perspectives: McEnery and Wilson (1996), and Stubbs (1996). Further empirical corpus studies and research in computational linguistics can be found in the journals *Literary and Linguistic Computing*, *International Journal of Corpus Linguistics*, *Computational Linguistics*.

The journal *Computer Assisted Language Learning* frequently has articles about corpora, concordancing, and other computer applications in the classroom. Other studies discussing pedagogical implications of corpus-based research include Biber, Conrad, and Reppen (1994), Carter and McCarthy (1995), Connor-Linton (1990), Johns (1986, 1994), Kennedy (1987, 1992), McEnery, Baker and Wilson (1995).

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Part IV

Methodology boxes

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Introduction

This section of the book presents ten methodology boxes. Each of these boxes addresses an important methodological area for corpus-based studies. These discussions center around three major topics:

1. Corpus design and compilation: Boxes 1 and 2
2. Computational analysis of corpora: Boxes 3, 4, and 5
3. Quantitative analyses and measures in corpus-based study: Boxes 6–10

The boxes present introductions to these topics, providing an overview of analytical techniques and focusing on methodological issues. However, these discussions do not include detailed "how-to" instructions for carrying out corpus-based studies. Thus, while the information in these boxes will help you understand corpus-based analytical techniques, you will need to learn more about these areas as you begin to carry out corpus-based research on your own.

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Methodology Box 1
Issues in corpus design

A corpus is not simply a collection of texts. Rather, a corpus seeks to represent a language or some part of a language. The appropriate design for a corpus therefore depends upon what it is meant to represent. The representativeness of the corpus, in turn, determines the kinds of research questions that can be addressed and the generalizability of the results of the research. For example, a corpus composed primarily of news reportage would not allow a general investigation of variation in English. Similarly, research based on a corpus containing a single type of conversation – such as conversations between teenagers – could not be generalized to conversation overall. Thus, whether you are designing a corpus of your own, choosing a corpus to use in a study, or reading others' corpus-based work, issues of representativeness in corpus design are crucial.

It is important to realize up front that representing a language – or even part of a language – is a problematic task. We do not know the full extent of variation in languages or all the contextual variables that need to be covered in order to capture all variation in texts. However, attention to certain issues will ensure that a corpus is as representative as possible, given our current knowledge of language. This methodology box introduces these issues, as well as some means for improving corpus design in the future.

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Approaches to sampling

Suppose you were asked to build a corpus to represent English. How would you decide what to include?

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relate to the number of texts from different categories, the number of samples from each text, and the number of words in each sample.

If too few texts are included, a single text can have an undue influence on the results of an analysis (see, for example, the discussion of the collocation *great burrow* in Chapter 2). Enough texts must be included in each category to encompass variation across speakers or authors. Investigating how many texts are needed for a category, Biber (1990) found that ten texts represented the categories of the LOB Corpus well for many grammatical features. However, most corpora include more texts than this in each category. (The importance of the number of texts per category does vary with different research designs; see also Methodology Box 8.)

The number of samples from each text also deserves attention because the characteristics of a text can vary dramatically internally. A clear example of this is with experimental research articles. As Chapter 6 shows, Introduction, Methods, Results, and Discussion sections have quite different patterns of language use. Thus, sampling that did not include all of these sections would misrepresent the language patterns found in research articles.

Finally, the number of words in each sample is important for providing a reliable count of features in a text. For many common grammatical features, Biber (1990) finds that counts are relatively stable across 1,000-word samples from a text. However, some grammatical features (such as subject position *that*-clauses discussed in Chapter 3) are so rare that they would require much larger samples for quantitative studies.

In addition, we note in Chapter 2 that lexicographic studies require particularly large corpora. Many words and collocations occur with low frequencies, and a corpus must be many millions of words, taken from many different texts, to enable investigations of their use. However, for all kinds of research, it is important to realize that size cannot make up for a lack of diversity.

Other considerations

In addition to the factors of design discussed above, there are other important considerations when creating or using a corpus.

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computer. Therefore, when you can write programs, you harness the power of the computer to fit your own research goals.

What are the advantages of writing your own programs?

There are several concrete advantages to writing your own computer programs rather than relying on available concordancing software. Perhaps most importantly, writing programs allows you to conduct analyses that are not possible with concordancers. For example, the complete analysis of *that*- and *to*-complement clauses in Chapters 3 and 4 would not have been possible with a concordancing package. Similarly, in the second part of this book, we explore the patterns of use among more than sixty different linguistic features. This scale of analysis would also be impossible with concordancing software.

Another advantage of writing your own programs is that you can do many analyses more quickly and accurately. In Chapter 3, for example, we carry out several investigations relating to nominalizations: their overall distribution, the distribution of each type (i.e., each nominalization suffix), and the use of individual nominalized words. All three of these investigations were carried out with a single computer program, with just one run through the corpus. In addition, the analyses were made more accurate by writing the program to exclude certain words that are not nominalizations and would have erroneously inflated the counts.

For certain analyses of conversation, too, programs can often increase speed and accuracy; for example, it is possible to ask the computer to keep track of counts of features for the different participants in a conversation. For analyses that require human judgments, programs can even be written to interact with the user – asking for judgments of certain items. (An example of an interactive program is discussed in Chapter 5.)

Still another advantage of programming is that you can tailor the output of the analysis to fit your research needs. At the same time that counts are compiled, you can have the computer make KWIC files coded with a variety of information. For example, KWIC files of *that*-complement clauses can be coded for the use or deletion of *that*, the type of subject in the matrix clause, and the verb taking

the complement. KWIC files can be adjusted to include different amounts of context. Counts can be calculated in raw, normed, and percentage forms. Output can even be produced for a graphic display, rather than counts or KWIC files (see Chapter 5).

Finally, when you write your own programs, there is no limit to the size of the corpus that can be analyzed. The program will analyze each text in the corpus sequentially, for as long as there are texts to analyze.

All of these advantages mean that writing your own computer programs opens up a wider variety of options in the research questions that you can investigate.

What do you need in order to write your own programs?

Fortunately (and contrary to many people's fears), programming for linguistic research questions does not require a special aptitude in computer science or mathematics. The basic tool you need is knowledge of a language that a computer understands. Many "programming languages" exist (e.g., Pascal, C), and it is simply a matter of learning how to use one so that you can tell the computer what to do. In addition, of course, when programming for corpus analysis, you need the linguistic knowledge that allows you to identify linguistic structures, so that you can give the computer accurate instructions. Computers follow whatever logic you tell them – so if you tell them to analyze something improperly, either because your linguistic analysis is faulty or because you didn't use the computer's language correctly, you will not get correct results.

It is easiest to learn to program in a course. However, not all courses are equally useful for the kinds of analyses that are needed in linguistics (i.e., analyses of words and word sequences). Therefore, a course in programming for linguistic analysis is likely to be most useful for you.

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Methodology Box 4 Characteristics of tagged corpora

The main use of an uncoded corpus is searching for a particular word or sequence of words. Concordancing software is designed especially to allow searches of this type, for example, to examine frequencies of words or collocations, or to find examples of certain words or structures. However, many linguistic investigations – including most of the analyses in this book – are not possible if we are restricted to simply searching for words. Even structures that conform to fairly regular morphological or syntactic patterns are not easy to study based on word searches.

Suppose, for example, that you wanted to investigate the use of passive voice. With an uncoded corpus, you might start by searching for any form of *be* plus a word ending in *-en*. This search pattern would find many passives – such as *was eaten* and *is taken* – but it would miss the passives ending in *-ed* (e.g., *been carried*, *was kicked*). Even if you expanded your search to include these passives, you would still miss all the irregular passives, of which there are many – e.g., *shown*, *torn*, *built*, *kept*, *sold*, *meant*, *brought*. Each ends with a different sequence of letters, making patterned searches impossible. In addition, you would miss instances that have intervening adverbs (e.g., *was completely eaten*). Conversely, this search pattern would fit structures that are not passives, such as *was green*, *is red*. As you expanded the search pattern to include more passives, you would also include more structures that are not passives. You would be left to do a great deal of editing and counting by hand – negating many of the advantages of the corpus-based approach.

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The solution to these problems is to use a corpus which includes grammatical information for each word – i.e., a grammatically "tagged" corpus. This methodology box discusses the kind of information that is included in tagged corpora. Methodology Box 5 then discusses the process of tagging and issues of accuracy in tagging.

Information in a grammatically tagged corpus

The following samples illustrate tagging for a sentence taken from news reportage in the LOB Corpus. Sample 1 comes from the tagged LOB Corpus. Sample 2 shows the same passage tagged by the tagger developed for multi-dimensional analyses, which we used for most analyses in this book.

Untagged sentence
A move to stop Mr Gaitskell from nominating any more labour life peers is to be made at a meeting of labour MPs tomorrow.

Tagging Sample 1
^ a_AT move_NN to_TO stop_VB \0Mr_NPT Gaitskell_NP from_IN nominating_VBG any_DTI more_AP labour_NN life_NN peers_NNS is_BEZ to_TO be_BE made_VBN at_IN a_AT meeting_NN of_IN labour_NN \0MPs_NPTS tomorrow_NR ._.

Tagging Sample 2
A ^at++++
move ^nn++++
to ^to++++
stop ^vbi++++
Mr ^npt++++
Gaitskell ^np++++
from ^in++++
nominating ^xvbg+++xvbg+
any ^dti++++
more ^ap++++
labour ^nn++++
life ^nn++++
peers ^nns++++
is ^vbz+bez+aux++
to ^to++++

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be ^vb+be+aux++
made ^vpsv++agls+xvbnx+
at ^in++++
a ^at++++
meeting ^nn+++xvbg+
of ^in++++
labour ^nn++++
MPs ^npts++++
tomorrow ^nr+tm+++
. ^.+clp+++
```

These two examples illustrate several important ways that tagged corpora vary. A first obvious difference is the layout. Sample 1 has a horizontal format with each tag attached to its word with an underscore. Thus a_AT signifies that the word is *a* and it is tagged as an indefinite article (AT). Sample 2 has a vertical format, with each word on a separate line, a space after the word, and the tag beginning with ^.

A more substantive way that the samples differ, however, is in the amount of information given by the tags. The + marks in Sample 2 separate different "fields" of information for each word, while Sample 1 includes just one field. In some cases, the information given in both samples is the same (e.g., *a* is tagged as *AT*, the indefinite article). However, Sample 2 includes more information about certain words. Consider the sequence *to be made*, for example. In Sample 1 the corresponding tags show: 1. the infinitive marker (*to*), 2. the infinitive form of *be* (*be*), and 3. a past participle verb form (*vbn*). In Sample 2 the corresponding tags show: 1. the infinitive marker (*to*); 2. the unconjugated form of a verb (*vb* in the first field) with the verb further identified as the copula *be* (*be* in the second field) used as an auxiliary verb (*aux* in the third tag field); and 3. a passive form of a verb (*vpsv* in the first field) used as an agentless passive (*agls* in the second field), also identified as a past participle form (*xvbnx* in the third field). Certain semantic information is also included in the second sample. The word *tomorrow*, for example, is identified as an adverbial noun (*nr*) in the first tag field, but also as referring to time (*tm*) in the second field. Tagged corpora thus vary with respect to the amount of information that they include about words. All taggers (as opposed

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to parsers, see "other annotations of corpora" below) focus primarily on grammatical class information, but different taggers will include differing amounts of grammatical information and some will also include some semantic and syntactic information.

Other annotations of corpora

Grammatical tagging is not the only type of annotation that can be added to a corpus, although it is the most common. There are also parsers – programs which add syntactic analysis to a corpus, identifying subjects, verbs, objects, as well as more complex syntactic information. Semantic features can also be annotated in corpora, as can prosodic features for spoken corpora. The appendix gives further information about publicly available programs that annotate corpora.

Further reading

Biber (1993c); Garside, Leech, and Sampson (1987); Karlsson, Voutilainen, Heikkilä, and Anttila (1995); Leech (1993); Leech, Garside, and Bryant (1994); Souter and Atwell (1994); McEnery and Wilson (1996).

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regular sequences. For example, if the sequence is a form of *be* + adverb + past participle, the rule-based component will analyze the past participle as a passive verb, with an intervening adverb. Finally, morphological analyzers make use of certain morphological characteristics – such as *ly* commonly occurring at the end of adverbs – to help analyze words that are not found in the dictionary.

Taggers rely on these various kinds of information to different extents. However, all taggers put together the different information to either definitely identify the tag for a word (in the case of unambiguous items) or to make the "best guess" based on all the information available.

Accuracy and editing

Given the complexity of natural language, it is sometimes difficult for a machine to make accurate decisions about tags. Accuracy will also vary with the amount of information included in tags (e.g., it is much easier to identify a word as a verb than to specify that is a past participle form of a verb used as a passive postnominal modifier). Many reports on taggers report accuracy in the mid to high 90 percent range; however, no automatic tagger is 100 percent accurate.

Occasional errors in tagging should not be taken as a major roadblock to research. Errors tend to occur with certain types of ambiguous structures. When carrying out a study based on a tagged corpus, it is essential to check and, if necessary, correct features that are important for your research. An interactive computer program designed specifically for checking tags can make this process much faster and more reliable (this type of interactive program is described in Chapter 5).

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Methodology Box 7
Statistical measures of lexical associations

The simplest way to identify collocate pairs is by their relative frequency – that is, by how commonly one pair, such as "large number," occurs relative to another pair, such as "large man." Such frequency information can give a sense of the most common collocational associations.

However, frequency information alone may present a biased measure of the strength of associations between words. More common words are more likely to occur in a collocate pair simply by chance. Therefore, an alternative way to judge the strength of associations between words is to use measures that account for the likelihood of words occurring together by chance – i.e., statistical measures.

Statistical measures can be used to analyze both the associations between collocate pairs and the differences between the collocations of particular words. Below we briefly review a common statistical test for each of these purposes. We explain the principles behind the tests, but for details about the statistical formulas, you should consult the articles listed under "Further reading."

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Mutual information score

The mutual information score or mutual information index gives a measure of the strength of association between two words. It focuses on the likelihood of two words appearing together within a particular span of words (the span is specified for the analysis, e.g., adjacent words, a window of three words, etc.).

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are more likely to occur with *large*. As with the frequency analyses in Chapter 2, the lists show the differences in the typical uses of *big* and *large*: *big* tends to be used to show physical size (e.g., as *big man, big toe, big boy, big dog, big house*) and also in the expression of non-importance (*big deal*), while *large* tends to be used to show amounts or parts (*large number(s), large proportion, large amount(s), large quantities*). Unlike frequency counts alone, however, these lists show that these patterns of use hold even when we account for the random chance of words occurring together.

Summary

Most concordancing programs include a statistical measure of word associations when they display a list of collocations. (Some, such as TACT, use *z*-scores rather than mutual information scores. The principles behind *z*-scores are similar, though the mathematical calculations vary.) *T*-scores are usually included with the more sophisticated concordancing packages, such as Corpus Bench.

For the use of all statistical measures, certain considerations are important. Prime among these is the frequency of the items that are studied. Statistical calculations are generally not reliable with very low frequencies, and for any statistical test, it is important that you make sure you have enough tokens for the analysis. In addition, it is important that you use a procedure that fits your research question. If you want to contrast the use of two words, for example, *t*-scores are appropriate, but they would not be appropriate for compiling a list of the most important collocates for a single word.

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Further reading

Biber (1993d); Church and Hanks (1990); Church, Gale, Hanks, and Hindle (1991); Nakamura and Sinclair (1995); Stubbs (1995).

The manuals of many concordancing programs also explain the statistical tests used in the package.

Methodology Box 8
The unit of analysis in corpus-based studies

One of the very first decisions required when carrying out a corpus-based analysis is to determine what your unit of analysis is. This is a crucial decision because it determines the object of your research; if you do not identify the unit of analysis properly, you may not be able to address the research questions that you have posed.

As you see in this book, corpus-based studies usually have one of two primary research goals: describing a linguistic structure and its variants (as in Part I of the book), or describing some group of texts (as in Part II of the book). Thus, the unit of analysis in corpus-based studies is typically one of two kinds: either occurrences of a linguistic feature or a text. These units of analysis are called the "observations" for the study. Thus, in a study that characterizes a linguistic structure, each observation is an occurrence of the structure in question. In a study that seeks to describe a group of texts, each observation is a text.

Suppose that we plan a study to analyze variants of *that*-complement clauses and to determine which contextual factors are most strongly associated with each variant. The study should be designed so that each occurrence of a *that*-complement clause is a separate observation. For example, two major variants of *that*-complement clauses are with and without the complementizer *that*, as in:

(1) I do not think that the situation is slipping out of control.

versus

(2) I don't think [] any of us would be willing to do that.

There are a number of contextual factors that might be influential in making the choice between these two variants. For example, it might be the case that the complementizer *that* is omitted more often with common matrix verbs, such as *think*, than with less common verbs, such as *show*. It also might be the case that the complementizer *that* is omitted more often with first-person pronouns as subject than with other subjects. Finally, it should be noted that register can also be a contextual factor for individual linguistic features; thus, it might be the case that the complementizer *that* is omitted more often in conversation than in other registers.

To investigate the relative influence of these contextual factors, we would code a large sample of *that*-clause constructions, where each clause constitutes a separate observation. The output of such an analysis might look like this:

present	indicates	noun	news
present	suggests	pro-she	news
omitted	said	pro-he	news
present	argued	noun	news
present	said	pro-I	news
present	implied	noun	news
omitted	thought	pro-I	news
present	think	pro-they	news
present	thinks	noun	news
omitted	think	pro-I	conv
omitted	said	pro-I	conv
present	felt	pro-I	conv
omitted	says	pro-I	conv
omitted	thought	pro-I	conv
omitted	think	pro-I	conv

Each line represents the information about a single observation – i.e., a single occurrence of a *that*-clause. Each column represents the values for a different variable. The first column shows whether the complementizer *that* is present or omitted; the second column records the matrix verb; the third column records the type of matrix-clause subject; and the fourth column records the register that the example was taken from. The first line in the output above, for example, would correspond to the following *that*-clause occurrence:

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Register could also be used as a variable in a cross-tabulation table. Here are the data from the observations above:

Comp	Register		
that	News	Conv.	Total
omitted	2	5	7
present	7	1	8
Total	9	6	15

From this distribution you should see that the data suggest that retention is favored in news, but omission is favored in conversation. (Again, of course, many more observations would be needed in an actual analysis!)

Given a large enough data set, we could also consider the influence of multiple factors at the same time. For example, we could contrast the influence of the matrix verb for *that*-clauses in news reportage with the influence of this factor in conversation. The following tables show the results based on 100 coded *that*-clauses from each register:

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Patterns for news reportage

Comp	Matrix verb		
that	THINK or SAY	other	Total
omitted	27	8	35
present	24	41	65
Total	51	49	100

Patterns for conversation

Comp	Matrix verb		
that	THINK or SAY	other	Total
omitted	48	39	87
present	3	10	13
Total	51	49	100

the size of the average differences among registers relative to the size of the differences among the texts within a register. (Methodology Box 9 explains more about the use of significance tests and the reporting of statistics.)

Further reading

Sankoff (1987), Scholfield (1995).

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Methodology Box 9
Significance tests and the reporting of statistics

Inferential statistics provide an important tool for assessing whether observed patterns are meaningful. There are many different statistical techniques, depending on the types of variables that are being compared. Statistical techniques can be used to measure the differences between groups (as with a *t*-test or ANOVA) or the extent of the relationship between variables (as with a correlation). Each technique can be used to produce a test of significance, assessing the likelihood that the observed difference or relationship could be due to chance, and a measure of strength, assessing the importance of the difference or relationship. This methodology box introduces you to some of the most common statistical techniques used in corpus-based studies.

A *t*-test is used to determine if a significant difference exists between two groups. The statistical procedure compares the distance between mean scores for the two groups relative to the amount of variation that exists within each group. The *t*-value is a score measuring the likelihood that the observed difference could be due to chance. To evaluate the significance associated with a score for *t*, it is necessary to also consider the number of observations analyzed in the study. A relatively small difference in mean scores can be significant if it is based on a large number of observations, while a relatively large difference might not be significant if it is based on few observations. In reporting the results for a *t*-test, the number of observations is reflected in the "degrees of freedom," which is the number of observations in a group minus one.

If a *t*-score represents a significant difference, the result is reported in terms of the probability level (*p*) being less than some pre-determined cut-off: either .05, .01, or .001. The value for *p* measures the likelihood of an error – i.e., the researcher incorrectly concludes there is a difference when in fact the two groups are the same. For example, a *p* of .01 means that there is only a one in a hundred chance that the observed difference was due to chance. The smaller the *p*-value, the less likely it is that the observed difference occurred due to chance.

Analysis of Variance (ANOVA) is used to compare more than two groups. This test is similar to a *t*-test in that it compares the extent of variation among groups to the variation within groups, to determine whether the observed differences among groups are significant. Instead of a *t*-score, ANOVA produces an *F*-score, which is similarly interpreted relative to the degrees of freedom. The final probability level is also reported using *p*, which is interpreted in exactly the same way as in a *t*-test.

Pearson correlation is used to measure the extent of the relationship between two numeric variables (rather than the difference between groups). The Pearson correlation coefficient, *r*, can be squared to provide a direct measure of the proportion of the variation in one variable that can be accounted for by knowing the values for some second variable. An r^2 value near 1.0 represents a near perfect correlation, with 100 percent of the variation in one variable being accounted for by knowing the second variable. An r^2 value near 0.0 represents the absence of any correlation at all.

Correlations can also be computed for *t*-tests and ANOVA. The correlations in this case provide a direct measure of strength or importance. For example, the r^2 value of .549 in Table 5.3 shows that 54.9 percent of the variation in the normed counts for present tense can be accounted for by knowing the register category of each text. This measure shows that the differences across registers in the use of present tense verbs are very important in addition to being statistically significant.

Finally, a chi-squared test is used to compare two "nominal" variables – when both variables represent groups rather than

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Methodology Box 10 Factor loadings and dimension scores

Several of the sample studies in the second part of this book use multi-dimensional analysis. As explained in more detail in Chapter 6, in MD analysis the co-occurrence patterns among a large number of linguistic features are identified with the statistical technique known as factor analysis. In a factor analysis, the correlations among a large number of variables (i.e., the linguistic features) are identified, and the variables that are distributed in similar ways are grouped together. Each group of variables is a factor – which is then interpreted functionally as a "dimension" of variation. In this methodology box we discuss the relationship between an individual linguistic item and the factor that it is part of, as well as the determination of a "factor score" for a given text with respect to a given factor.

Chapter 6 (see Table 6.2) presented five factors (the dimensions) found in the MD analysis of English. Part of the first factor is reproduced here for reference:

Dimension 1: "Involved versus informational production"	
private verbs	.96
that-deletion	.91
contractions	.90
present tense verbs	.86
2nd-person pronouns	.86
. . .	
possibility modals	.50
non-phrasal coordination	.48
wh-clauses	.47
final prepositions	.43

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nouns	−.80
word length	−.58
prepositions	−.54
type/token ratio	−.54
attributive adjs.	−.47

Next to each feature which is included on this factor you see a number. This is the factor "loading." It gives a measure of the strength of the relationship between the particular linguistic feature and the factor. In other words, it shows how representative a particular linguistic feature is of the construct underlying the factor. Loadings range from +1 to −1. A loading closer to |1| reflects greater representativeness of the factor, with a value of exactly 1 reflecting a perfect correlation. As the loading moves away from 1 and closer to 0, the strength of the association diminishes. Thus, for the features on Dimension 1, private verbs and nouns (with their factor loadings of .96 and −.80) are more representative of the factor than possibility modals (.50) or attributive adjectives (−.47).

Every feature has some loading on each factor. Because features with larger loadings are more representative of the factor, they are also more useful in the functional interpretation of the factor. A cut-off score is used to eliminate features that do not contribute meaningfully to the interpretation of a factor. Typically a feature with a value of less that |.30| is not considered in the interpretation of the factor.

Factor scores (or dimension scores) provide an overall characterization of a text with respect to all the salient features grouped on a factor. A factor score can be computed for each text by summing the frequencies of the features having salient loadings on that factor. In the study of spoken and written registers (described in Chapter 6, Section 6.3), only features with loadings greater than |.35| on a factor were considered important enough to be used in the computation of factor scores. For example, the Factor 1 score for each text is computed by adding together the frequencies of private verbs, *that*-deletions, contractions, present tense verbs, etc. – the features with positive loadings on Factor 1 – and then subtracting the frequencies of nouns, word length, prepositions, etc. – the features with negative loadings.

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In all MD studies, frequencies are standardized to a mean of 0.0 and a standard deviation of 1.0 before factor scores are computed. This process translates the scores for all features to scales representing standard deviation units. Thus, regardless of whether a feature is extremely rare or extremely common in absolute terms, a standard score of +1 represents one standard deviation unit above the mean score for the feature in question. That is, standardized scores measure whether a feature is common or rare in a text relative to the overall average occurrence of that feature. The raw frequencies are transformed to standard scores so that all features on a factor will have equivalent weights in the computation of dimension scores. If this process was not followed, extremely common features would have a much greater influence than rare features on the dimension scores.

The methodological steps followed to standardize frequency counts and compute dimension scores are described more fully in Biber (1988: 93–97).

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Appendix: commercially available corpora and analytical tools

This appendix lists resources that can be useful as you pursue your own corpus-based investigations. The appendix consists of three main sections: corpora, analytical tools, and on-line resources. The list of corpora is divided into categories (e.g., written, spoken, historical), with corpora arranged alphabetically within each category. This list also notes whether the corpus is composed of American English (AE) or British English (BE) texts, along with the approximate number of words. The list of tools is also arranged alphabetically, with a brief description of each tool. For both corpora and tools, addresses to contact are provided at the end of the entry.

The World Wide Web is also becoming a productive source of corpora and tools, and several useful nodes on the Web are listed under "On-line sources." New sources are constantly being added, so interested readers should periodically search the web for new sites.

Corpora

Written

ACL/DCI (Association of Computational Linguistics/Data Collection Initiative): inludes 63 million words written AE; 40 million words from the *Wall Street Journal*; 23 million words from scientific abstracts. Contact: Department of Linguistics, University of Pennsylvania, Philadelphia, PA 19104 USA

Brown: c. 1 million words of written AE; many registers. Contact: ICAME, Norwegian Computing Centre for the Humanities, Harald Hårfagres gate 31, N-5007 Bergen, Norway (e-mail: icame@hd.uib.no)

Freiburg: c. 1 million words written BE; parallel to the LOB, but from material published in 1991. Contact: Christian Mair, Englisches Seminar I, Institut für Englische Sprache und Literatur, Albert-Ludwigs Universität, D-7800 Freiburg, Germany

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Historical

ARCHER Corpus: c. 2 million words BE and AE, 1650–1990; written and speech-based registers. Contact: Douglas Biber, Department of English, Northern Arizona University, Flagstaff, AZ 86011–6032, USA (e-mail: douglas.biber@nau.edu)

Helsinki Corpus: c. 1.5 million words from 850–1710; many registers. Contact: ICAME, Norwegian Computing Centre for the Humanities, Harald Hårfagres gate 31, N-5007 Bergen, Norway (e-mail: icame@hd.uib.no)

Helsinki Corpus of Older Scots: 830,000 words from 15 registers 1450–1700. Department of English, University of Helsinki, Portania 311, 00100 Helsinki, Finland

Innsbruck Computer Archive of Middle English Texts (ICAMET): c. 2 million words of Middle English prose from 1100–1500 texts arranged alphabetically. Contact: e-mail: manfred.markus@uibk.ac.at

Newdigate Newsletter: 750,000 words of manuscript newsletters from 1674–92. Contact: ICAME, Norwegian Computing Centre for the Humanities, Harald Hårfagres gate 31, N-5007 Bergen, Norway (e-mail: icame@hd.uib.no)

Zurich Corpus of English Newspapers (ZEN): London newspapers from the mid 1660s to the beginning of the twentieth century. Contact: Dr. Udo Fries, Englisches Seminar, University of Zurich, Platten str 47, CH-8032 Zurich, Switzerland (e-mail: ufries@es.unizh.ch)

Varieties of English

Hong Kong University of Science and Technology (HKUST) Learner Corpus: c. 6 million words (with on-going collection) of written undergraduate assignments and "A" level Use of English scripts from the Hong Kong Examination Authority. Contact: Gregory James, Language Center, Hong Kong University of Science and Technology, Clear Water Bay, Hong Kong

International Corpus of English (ICE): A collection of corpora, each representing spoken and written registers from a national variety of English. Each corpus c. 1 million words – in-progress collection. Contact: Survey of English Usage, University College London, Gower Street, London WC1E 6BT UK

International Corpus of Learner English (ICLE): c. 1 million words of written English texts from nine different language backgrounds: Chinese, Czech, Dutch, Finnish, French, German, Japanese, Spanish, and Swedish. Contact: Sylviane Granger, University of Louvain. B-1348 Louvain-La-Neuve, Belgium (e-mail: granger@etan.ucl.ac.be)

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Analytical tools

Conc 1.7: A freeware Macintosh concordancing program limited to small texts. Many search options are available and results can be exported to a file. This software can be downloaded from the Web site for the Summer Institute of Linguistics. Contact: Summer Institute of Linguistics, 7500 West Camp Wisdom Road, Dallas TX 75236, USA

Corpusbench: Word counts, concordancing, simple grammatical and morphological analyses (e.g., past tense "ed"), mutual information index. Can handle large corpora. Need to construct a text database. Contact: Textware Direct, Hörscholmsgrade. 20 2, DK-2200 Kϕbenhaven N, Copenhagen, Denmark

FreeText Browser: A fast concordancing program for Macintosh computers. This concordancer has no file size limitations and works quickly. Unfortunately, there is no method for printing or extracting files. Contact: FreeText Browser, PO Box 598, Kensington, MD 20895, USA (e-mail: science@nems.dt.navy.mil)

International Corpus of English Utility Progam (ICEUP): For use only with the International Corpus of English. Contact: Survey of English Usage, University College London, Gower Street, London WC1E 6BT UK

LEXA: A powerful shareware program for concordancing and simple tagging. Contact: ICAME, Norwegian Computing Centre for the Humanities, Harald Hårfagres gate 31, N-5007 Bergen, Norway (e-mail: icame@hd.uib.no)

Longman Mini Concordancer: For use with ASCII texts less than 50,000 words. Provides frequencies, KWICs, and some text statistics. Contact: Longman Group UK, Longman House, Burnt Mill, Harlow, Essex, CM20 2JE UK

Nijmegen Linguistic DataBase Software (LDB): Browsing, concordancing, and syntactic pattern searches for use with the Nijmegen Corpus. It can also be used with other parsed corpora that have been adapted for use with the LDB. Contact: TOSCA group, Department of Language and Speech, University of Nijmegen, Erasmusplein 1, NL-6525 HT Nijmegen, The Netherlands. (e-mail: hvh@lett.kun.nl)

MicroConcord: Word counts, concordancing, simple syntactic analyses, some morphological analyses (e.g., past tense "ed"). Can be used with a variety of languages and alphabets. Works on ASCII files and those from major word processing programs. Site licenses are available. Contact: Athelstan, PO Box 8025, La Jolla CA 92038–8025 USA (Phone: (800) 598–3880)

Micro-OCP: A concordancing program with many options for indices and statistics. Frequency lists and statistics can be saved to an output file.

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Can be used with a variety of languages. Contact: Electronic Publishing, Oxford University Press, 200 Madison Ave., New York, NY 10016, USA (FAX: (212) 725-2972)

TACT: A shareware progam capable of frequency counts, concordancing, and collocations. Limited as to the size of the corpus. Any ASCII text can be used to construct a text database. Contact: Center for Computing in the Humanities, Room 14297A, Robarts Library, University of Toronto, Toronto, Ontario M5S 1A5 Canada. (e-mail: cch@epas.utoronto.ca). TACT can be purchased in a package with a variety of literary texts from the Modern Language Association, 10 Astor Place, New York, NY 10003, USA.

WordCruncher: A concordancing program that also produces frequency lists and collocations. Texts must be prepared specifically for use with this program. Corpora in WordCruncher format are available from ICAME. Contact: Johnston & Company, P.O. Box 446, American Fork, UT 84003 USA.

ON-LINE SOURCES: These sources can be accessed via the World Wide Web (http), FTP, or Gopher.

ALEX: A Catalogue of Electronic Texts on the Internet http://www.lib.ncsu.edu/stacks/alex-index.html gopher: vega.lib.ncsu.edu/00/library/alex/about%20alex

Association for Computational Linguistics Special Interest Group on the Lexicon: includes information about corpora particularly for lexical and language acquisition studies http://www. clres.com/siglex.html

BNC: Information about the BNC http://info.ox.ac.uk/bnc/

CobuildDirect World Wide Web Service. http: //titania.cobuild.collins.co.uk/

Corpus linguistics: a rich source of corpus linguistics information with links to other useful sites. http://www.ruf.rice.edu/~barlow/corpus.html

Electronic Newsstand. http: //www.enews.com/ gopher: gopher.enews.com:2100/11/magazines

Georgetown Catalogue of Projects in Electronic Texts. gopher: gopher-test.georgetown.edu/11/computing/humanities/

Gutenburg Etext Wwide Web Home Page. http: //jg.cso.uiuc.edu/pg/welcome.html

HotWired (registration required). http: //www.wired.com/

International Computer Archive of English (ICAME). http://www.hd.uib.no

Linguistic Data Consortium: information about the LDC, including catalogs of corpora. http://www.ldc.upenn.edu

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On-line Books Page. http: //www.cs.cmu.edu:8001/web/books.html
Oxford Text Archives. http: //ota.ox.ac.uk/tei/ota.html FTP: ota.ox.ac.uk
Path: /ota/textarchive.list
Summer Institute of Linguistics. http://www.sil.org
The English server. http://english-server.hss.cmu.edu

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This book is about investigating the way people use language in speech and writing. It introduces the corpus-based approach to linguistics, based on analysis of large databases of real language examples stored on computer.

Each chapter focuses on a different area of linguistics, including lexicography, grammar, discourse, register variation, language acquisition, and historical linguistics. The chapters follow the same progression of topics, beginning with the identification of important research questions, the steps followed for corpus-based analysis, and finally the interpretation of the findings. Example analyses are presented to provide concrete descriptions of the research methods and advantages of corpus-based techniques.

Ten methodology boxes provide clear and concise explanations of the issues involved in doing corpus-based research and reading corpus-based studies, and there is a useful appendix of resources for corpus-based investigation.

This lucid and comprehensive introduction to the subject will be welcomed by a broad range of readers, from undergraduate students to professional researchers.

Douglas Biber is Professor in the Applied Linguistics Program in the Department of English at Northern Arizona University. He is the author of *Variation Across Speech and Writing* (1991) and *Dimensions of Register Variation* (1995). Susan Conrad is Assistant Professor in the Program in Linguistics in the Department of English at Iowa State University. Randi Reppen is Director of the Program in Intensive English in the Department of English at Northern Arizona University.

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